



Republic of Zambia

MINISTRY OF EDUCATION

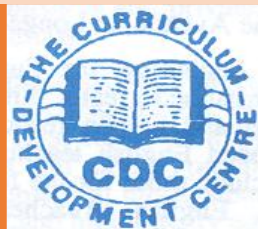
ORDINARY LEVEL

PHYSICS

TEACHING MODULE

Form 2

TERM 1 & 2



DEVELOPED BY THE CURRICULUM DEVELOPMENT CENTRE
P.O. BOX 50092
LUSAKA

2026

**PHYSICS
TEACHING MODULE
FORM 2
TERM 1 & 2**

© Curriculum Development Centre, 2026.

All rights reserved. No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording or otherwise without prior written permission of the copyright owner.



Authors

Bridget Changwe : Lusenga Secondary School, Chitambo
Mbewe Samuel Chingaipe : Chizongwe Provincial STEM School, Chipata
Mvula Anderson Kambani : Mufulira College of Education, Mufulira

Coordinator

Ireen Bwembya (PhD) : Directorate of Curriculum Development, Lusaka

Supervisors

Kenneth Mudenda : Directorate of Curriculum Development, Lusaka
Njapau Samson : Directorate of Curriculum Development, Lusaka

Vision

Quality, life- long education for all which is accessible, inclusive and relevant to individual, national and global needs which aligns with the 8 NDP and SDG 4.

Preface

The Physics Teaching Module for Forms 2 is designed to guide the teachers on how to deliver Competence-based lessons to facilitate the understanding of Physics concepts, fostering a deep appreciation for the role of Physics in everyday life and its applications in various fields. This module aims to develop a solid foundation in Physics and cultivate critical thinking, analytical skills, and problem-solving strategies. Subsequently, apply Physics concepts to real-world problems and emerging technologies through the engagement of learners in hands-on, hearts-on and minds-on practical activities and simulations to reinforce theoretical understanding.

This Physics Teaching Module for Form 2 intends to create a stimulating and supportive learning environment where teachers can prepare lessons with a profound understanding of Physics. The module will guide the teachers to support learners grow intellectually and personally by preparing them for professions in science and technology as well as for higher education by inculcating curiosity, critical thinking, and practical skills.

It is hoped that this module will inspire teachers to prepare lessons which are competence based with the fascinating world of Physics and appreciate its significance in shaping the future.

Kelvin Mambwe, (PhD)
Permanent Secretary- Educational Services
MINISTRY OF EDUCATION

Acknowledgement

This Physics Teaching Module is designed to provide the teacher with a scope and sequence of topics considered necessary to be offered at Secondary School level in Form 2 of term one (1) and term two (2). This is with a view of providing guidance to the teaching and learning of this unique, but yet existing blend of concepts of Physics for teachers and other experts in the field to appropriately offer relevant lessons at Secondary ordinary level.

Many thanks go to individuals, institutions and Organisations that participated in the successful development of this module. These include; teachers, lecturers from colleges, public universities in Zambia and in particular, the University of Zambia. Sincere gratitude also goes to the Directorate of Secondary Education and National Science Centre in the Ministry of Education for their support and collaboration during the consultation period.

Last but not the least, the commitment and hard work of all the staff at the Curriculum Development Centre in ensuring that this module comes to reality is recognised.

Charles Ndakala, (Dr.)
Director – Curriculum Development
MINISTRY OF EDUCATION

Table of Contents

Preface	vi
Acknowledgements	vii
How to use this Module	x
Introduction	xi
Suggested Teaching and Learning Materials	xi
Learning Environment Set Up	xii
Safety in the Learning Environment.....	xii
Teaching Methodology	xii
Learning Activities.....	xiii
Time Allocation	xiii
Icons used in this Module	xiii
Assessment	xiv
Key Competences.....	xiv
PHYSICS TEACHING MODULE TERM 1	1
TOPIC 2.1: MECHANICS 2	2
SUB-TOPIC 2.1.1: WORK, ENERGY AND POWER	2
SUB-TOPIC 2.1.2: LINEAR MOMENTUM	22
SUB-TOPIC 2.1.3: SIMPLE MACHINES	35
SUB-TOPIC 2.1.4: PRESSURE	46

PHYSICS TEACHING MODULE TERM 2	65
TOPIC 2.2: THERMAL PHYSICS.....	66
SUB-TOPIC 2.2.1 SIMPLE KINETIC THEORY OF MATTER	66
SUB-TOPIC 2.2.2: MEASUREMENT OF TEMPERATURE	86
SUB-TOPIC 2.2.3: EXPANSION OF SOLIDS, LIQUIDS AND GASES.....	105
SUB-TOPIC 2.2.4 INTERNAL COMBUSTION ENGINE (I.C.E)	114
SUB-TOPIC 2.2.5: HEAT TRANSFER.....	126
SUB-TOPIC 2.2.6: MEASUREMENT OF HEAT	140
BIBLIOGRAPHY	152
Appendix: DETAILED SPECIFICATIONS -Apparatus/Equipment	Error! Bookmark not defined.

How to use this Module

To effectively use this Physics Teaching Module for Form 2:

- Read and familiarise yourself with the module's content, learning activities, and assessment guides.
- Plan your lessons in advance, using the module's suggested teaching and learning activities.
- Use a variety of teaching methods, including demonstrations, discussions, group work, and hands-on experiments.
- Encourage active learning by asking open-ended questions, promoting critical thinking, and fostering problem-solving skills.
- Assess learners learning regularly, using the module's suggested assessment strategies and tools.
- Provide feedback and support to learners, helping them to identify areas for improvement and develop their skills.
- Integrate technology into your teaching, using multimedia resources and interactive simulations to enhance student engagement and understanding.
- Monitor student progress and adjust your teaching strategies as needed, to ensure that all students meet the learning objectives

By following these steps, you can effectively use this Physics Teaching Module for Form 2 to support your teaching and promote learning

Introduction

The **Physics Teaching Module for Form 2** provides guidance to teachers on how to deliver competence-based lessons on key Physics concepts in line with the syllabus. The module focuses on **Mechanics II and Thermal Physics**, which help learners understand motion, forces, energy, heat, and temperature, and how these concepts explain many natural and technological processes. It supports teachers in preparing lessons that help learners understand the natural, artificial, and technological environment while developing interest in Physics.

The module promotes learner-centred teaching approaches that encourage inquiry, curiosity, experimentation, and problem-solving. Through carefully designed learning activities, teachers are supported in helping learners develop critical thinking, analytical reasoning, innovation, and practical scientific skills. These competences enable learners to apply scientific principles to everyday situations and technological advancements.

The study of Physics Concepts in **Mechanics** are widely applied in engineering, transport, construction, and mining industries, while **Thermal Physics** plays an important role in energy production, manufacturing, food processing, and environmental management. In Zambia, these applications support key sectors such as **mining, energy, infrastructure development, agriculture, and industrial production**, which are essential for economic growth and sustainable development. Through this module, learners are expected to develop a strong understanding of Physics principles and acquire the knowledge and skills necessary for further studies and careers in science, technology, and related fields.

Suggested Teaching and Learning Materials

The study of Physics content requires hands-on experiences, visual aids, and interactive resources to strengthen deep understanding and appreciation. To support the teaching and learning of Physics in Form 1, the suggested teaching and learning materials are meant to:

- Enhance learner engagement and motivation in Physics
- Develop practical skills and laboratory techniques
- Promote critical thinking, problem-solving, and analytical skills
- Support differentiated instruction and inclusive learning

The suggested teaching and learning materials are either artificial or natural. By utilising these suggested teaching and learning materials, teachers can create a dynamic and supportive learning environment that promotes academic excellence, creativity, and scientific literacy in Physics for Form 1 learners.

Learning Environment Set Up

To create an effective learning environment for teaching and learning Physics it is important to deepen understanding of the concepts and application in real life context. The suggested learning environment set-up aims to create a safe, inclusive, and engaging space.

- **Natural Environment:** A natural learning environment is a setting where learners explore and learn naturally, often without explicit instruction or formal teaching, such as in school surroundings
- **Man-Made Environment:** Man-made learning environments are intentionally designed safe spaces, such as classrooms, laboratories, and libraries, designed for formal instruction, hands-on activities, and games and songs
- **Technological Learning Environment:** Access educational apps, and software for learning, including game-based platforms, virtual platforms, and simulations, to engage learners and promote learning

Safety in the Learning Environment

Safety in the learning environment is a requirement in learning Physics, as learners are exposed to potential hazards during hands-on experiments. Collaboration between teachers, and learners, is important to create a responsible learning environment that promotes scientific inquiry. Guidelines for maintaining a safe learning environment include laboratory safety rules (protocols), risk identification, personal protective equipment, emergency response plans, storage, and disposal, and learner responsibilities. Prioritising safety minimises risks, prevents accidents, and ensures a positive learning experience.

Teaching Methodology

The effective teaching methodologies in STEM Physics include:

- **Conducting experiments:** demonstrate key principles and encourage curiosity among learners.
- **Collaborative learning:** Pair learners to work together, promoting peer-to-peer teaching, discussion, and problem-solving.
- **Conceptual learning:** Connect Physical concepts to everyday life, industry, or current events, making learning relevant and meaningful.
- **Differentiated instructions:** Tailor teaching to meet diverse learning styles, abilities, and interests of different learners.
- **Feedback and Reflection:** Encourage learners to reflect on their learning, providing constructive feedback to guide improvement.
- **Inquiry-based learning:** encourage learners to explore, investigate, and discover Physics concepts through hands-on experiments and activities.
- **Integration of Technology:** Use digital tools, simulations, and visualisations to enhance engagement, understanding, and analysis.

- **Problem-based learning:** Present real-world problems or case studies, requiring learners to apply Physics principles to develop solutions.
- **Project-based learning:** Assign open-ended projects, allowing learners to design, conduct, and present research or applications of Physics concepts.

By implementing these methodologies, a teacher can create an engaging, inclusive, and effective STEM Physics learning environment.

Learning Activities

Learning activities are intentional educational experiences aimed at promoting learning, engagement, and achievement among learners. Facilitated by teachers, they help acquire new knowledge, skills, attitudes, and behaviour change. To create an inclusive environment, teachers should use a "**hook**" or problem posing or scenario or key question or case studies to introduce new learning activities in an interactive and interesting way.

Time Allocation

The standard minimum teacher - learner contact time for Physics at Secondary School Level is four (4) hours per week, translating into Six (6) periods. The duration for a single period is forty (40) minutes. The contact time at Secondary School Level is planned in such a way as to give ample time for practical activities.

Icons used in this Module

This module utilises icons as visual symbols or graphics to represent instructions, enhancing the learning experience for learners. Icons categorise and organise instructions, making navigation easier for teachers.

				 Introduction
Learners Activities	Assessment Guide	Key Terms	Skills Developed	
				 Learners' Tasks
Discussion	Key Learning Points	Tips	General Competences	

Assessment

This assessment shall include a variety of methods to evaluate the competences of learners in terms of knowledge, skills, and general understanding of scientific concepts. The assessment approach will integrate both formative and summative components to support ongoing competence development and final achievement measurement. Summative assessment will be used to evaluate learners' learning at the end of the O level Physics course to measure their achievements against specific competences through Final Examinations. In order to help teachers and learners identify areas where learners need more support or revision, formative assessment will be used to track learner progress and knowledge throughout the teaching and learning process.

However, assessments shall follow the following pattern:

- School Based Assessment (SBA) shall comprise of assignments, projects, practical work, research and end of term tests during the period of study and as guided by the Examinations Council of Zambia (ECZ). This shall carry 30% of the total marks.
- Summative assessment shall carry 70% of the total marks.

The Examinations Council of Zambia (ECZ) shall prepare detailed procedures or guidelines on how SBA will be conducted by the teachers and the management of the assessment results. The standardised national examination shall be administered at the end of Form 4 by the Examination Council of Zambia.

Key Competences

In Physics the following key competences are the fundamental abilities and qualities that will enable individual learners to:

- Manage their own learning and knowledge.
- Interact with others and solve problems.
- Contribute to society and the economic development of the country.
- Adapt to change and navigate through emerging issues in the environment.

KEY COMPETENCE	DESCRIPTOR
Analytical Thinking	To analyse and interpret data, making evidence-based conclusions.
Collaboration	To work together, promoting peer-to-peer teaching, discussion, and problem-solving.
Communication	To communicate scientific information effectively, both orally and in writing.
Creativity and Innovation	To create new ideas and products by applying processes and introducing new techniques that can add value.
Critical Thinking	To enhance learners' ability to think critically and solve problems through logical reasoning based on conclusions.
Digital literacy	Using a broad range of Information and Communication Technologies such as a cell phone, computer and calculator in specific contexts.
Environmental Sustainability	To apply physical principles to understand and mitigate the environmental impact of human activities.
Problem Solving	To use scientific knowledge, critical thinking, and analytical skills to develop a robust problem-solving mindset, enabling learners to tackle complex challenges and drive innovation in various fields.

PHYSICS TEACHING MODULE

TERM 1

TOPIC 2.1: MECHANICS 2



INTRODUCTION

Mechanics is the branch of Physics that deals with the motion of objects and the forces acting upon them. It is all about understanding how objects move, respond to forces and interact with their environment. It is divided into kinematics (describing motion) and dynamics (explaining why objects move). **Mechanics 2** comprises of four (4) sub-topics; Work, Energy and Power, Linear Momentum, Simple Machines and Pressure.



Key Competence(s):

- Analytical Thinking
- Collaborative
- Communication
- Critical Thinking
- Problem Solving
- Real world application
- Environmental sustainability
- Creativity and innovation

SUB-TOPIC 2.1.1: WORK, ENERGY AND POWER



INTRODUCTION

Work, energy and power are fundamental concepts that describe the relationships between forces, motion and energy transfer. The study of Work, Energy and Power helps learners understand how machines operate, how energy is transferred and how different tasks are performed efficiently in everyday life. This sub-topic has nine learning activities as sequenced in the 2024 Form 1-4 Physics syllabus, page 10.

Specific Competence

- Create a system that applies work, energy, and power concepts.



Key terms

- **Work** is the product of the force applied and the distance moved in the direction of the force.

- **A joule** is the amount of work done when a force of 1N moves an object through a distance of 1m in the direction of the force.
- **Energy** is the ability or capacity to do work.
- **Mechanical energy** is the sum total of an object's kinetic energy (KE) and potential energy (PE).
- **Potential energy** is the energy possessed by a body due to its position or condition.
- **Gravitational Potential energy** is the work done in raising an object against the earth's gravitational attraction.
- **Elastic Potential Energy** Is the energy possessed by all elastic materials such as stretched spring, wire or rubber band.
- **Kinetic energy** is the energy possessed by a body due to its velocity or motion.
- **Power** is the rate of doing work.
- **A watt** is the rate of transfer of energy of one joule per second.

Learning activities

This sub-topic has nine learning activities from the syllabus focusing on learning through hands-on activities and problem-solving tasks.



Activity 2.1.1.1: Developing a system that maximises the mechanical energy (*a simple pendulum, inclined plane, and water tank...*)

This activity is designed to help learners develop a system that maximises the mechanical energy. In this activity, learners will design and build a system that maximises the mechanical energy.

HOOK: Imagine you are told to design a simple swing using local materials that converts potential energy to kinetic energy efficiently and safely. How would you develop this system?"

Suggested Teaching and Learning Materials

Natural Materials: Wood (branches, logs), rope, stones or rocks, water, ...

Artificial Materials: Metal wires or nails, synthetic rope/metallic chain, plastic buckets or containers, old tires or rubber bands, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce the concept of mechanical energy (e.g. in a simple pendulum, inclined plane, and water tank...) through a scenario
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Guide learners to identify local resources
- ❖ Guide learners in designing and testing their systems
- ❖ Provide feedback on energy conversion
- ❖ Ensure safety



Learners' Tasks

- ❖ Researching and Brainstorming on the given scenario of mechanical energy (e.g in a simple pendulum, inclined plane, water tank, ...)
- ❖ Identifying local materials
- ❖ Designing and testing the system
- ❖ Getting feedback on energy conversion
- ❖ Ensuring safety protocols
- ❖ Presenting their findings



Assessment Guide

- ❖ Evaluate system design
- ❖ Assess understanding of energy conversion
- ❖ Observe demonstration of problem-solving and critical thinking skills



Skills Developed

- ❖ Problem-solving: Designing systems for maximising mechanical energy
- ❖ Critical thinking: Analysing energy conversions
- ❖ Collaboration: Working in groups as they were designing and identifying materials to use
- ❖ Communication: Presenting their findings
- ❖ Application: Real-world energy optimisation



Key Learning Points

- ❖ Mechanical Energy = KE + PE
- ❖ Systems can maximise energy through:

- i. Pendulum: Optimal length (best possible length for maximum efficiency), amplitude (best possible amplitude for maximum efficiency)
 - ii. Inclined plane: Optimal angle (best possible angle for maximum efficiency)
 - iii. Water tank: Optimal height (best possible height for maximum efficiency)
- ❖ Energy conversions (KE to PE or PE to KE) affect efficiency



Activity 2.1.1.2: Determining the work done on an object by a force.

This activity is designed to help learners determine the work done on an object by a force by carryout an experiment. This will help explain many everyday situations.

HOOK: A learner designed a simple pulley system to lift a 50 kg bag of maize to a height of 2 meters using a rope and a wooden beam. How would you determine the work done on a bag?

Suggested Teaching and Learning Materials

Natural Materials: Wood (for ramps, levers, or pulleys), rocks or stones (as weights), rope (from plants), ...

Artificial Materials: Metal (weights or hooks, pulleys, levers, ...), spring scales (for measuring force), measuring tapes or rulers, ...

Learning environment set up

- Natural Environment such as school surroundings, forest, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom, ...

Teacher's Role(s)

- ❖ Introduce an activity through a hook
- ❖ Put learners in small groups
- ❖ Provide materials to the learners in groups
- ❖ Guide learners to design experiments to determine the work done on an object by force
- ❖ Guide learners in measuring force and distance
- ❖ Facilitate the determination of work done



Learners' Tasks

- ❖ Designing experiments to determine the work done on an object by force apply forces
- ❖ Measuring force (using spring scales) and distance (using a meter rule)
- ❖ Determining work done using the formula, Work done = Force x distance.
- ❖ Comparing results and discuss real-world applications



Assessment Guide

- ❖ Review calculations of work done ($W = F \times d$; $W = F \times d \times \cos \theta$; $F \times d \times \sin \theta$). Explain the conditions how these formulae apply.
- ❖ Observe demonstration of problem-solving in groups



Skills Developed

- ❖ Analytical thinking: Calculating work done with correct units
- ❖ Problem-solving: Applying $W = F \times d \times \cos \theta$
- ❖ Critical thinking: Considering force and direction
- ❖ Application: Real-world examples (e.g., lifting, pushing)



Key Learning Points

- ❖ Work is done when force moves an object in the direction of action of force
- ❖ Work done (W) = $F \times d$
- ❖ Units: Joules (J) = Nm
- ❖ θ is the angle between force and displacement
- ❖ Work is scalar quantity, can be positive (+), negative (-), or zero (0)
- ❖ Examples: lifting weights over a distance, pushing carts over a distance, ...



Activity 2.1.1.3: Investigating the relationship between force and displacement through

experimenting, $W = Fs$.

This activity is designed to help learners investigate the relationship between force and displacement through experiment.

HOOK: “A rubber band and a ruler were used to investigate the relationship between stretch (displacement) and applied force. What pattern do you observe?”

Suggested Teaching and Learning Materials

Natural Materials: Rulers (wooden) or sticks, stones, ...

Artificial Materials: Spring scales (for measuring force), rulers, rubber bands, measuring tapes, ...

Learning environment set up

- Natural Environment such as school surroundings, forest, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce an activity through a hook
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Guide learners to investigate the relationship between force and stretch during an experiment.
- ❖ Encourage data collection through measurements
- ❖ Encourage data tabulation and checking the pattern
- ❖ Encourage plotting the graph and checking the trends
- ❖ Facilitate discussions on the relationship between force and displacement
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Investigating the relationship between force and stretch using an experiment
- ❖ Collecting data through measurements
- ❖ Tabulating and checking the data patterns
- ❖ Plotting the graph and checking the trends
- ❖ Discussing the relationship between force and displacement
- ❖ Present their findings



Assessment Guide

- ❖ Observe group experiments (setup, data collection, tabulation)
- ❖ Review calculations of work done ($W = F \times s$)
- ❖ Evaluate understanding of force-displacement graphs



Skills Developed

- ❖ Scientific inquiry: Investigating the relationship between applied force and displacement through an experiment
- ❖ Analytical thinking: Evaluating data patterns and trends
- ❖ Critical thinking: Setting up an experiment to investigate the relationship between force and displacement
- ❖ Collaboration: Working in groups to investigate the relationship between force and displacement
- ❖ Communication: Presentation of findings



Key Learning Points

- ❖ $W = F \times s$ (when F is constant and parallel to s)
- ❖ Work = Area under Fs graph
- ❖ Units: Joules (J) = Nm
- ❖ Understanding applied force in relation to stretch



Activity 2.1.1.4: Conducting experiments to measure and calculate mechanical energy (kinetic and potential) in different systems, $E_p = mgh$ and $E_k = \frac{1}{2}mv^2$.

This activity is designed to help learners conduct experiments to measure and calculate mechanical energy (kinetic and potential) in different systems, $E_p = mgh$ and $E_k = \frac{1}{2}mv^2$.

HOOK: Picture this: A bunch of boys are kicking a football ⚽ on top of a steep hill 🏔️. One of them nails a powerful kick, and the ball rolls down the slope like a wild ride 🎢.
Discuss what happens to its 'energy of movement' and 'energy of position?'

Suggested Teaching and Learning Materials

Natural Materials: Stone, ramps (wooden planks, inclined paths), ...

Artificial Materials: Stopwatches, ramps, measuring tapes, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce an activity through a hook
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Guide learners on how to conduct an experiment to measure and calculate mechanical energy.
- ❖ Guide on measurement and data collection
- ❖ Facilitate the calculations of mechanical energy
- ❖ Facilitate discussions on energy conversions
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Conducting an experiment to measure and calculate mechanical energy
- ❖ Measuring and collecting data
- ❖ Calculating mechanical energy
- ❖ Discussing on energy conversions
- ❖ Presenting their findings



Assessment Guide

- ❖ Observe experiment setup and data collection
- ❖ Review calculations of $KE = \frac{1}{2}mv^2$, $PE = mgh$
- ❖ Evaluate understanding of energy conversions
- ❖ Use of appropriate units (Joules)



Skills Developed

- ❖ Scientific inquiry: Conducting an experiment
- ❖ Analytical thinking: Calculating KE, PE, energy values from collected data
- ❖ Data analysis: Interpreting energy conversions (mechanical energy)



Key Learning Points

- ❖ $KE = \frac{1}{2}mv^2$ where m =mass and v = velocity
- ❖ $PE = mgh$ where m = mass, g = gravity and h = height
- ❖ Energy conversions: KE to PE or PE to KE.
- ❖ Systems of mechanical energy: Pendulums, ramps, falling objects
- ❖ Energy conversion application: Elevated water tank system



Activity 2.1.1.5: Demonstrating the law of conservation of energy in mechanical energy.

This activity is designed to help learners demonstrate the law of conservation of energy in mechanical energy

HOOK: During a science lesson, a teacher takes learners outside and ties a small stone to a strong string. The teacher begins to swing the stone in a vertical circle and asks the learners to carefully observe the motion and think about what is happening to the energy of the stone during its movement

Suggested Teaching and Learning Materials

Natural Materials: Stones or fruits, string or rope, tree branches (for pendulums), ...

Artificial Materials: Stopwatches, string or rope, measuring tapes, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce an activity through a hook
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Guide learners to design pendulum experiments
- ❖ Encourage observation and discussion
- ❖ Guide learners on measurement of mass of the bob, height and time
- ❖ Facilitate calculations of PE maximum and KE maximum

- ❖ Instruct learners to compare calculated PE maximum and KE maximum
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Designing a simple pendulum experiment
- ❖ Measuring of mass of the bob, height and time
- ❖ Calculating PE maximum and KE maximum from measured data
- ❖ Comparing calculated PE maximum and KE maximum
- ❖ Presenting their findings



Assessment Guide

- ❖ Observe experiment setup (e.g., pendulum, ...)
- ❖ Review calculations of KE maximum and PE maximum
- ❖ Evaluate understanding of energy conservations
- ❖ Use of appropriate units (Joules)



Skills Developed

- ❖ Scientific inquiry: Designing experiments
- ❖ Analytical thinking: Calculating KE maximum and PE maximum
- ❖ Data analysis: Comparing calculated PE maximum and KE maximum
- ❖ Critical thinking: Explaining energy conservations



Key Learning Points

- ❖ Law of Conservation of Energy: Total energy is constant
- ❖ $KE + PE = \text{Constant}$ (ideal systems)
- ❖ Energy conversions: KE to PE or PE to KE (e.g., pendulum)
- ❖ Real systems of energy conversations: Water tank supply system, ...



Activity 2.1.1.6: Exploring how machines can change the amount of mechanical energy required to perform a task, such as cranes.

This activity is designed to help learners explore how machines can change the amount of mechanical energy required to perform a task, such as cranes.

HOOK: During harvest time, imagine carrying a 90Kg bag of maize, lifting and loading the bag onto a truck by hand. Imagine also the same task being done using a machine such as a crane, pulley, or lifting device.

- a. How does it make work easier in farms and also at storage facilities?
- b. Where else does this situation apply?

Suggested Teaching and Learning Materials

Natural Materials: Sticks, rocks, strings or ropes, bamboo, ...

Artificial Materials: Cardboard, pulleys, levers, weights, crane, rope, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce an activity through a hook
- ❖ Put learners in small groups
- ❖ Ask the learners to look for sticks, rocks and bamboo and later provide other materials (Cardboard, pulleys, levers, weights, rope) to the learners
- ❖ Facilitate groups to explore how machines can change the amount of energy required to perform a task e.g. a crane lifting loads
- ❖ Ask guiding questions (e.g., "How does using a pulley affect effort?")
- ❖ Facilitate sharing of findings
- ❖ Ensure safety, encourage adjustments



Learners' Tasks

- ❖ Looking for sticks, rocks and bamboo
- ❖ Collecting other materials (Cardboard, pulleys, levers, weights, rope)
- ❖ Exploring how machines can change the amount of energy required to perform a task e.g. a crane lifting loads
- ❖ Responding to guiding questions (e.g., "How does using a pulley affect effort?")
- ❖ Sharing of findings through a presentation or a discussion.
- ❖ Following safety protocols



Assessment Guide

- ❖ Observe group work and ask guiding questions
- ❖ Assess ability to explore and explain machine's effect on effort
- ❖ Evaluate presentations on machine effectiveness



Skills Developed

- ❖ Inquiring: Exploring how machines can change the amount of mechanical energy
- ❖ Critical thinking: Analysing effort vs. load
- ❖ Collaboration: Working in groups
- ❖ Practical skills: Using machines such as a crane
- ❖ Communication: Presenting their findings



Key Learning Points

- ❖ Machines can change the amount of mechanical energy needed for tasks to be performed
- ❖ Pulleys and levers reduce effort for heavy loads
- ❖ Mechanical advantage of a machine helps perform tasks easily in daily life (like construction cranes)



Activity 2.1.1.7: Calculating the efficiency of energy conversion using the appropriate

formula, $\text{efficiency} = \frac{\text{energy output}}{\text{energy input}} \times 100\%$.

This activity is designed to help learners calculate the efficiency of energy conversion using the

appropriate formula, $\text{efficiency} = \frac{\text{energy output}}{\text{energy input}} \times 100\%$.

HOOK: You light a bulb that consumes X electrical energy to produce Y heat and Z light energy.

How would you determine the efficient of the bulb?

Suggested Teaching and Learning Materials

Natural Materials: Sunlight

Artificial Materials: Batteries, bulbs, resistor, connecting wires, solar panels (optional), ammeter, voltmeter, switch, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce an activity through a scenario
- ❖ Put learners in small groups
- ❖ Facilitate the discussion on the formula: $Efficiency = \frac{\text{Useful energy (output)}}{\text{Supplied energy (input)}} \times 100\%$
- ❖ Guide each group to set up a simple circuit (e.g., bulb or any load resistor connected to an electric source such as a battery, solar panel, power pack ...) in each group
- ❖ Guide learners to measure the input and output voltage, and current
- ❖ Ask probing questions such as: "How do we measure input and output energy?"
- ❖ Support calculations and ensure safety



Learners' Tasks

- ❖ Discussing the formula: $Efficiency = \frac{\text{Useful energy (output)}}{\text{Supplied energy (input)}} \times 100\%$
- ❖ Setting up a simple circuit (e.g., bulb or any load resistor connected to an electric source such as a battery, solar panel, power pack ...) in each group
- ❖ Measuring the input and output voltage, and current
- ❖ Responding to probing questions such as: "How do we measure input and output energy?"
- ❖ Calculating and observing safety protocols



Assessment Guide

- ❖ Review calculations for accuracy
- ❖ Evaluate understanding of efficiency concept
- ❖ Observe group work, ask open ended questions
- ❖ Application of efficiency in everyday life



Skills Developed

- ❖ Critical thinking: Applying formula for efficiency of energy conversion

- ❖ Problem-solving: Measuring energy values
- ❖ Analytical thinking: Comparing the output energy and the input energy to calculate efficiency
- ❖ Innovation: Setting up a circuit to determine efficiency
- ❖ Communication: Presenting their findings



Key Learning Points

- ❖ $Efficiency = \frac{\text{Useful energy (output)}}{\text{Supplied energy (input)}} \times 100\%$
- ❖ No system is 100% efficient due to energy losses
- ❖ Efficiency helps to evaluate energy conversion effectiveness
- ❖ Examples of conversion devices: bulbs (light vs. heat), solar panels (light to electricity), ...



Activity 2.1.1.8: Solving mathematical problems involving power developed by mechanical energy systems

This activity is designed to help learners solve mathematical problems involving power developed by mechanical energy systems using scenario based questions.

HOOK: Our school needs a pump to lift 300kg of water from a 15m deep well to irrigate crops. How would you determine the power required for the pump in watts?

Suggested Teaching and Learning Materials

Natural Materials: Weights (rocks), strings, ...

Artificial Materials: Pulleys, stopwatches, calculators, worksheet, rulers, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce the activity using a scenario question on mechanical system (water pump)
- ❖ Introduce learners to the use of the formula, $Power = \frac{\text{Work done}}{\text{time}}$
- ❖ Put learners in small groups

- ❖ Provide the worksheet with scenario-based problems on mechanical systems
- ❖ Guide groups to solve scenario-based problems on mechanical systems
- ❖ Ask a probing question: "How would you calculate work and power?"
- ❖ Provide feedback to learners



Learners' Tasks

- ❖ Discussing scenario question on mechanical system (water pump)
- ❖ Asking questions on the formula, $Power = \frac{Work\ done}{time}$
- ❖ Solving scenario-based problems on mechanical systems
- ❖ Responding to probing question such as, "How would you calculate work and power?"
- ❖ Receiving feedback



Assessment Guide

- ❖ Review calculations on scenario-based problems on mechanical systems
- ❖ Evaluate problem-solving
- ❖ Observe group work



Skills Developed

- ❖ Critical thinking: Applying formulas in a scenario-based question
- ❖ Problem-solving: Calculating power
- ❖ Analytical thinking: Math calculations
- ❖ Collaboration: Solving scenario-based problems in groups
- ❖ Application: Real-world problems



Key Learning Points

- ❖ $Power = \frac{Work\ done}{time}$ (Watts = Joules/second)
- ❖ Examples of mechanical systems: pumps, cranes, engines, human power



Activity 2.1.1.9: Calculating the efficiency of power using the appropriate formula, efficiency

$$= \frac{\text{Power output}}{\text{Power input}} \times 100\%.$$

This activity is designed to help learners calculate the efficiency of power using the appropriate formula, efficiency = $\frac{\text{Power output}}{\text{Power input}} \times 100\%$.

HOOK: Imagine you are using a generator that consumes a large amount of fuel but produces only a limited amount of electricity. How would you determine its efficiency in terms of input power from fuel combustion with the output electrical power it produces?

Suggested Teaching and Learning Materials

Natural Materials: Fuel (e.g., petrol), ...

Artificial Materials: Generator, bulbs, stopwatch, calculator, multimeter, ...

Learning environment set up

- Natural Environment such as school surroundings, ...
- Artificial Environment such as the laboratory or any room or place where learners can be such as mobile laboratory, classroom,

Teacher's role(s)

- ❖ Introduce the activity through a scenario question
- ❖ Put learners in small groups
- ❖ Explain the formula, $Power = \frac{\text{useful power (output)}}{\text{power input}} \times 100\%$
- ❖ Guide groups to set up a simple test (e.g., generator → bulbs)
- ❖ Ask a probing question such as: How would you measure input and output power?
- ❖ Support calculations and ensure safety
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Discussing the formula, $Power = \frac{\text{useful power (output)}}{\text{power input}} \times 100\%$
- ❖ Setting up a simple test (e.g., generator → bulbs)
- ❖ Responding to a probing question such as: How would you measure input and output power?
- ❖ Calculating efficiency
- ❖ Observing safety protocols
- ❖ Presenting their findings



Assessment Guide

- ❖ Review calculations
- ❖ Evaluate understanding of efficiency
- ❖ Observe if learners are working collaboratively in groups



Skills Developed

- ❖ Critical thinking: Applying formulas
- ❖ Problem-solving: Calculating efficiency
- ❖ Collaboration: Working in groups
- ❖ Application: Real-world problems



Key Learning Points

- ❖ $Power = \frac{\text{useful power (output)}}{\text{power input}} \times 100\%$
- ❖ Energy losses reduce efficiency (heat, friction)
- ❖ Examples of sources of power: generators, engines, ...

EXPECTED STANDARD

SAMPLE QUESTIONS

Question 1

A small-scale farmer in Luapula Province wants to lift 50 kg bags of maize from the ground to a 4 m high storage platform using a simple pulley system powered by human effort (or a small petrol engine as backup).

- a) Design a basic pulley system (specify type: fixed, movable, or block-and-tackle) that minimises the effort force needed. Calculate the minimum effort force required (assume $g = 10 \text{ N/kg}$, ignore friction).
- b) Explain the energy changes that occur when lifting one bag, and use the principle of conservation of energy to show that total energy is conserved (even if the system is not 100% efficient).
- c) If the farmer lifts one bag in 8 seconds using a steady force, calculate the power developed. Analyse how this system applies work, energy, and power concepts to improve farming efficiency.

Question 2

A community in Eastern Province is building a hand-operated water pump to draw water from a 15 m deep borehole for irrigation and domestic use.

- a) Create a simple lever-pump system (e.g., using a long handle as a class 1 lever) and calculate the work done to lift 10 kg of water through 15 m vertically ($g = 10 \text{ N/kg}$).
- b) If the pump has an efficiency of 60%, calculate the actual work input required by the operator and explain where the lost energy goes.
- c) Analyse how power considerations influence the design (e.g., handle length, pumping rate) to make it sustainable for women and youth users.

Question 3

A bicycle repair shop in Kitwe is designing a pedal-powered phone/laptop charging station using a dynamo attached to the rear wheel.

- a) Describe the system: how pedalling converts energy to electrical energy. Calculate the kinetic energy of a 70 kg rider + 15 kg bike travelling at 5 m/s (assume all converted via dynamo).
- b) If the dynamo produces 10 W of useful electrical power at that speed, calculate the time to charge a 3600 J phone battery (ignore losses first, then adjust for 40% overall efficiency).
- c) Analyse the impact of this system on rural households with no grid electricity.

Question 4

A school in Ndola wants to build a playground seesaw that doubles as an energy demonstration device, showing energy transfers.

- a) Design the seesaw (class 1 lever) with two 50 kg students. One student (A) sits 1.5 m from the pivot; where should student B sit for balance? Calculate work done by gravity when student A descends 0.8 m.
- b) Explain energy conservation as one side goes down (GPE lost = GPE gained on other side + kinetic energy + losses).
- c) If the motion takes 2 seconds, calculate average power involved in the transfer. Analyse educational value.

Question 5

A Zambian entrepreneur is developing a solar-powered maize sheller for cooperatives, but includes a hand-crank backup for cloudy days.

- a) For the hand-crank: design a gear system to increase rotational speed. If operator applies 80 N force at 0.3 m radius crank, calculate work per revolution.
- b) Assume 20 revolutions to shell 5 kg maize, lifting kernels 0.2 m on average. Calculate useful energy and efficiency if total work input = 1500 J.
- c) Analyse how work, energy, and power concepts are applied to make the machine practical and efficient.

SAMPLE QUESTIONS ANSWERS

Question 1

- a) A single movable pulley (or block-and-tackle with mechanical advantage 2) halves the effort. Effort force = weight / 2 = $(50 \times 10) / 2 = 250$ N (distance rope pulled = 8 m for 4 m lift).
- b) Energy changes: Chemical energy (from farmer's muscles), kinetic energy (of bag during lift), gravitational potential energy (GPE = $mgh = 50 \times 10 \times 4 = 2000$ J) at top. Some energy lost as heat/sound due to friction/muscle inefficiency. Conservation: Input energy (work done by farmer) = GPE gained + losses.
- c) Work done = 2000 J (against gravity). Power = work/time = $2000 / 8 = 250$ W. The system applies work (force $\times$ displacement), converts chemical $\rightarrow$ mechanical $\rightarrow$ potential energy, and calculates power to show rate of energy transfer, helps farmer work faster, reduce fatigue, store more maize safely.

Question 2

- a) Work done against gravity = $mgh = 10 \times 10 \times 15 = 1500 \text{ J}$ per stroke.
- b) Efficiency = useful output / input $\times 100\%$. Input work = $1500 / 0.60 = 2500 \text{ J}$. Lost energy = 1000 J (mainly friction in pump seals/pipes + heat in muscles).
- c) Longer handle increases mechanical advantage, lower effort force but more distance moved, same work. Power = work/time; slower pumping lowers power demand (less fatigue). System applies work (to overcome gravity/friction), energy conversion (muscular, potential), power to match human capability, improves water access, reduces drudgery.

Question 3

- a) Pedalling $\rightarrow$ kinetic energy of bike/rider $\rightarrow$ rotational $\rightarrow$ dynamo $\rightarrow$ electrical energy. $KE = \frac{1}{2}mv^2 = \frac{1}{2} \times 85 \times 25 = 1062.5 \text{ J}$.
- b) Time (ideal) = energy / power = $3600 / 10 = 360 \text{ s}$ (6 min). With 40% efficiency, input power needed $\approx 25 \text{ W}$, actual pedalling time longer ($\sim 15 \text{ min}$).
- c) Applies work (force on pedals $\times$ distance), energy conversion (KE $\rightarrow$ electrical), power (rate of charging). Provides sustainable charging, supports education (night study), small businesses (phone money), reduces kerosene use.

Question 4

- a) Moments balance: $50 \times 10 \times 1.5 = 50 \times 10 \times d$, $d = 1.5 \text{ m}$ (symmetrical). Work by gravity on descending student = $mgh = 50 \times 10 \times 0.8 = 400 \text{ J}$.
- b) GPE lost by A = 400 J , converted to GPE gain by B + KE of both + friction/heat losses. Total mechanical energy conserved if no losses.
- c) Power $\approx$ work/time = $400 / 2 = 200 \text{ W}$ (average over transfer). Demonstrates work (force $\times$ distance), energy forms/conversion/conservation, power (rate), great for teaching concepts interactively.

Question 5

- a) Work per revolution = force $\times$ distance = $80 \times (2\pi \times 0.3) \approx 80 \times 1.885 = 150.8 \text{ J}$.
- b) Useful energy = $mgh = 5 \times 10 \times 0.2 = 100 \text{ J}$ (plus some kinetic). Efficiency = $100 / 1500 \times 100\% \approx 6.7\%$ (low due to friction; gears help).
- c) Applies work (torque $\times$ angle), energy conversion (muscular $\rightarrow$ rotational $\rightarrow$ mechanical), power (sustained rate). Backup ensures reliability; gears optimise power input vs output, reduces manual labour, increases shelling speed, boosts cooperative income.

SUB-TOPIC 2.1.2: LINEAR MOMENTUM



INTRODUCTION

Linear momentum is the product of an object's mass and velocity. It's a vector quantity, meaning it has both magnitude and direction. It is calculated using the formula:

$p = mv$ where p = linear momentum, m =mass (kg) and v = velocity (m/s).

Specific competence

- Apply the principle of linear momentum in everyday life.



Key terms

- Momentum is a measure of an object's tendency to keep moving in a straight line.
- Linear Momentum (p): The product of an object's mass (m) and velocity (v). ($p = mv$)
- Conservation of Momentum: Total momentum remains constant in a closed system (no external forces).
- Impulse (J): Change in momentum, equals force $\times$ time. ($J = \Delta p$ or $J = F \times \Delta t$)
- Elastic Collision: Momentum and kinetic energy conserved; objects bounce apart.
- Inelastic Collision: Momentum conserved; kinetic energy not conserved; objects may stick together.
- Vector Quantity: Momentum has magnitude and direction.
- Units: kgm/s (kilogram meters per second).

Learning activities

This sub-topic has five learning activities from the syllabus focusing on learning through hands-on activities and problem-solving tasks.



Activity 2.1.2.1: Exploring linear momentum in real world scenarios (*sport, transportation, car accidents analyses, safety features built in cars and engineering, ...*)

This activity is designed to help learners explore linear momentum in real world scenarios.

HOOK: You are a traffic police officer investigating a case of collision between a moving car (1200 kg) and a stationary truck (4000 kg). Both vehicles moved together at 20 km/h after impact. The speed limit in the area is 40km/h. Analyse the accident using momentum principles to determine if speeding was a factor?

Suggested Teaching and Learning Materials

Natural Materials: Wood (sports equipment), human body (impact studies), ...

Artificial Materials: Airbags, crumple zones, safety helmets, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce linear momentum ($p = mv$) and its conservation in collisions.
- ❖ Put learners in small groups
- ❖ Assign scenarios to groups based on sport, transportation, car accidents analyses, safety features built in cars and engineering, ...
- ❖ Circulate, ask guiding questions (e.g., How does crumple zone reduce impact force?).
- ❖ Help groups prepare posters/presentations.
- ❖ Guide class discussions.
- ❖ Provide feedback and summarise
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Exploring scenarios provided such as sport, transportation, car accidents analyses, safety features built in cars and engineering, ...
- ❖ Responding to guiding questions (e.g., How does crumple zone reduce impact force?).
- ❖ Preparing posters/presentations.
- ❖ Discussing on the scenarios provided
- ❖ Receiving feedback and summary
- ❖ Presenting to the class their findings.



Assessment Guide

- ❖ Clarity, accuracy, application of linear momentum concepts.
- ❖ Depth of analysis on the principles of linear momentum.
- ❖ Engagement and teamwork.



Skills Developed

- ❖ Problem-solving: Applying linear momentum (mass and velocity) concepts to real-world problems.
- ❖ Critical Thinking: Analysing safety features using linear momentum principles.
- ❖ Collaboration: Working in teams to discuss and explore.
- ❖ Communication: Explaining concepts clearly and presenting their findings.



Key Learning Points

1. Momentum ($p = mv$): Product of mass and velocity; vector quantity. Its unit is Kgm/s
2. Conservation of Momentum: Total momentum before = total momentum after (in a closed system).
3. Impulse-Momentum: ($F \times \Delta t = m(v-u) = \Delta p$); reducing force by increasing time.
4. Real-World Applications:
 - ❖ Car Safety: Crumple zones increase collision time, reducing impact force ($F = \Delta p / \Delta t$).
 - ❖ Sports: Follow-through in golf/tennis increases contact time, improving momentum transfer.
 - ❖ Engineering: Design of safety helmets, airbags rely on momentum principles.



Activity 2.1.2.2: Exploring the concept of momentum.

This activity is designed to help learners explore the concept of momentum.

HOOK: You have designed a sports game where players collide with each other. How would you explain the player's momentum change when they catch a heavy medicine ball while running?

Suggested Teaching and Learning Materials

Natural Materials: Rolling balls (e.g., stone, fruit), athletes, ...

Artificial Materials: Toy cars, ramps, crash test dummies, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introducing the concept momentum ($p = mv$) using a hook
- ❖ Put learners in small groups
- ❖ Assign scenarios (sports, traffic, safety) for them to explore in groups.
- ❖ Guide hands-on activities (e.g., collisions).
- ❖ Ask learners present their findings.
- ❖ Ask learners to link momentum to real life.



Learners' Tasks

- ❖ Exploring the concept of momentum using the scenarios such sports, traffic, safety, ...
- ❖ Performing hands-on activities on momentum such as elastic and inelastic collisions
- ❖ Presenting their findings.
- ❖ Linking momentum to real life.



Assessment Guide

- ❖ Clarity on concept of momentum
- ❖ Exploring the concepts of momentum using scenarios.
- ❖ Teamwork.



Skills Developed

- ❖ Analytical Thinking: Link momentum to real life situations.
- ❖ Problem-Solving: Apply momentum to real life scenarios.
- ❖ Collaboration: Team discussions.



Key Learning Points

1. Momentum ($p = mv$): Product of mass and velocity.
2. Conservation: Total momentum conserved in closed systems.

3 Real-life applications: Car safety, sports (e.g., tackling), ...



Activity 2.1.2.3: Solving numerical problems involving linear momentum, $p = m_1u_1 + m_2u_2$.

This activity is designed to help learners solve numerical problems involving linear momentum, $p = m_1u_1 + m_2u_2$ using scenario questions.

HOOK: You are on a **Zambian safari**, and a **200 kg Buffalo** charges at **15 m/s** towards a **500 kg parked Land Cruiser**. How would you determine the total momentum of the system before impact?

Suggested Teaching and Learning Materials

Natural Materials: None directly used.

Artificial Materials: Calculators, whiteboard, example cards, worksheet, chalkboard, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the formula for total linear momentum, $p = m_1u_1 + m_2u_2$.
- ❖ Put learners in small groups
- ❖ Provide worksheets containing scenario questions on linear momentum.
- ❖ Guide learners in solving linear momentum numerical problems provided in the worksheet.
- ❖ Discuss solutions and provide feedback
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Receiving scenario questions on linear momentum.
- ❖ Solving linear momentum numerical problems provided in the worksheet using the formula, $p = m_1u_1 + m_2u_2$.
- ❖ Discussing solutions and providing feedback

- ❖ Presenting their findings



Assessment Guide

- ❖ Correct application of $p = m_1u_1 + m_2u_2$
- ❖ Clear working steps in solving numerical problems
- ❖ Group Participation



Skills Developed

- ❖ Problem-Solving: Apply linear momentum formula
- ❖ Analytical Thinking: Break down numerical scenario problems.
- ❖ Collaboration: Working together in small groups to solve linear momentum numerical problems
- ❖ Communication: Discussing and presenting their findings
- ❖ Mathematical Skills: Calculating total linear momentum using such as, $p = m_1u_1 + m_2u_2$



Key Learning Points

1. Formula: ($p = m_1u_1 + m_2u_2$) for total linear momentum.
2. Vector Nature: Momentum direction matters (+/- signs).
3. Units: kgm/s for momentum.



Activity 2.1.2.4: Demonstrating the law of conservation of momentum (*elastic and inelastic collision*), $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

This activity is designed to help learners demonstrate the law of conservation of momentum (*elastic and inelastic collision*), $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$.

HOOK: You are in a Physics lab, and you collide a moving trolley of mass 2 kg moving at 3 m/s with a stationary trolley of mass 1 kg.

- i. If they stick together, how would you determine their final velocity?
- ii. If they do not stick together, how would you determine the velocity of a 1kg trolley if a 2kg trolley finally moved at 1 m/s?

Suggested Teaching and Learning Materials

Natural Materials: Oranges, guavas, stones, dry clay balls, air track (low friction), ...

Artificial Materials: Marbles, trolleys, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the law of conservation of momentum through a scenario
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Guide learners on how to demonstrate elastic and inelastic collisions using an experiment.
- ❖ Ask learners to discuss calculations on the law of conservation of linear momentum
- ❖ Ask learners to analyse the results calculated and draw a conclusion
- ❖ Provide feedback
- ❖ Ask learners to present their findings.



Learners' Tasks

- ❖ Demonstrating elastic and inelastic collisions using an experiment.
- ❖ Discussing calculations on the law of conservation of linear momentum
- ❖ Analysing the results calculated and drawing a conclusion
- ❖ Paying attention to the feedback provided



Assessment Guide

- ❖ Correct setup of apparatus and data collection
- ❖ Accurate use of calculations using equation, $(m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2.)$
- ❖ Clear and concise presentations of the law of conservation of momentum



Skills Developed

- ❖ Practical Skills: Handling equipment, conducting experiments.
- ❖ Analytical Thinking: Applying data collected on the law of conservation of momentum.
- ❖ Collaboration: Teamwork in groups.



Key Learning Points

1. Law: ($m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$) (no external force).
2. Elastic Collisions: Momentum is conserved, KE conserved; objects bounce apart.
3. Inelastic Collisions: Momentum is conserved, KE not conserved; objects may stick together.
4. Applications: Car crashes, sports (e.g., billiards), rocket propulsion.
5. Vector Nature: Momentum direction matters (+/- signs).

Additional Notes

Closed System: No external forces act on the colliding objects.

Real-World Examples:

- ❖ Cars colliding and sticking (inelastic).
- ❖ Pool balls hitting (elastic).
- ❖ Common Mistakes: Forgetting vector directions in calculations.



Activity 2.1.2.5: Demonstrating an understanding of the effects and consequences of excessive speeding.

This activity is designed to help learners demonstrate an understanding of the effects and consequences of excessive speeding.

HOOK: You are advising a friend on safe driving in Lusaka. How would you explain how over speeding reduce reaction time and increases stopping distance, potentially leading to more severe accidents?

Suggested Teaching and Learning Materials

Natural Materials: None specific.

Artificial Materials: Car crash videos, speed graphs, impact models, crash test dummies.

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using scenario in a hook
- ❖ Show cases of real-life crash videos
- ❖ Assign scenarios (urban, highway, rural speeding impacts) to the groups
- ❖ Guide discussions on effects on safety, legal consequences, and environment.
- ❖ Ask learners to do research on consequence of excessive speeding
- ❖ Ask learners to present findings.



Learners' Tasks

- ❖ Viewing videos on cases of real-life crashes
- ❖ Discussing scenarios (urban, highway, rural speeding impacts)
- ❖ Discussing the effects on safety, legal consequences, and environment.
- ❖ Researching on the consequence of excessive speeding
- ❖ Presenting findings.



Assessment Guide

- ❖ Clarity on effects and consequence of over speeding
- ❖ Analysing the impact of over speeding.
- ❖ Creativity, ability to simulate impact of over speeding.



Skills Developed

- ❖ Critical Thinking: Link speed to crash severity
- ❖ Communication: Present impacts over speeding effectively.
- ❖ Collaboration: Doing research in groups.



Key Learning Points

1. Speeding Risks: Higher speed has a greater impact force and damage risk.
2. Consequences of over speeding: Injuries, car crash, fatalities, fines, environmental harm, ...

3. Safety Measures: Speed limits, seatbelts, road design, speed alerts, ...
4. Responsibility: Drivers control risk; obey traffic laws/rules, education on safety rules, ...
5. Physics Link: Kinetic energy, $KE = \frac{1}{2} mv^2$ rises rapidly with over speeding. Over speeding increases the momentum and the difficulty to control a moving object.

EXPECTED STANDARD

SAMPLE QUESTIONS

Question 1

During a road traffic accident investigation in Kabwe, police analyse a collision between a 1200 kg minibus travelling at 15 m/s and a stationary 800 kg car at a junction. After the impact, the vehicles lock together and move at 9 m/s in the original direction of the minibus.

- a) Using the principle of conservation of linear momentum, show that the post-collision velocity is consistent with the given data.
- b) Explain why this is an example of an inelastic collision and calculate the total kinetic energy lost.
- c) Analyse how understanding linear momentum helps improve road safety features in vehicles used on Zambian roads (e.g., minibuses).

Question 2

A fisherman in Lake Kariba uses a dugout canoe. When he throws a 2 kg fish horizontally backwards at 5 m/s relative to the canoe, the canoe (mass 80 kg including fisherman) moves forward.

- a) Apply conservation of linear momentum to explain why the canoe moves forward and calculate its velocity (assume initial momentum zero, ignore water resistance).
- b) If the fisherman throws the fish in 0.4 s, calculate the average force on the fish and on the fisherman/canoe system.
- c) Analyse how this principle applies to everyday activities like stepping off a small boat onto a jetty or pushing off in ice skating.

Question 3

In a school football match in Mansa, a 60 kg player running at 4 m/s tackles a stationary 55 kg player head-on. After collision, the tackler rebounds at 1 m/s backward.

- a) Use conservation of momentum to calculate the velocity of the tackled player after the collision (assume elastic for simplicity, but discuss realism).
- b) Calculate the change in momentum for each player and explain the impulse involved.
- c) Analyse why understanding momentum helps coaches and players reduce injury risk in contact sports like football or rugby in Zambian schools.

Question 4

A rocket-propelled fireworks seller in Kasama launches small rockets. The rocket (mass 0.5 kg empty) ejects 0.2 kg exhaust gases backward at 40 m/s relative to rocket.

- a) Apply conservation of momentum to find rocket velocity just after exhaust ejection (assume starts from rest).
- b) Explain why rockets work in space (no air) but fireworks need initial launch.
- c) Analyse how this principle is applied in modern transportation (e.g., jet engines in aircraft flying to/from Zambian airports).

Question 5

In a Zambian mine transport system near Solwezi, a 2000 kg loaded truck moving at 8 m/s collides with a stationary 1500 kg empty truck. They couple together after impact.

- a) Calculate the common velocity after collision using conservation of momentum.
- b) Discuss whether momentum is conserved if external forces (friction) act during collision.
- c) Analyse the importance of understanding linear momentum in designing safer mining haulage roads and vehicle braking systems in Zambia.

SAMPLE QUESTIONS ANSWERS

Question 1

- a) Momentum before = momentum after (isolated system, no external forces in horizontal direction).

$$\text{Momentum before} = (1200 \times 15) + (800 \times 0) = 18\,000 \text{ kg m/s}$$

$$\text{Momentum after} = (1200 + 800) \times v = 2000 v$$

$$\text{So, } 18\,000 = 2000 v, v = 9 \text{ m/s (matches given value).}$$

- b) Inelastic because kinetic energy is not conserved (vehicles stick together). KE before = $\frac{1}{2} \times 1200 \times 225 = 135\,000 \text{ J}$ (car has 0). KE after = $\frac{1}{2} \times 2000 \times 81 = 81\,000 \text{ J}$. Lost KE = 54 000 J (converted to heat, sound, deformation).

c) Principle explains why head-on collisions are dangerous (high momentum transfer). Safety features like crumple zones increase collision time, reduce force (impulse = change in momentum = $F \times \Delta t$), seatbelts/airbags spread force over body, reducing injury. In Zambia, enforcing these in public transport reduces fatalities.

Question 2

a) Total initial momentum = 0.

Momentum after: fish + canoe = 0

$(2 \times -5) + (80 \times v) = 0$, $-10 + 80v = 0$, $v = 0.125$ m/s forward. Recoil due to conservation, action-reaction (Newton's third law).

b) Impulse on fish = change in momentum = $2 \times 5 = 10$ Ns (from rest to 5 m/s). Average force = impulse / time = $10 / 0.4 = 25$ N backward on fish. Equal/opposite force (25 N forward) on fisherman/canoe.

c) Principle: when you push backward on boat/ice, you gain forward momentum. Explains why jumping off a small boat pushes it away, common in fishing or crossing rivers in rural Zambia.

Question 3

a) Momentum before = $60 \times 4 + 55 \times 0 = 240$ kg m/s

Momentum after = $60 \times (-1) + 55 \times v = -60 + 55v$

$240 = -60 + 55v$, $55v = 300$, $v \approx 5.45$ m/s forward. (Real tackle inelastic, some energy lost.)

b) Δp tackler = $60 \times (-1 - 4) = -300$ kg m/s (loss). Δp tackled = $55 \times 5.45 \approx +300$ kg m/s (gain). Impulse = Δp (force $\times$ time from collision).

c) High momentum change, high force if short time, injury (e.g., concussions). Tackling technique (low body, longer contact) increases Δt , reduces F . Helmets/pads help. Coaches teach safe techniques to apply principle.

Question 4

a) Initial momentum = 0.

After: rocket ($0.5 \times v$) + gases (0.2×-40) = 0

$0.5v - 8 = 0$, $v = 16$ m/s forward. Recoil propels rocket.

b) Conservation holds in vacuum, exhaust momentum backward = rocket forward. Fireworks use gunpowder explosion for initial impulse + air resistance/gravity.

c) Jet engines eject high-speed gases backward, plane forward momentum. Explains efficient long-distance flights (e.g., to Johannesburg). Principle key to aviation safety/efficiency in Zambia.

Question 5

a) Momentum before = $2000 \times 8 + 1500 \times 0 = 16\,000 \text{ kg m/s}$

After = $(2000 + 1500) \times v = 3500 v$

$v = 16\,000 / 3500 \approx 4.57 \text{ m/s}$.

b) Approximately conserved if collision time short (external impulse small compared to collision impulse). Friction reduces total slightly but often ignored for approximation.

c) Explains rear-end collision risks on mine roads. Brakes increase stopping time, reduce force on occupants. Speed limits, distance keeping, and vehicle design (crumple zones) apply principle to prevent injuries/fatalities in heavy transport.

SUB-TOPIC 2.1.3: SIMPLE MACHINES



INTRODUCTION

Simple machines are mechanical devices that makes work easier. These include tools such as, screw driver, scissors, wheelbarrow, bottle opener, wrench, hoe, cooking sticks, stapler, tweezers, seesaw ... In this sub-topic, learners will study different types of simple machines and how they operate.

Specific Competence: Build simple machines to solve real life problems.



Key terms:

- ❖ Load (L): The weight of an object being moved by a simple machine.
- ❖ Effort (E): Is the force applied in moving the load.
- ❖ Mechanical advantage (MA): Is the ratio of the load to effort, $MA = L/E$
- ❖ Velocity ratio (VR): Is the ratio of distance moved by effort (d_E) to the distance moved by load (d_L), $VR = d_E/d_L$
- ❖ Efficiency (η): Is the ratio mechanical advantage to the velocity ratio expressed as a percentage, $\eta = MA/VR \times 100\%$

Learning Activities

The four learning activities in this sub-topic introduces learners to explore various simple machines by designing and building them. Various calculations will be done using the designed simple machines to calculate Mechanical advantage, Velocity ratio and efficiency.



Learning Activity 2.13.1: Building simple machines to solve real world problems

In this activity, learners will build simple machines which include levers, pulley, inclined plane/wedge, screw, wheel and axle and gears to solve real life problems by using the available materials provided.

HOOK: During the construction of a classroom, workers find a large stone blocking the foundation area. They try pushing it but it does not move easily. Give suggestions on how the stone can be removed easily.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** A stone, wooden stick, wooden board, fiber, clay, woodblocks...
- ❖ **Artificial Materials:** A heavy book, ruler, desk or table, pencil/eraser (fulcrum), cardboard, toy car, ball, string, thread roll, hanger, bucket or cup, bottle tops, cardboard circles, toy

wheels, plastic knife, paper, door, screws, bolts, jar lids, two gears of different sizes, gear board, handle or crank, weights, ...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's Roles:

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups.
- ❖ Provide the learners with the materials.
- ❖ Ask learners to identify real life problems that require the use of simple machines
- ❖ Have learners build simple machines to solve identified problems
- ❖ Have learners apply the constructed simple machines on the identified problems
- ❖ Ask learners probing questions.
- ❖ Provide constructive feedback and clear misconceptions.
- ❖ Ask learners to report their findings on the tasks



Learner's Task

- ❖ Identifying real life problems that require the use of simple machines
- ❖ Building simple machines to solve identified problems
- ❖ Applying the constructed simple machines on the identified problems
- ❖ Responding to probing questions.
- ❖ Receiving constructive feedback and clarity on misconceptions.
- ❖ Reporting their findings on the tasks



Assessment Guide

- ❖ Check if the learners are correctly identifying real life problems
- ❖ Evaluate learners' choice of a simple machines built to solve real life problems
- ❖ Check creativity and innovation



Skills Developed

- ❖ Collaborating: Working in pairs or small groups as they build simple machines by sharing tasks and responsibilities.

- ❖ Critical thinking: Identifying real life problems
- ❖ Creativity and innovation: Designing simple machines to solve real world problems.
- ❖ Analytical thinking: Analysing and logical reasoning of the built simple machine and how it makes work easier.
- ❖ Communicating: Explaining and presenting their ideas and designs clearly.



Key Learning Points

- ❖ A simple machine is a mechanical device that makes work easier.
- ❖ Load (L): The weight of an object being moved by a simple machine.
- ❖ Effort (E): Is the force applied in moving the load.



Learning Activity 2.13.2: Demonstrating the application of the various types of simple machines (lever, pulley, inclined plane/wedge, screw, wheel and axle, and gears).

In this activity learners will demonstrate the application of various types of simple machine. Simple machines do not reduce or increase the amount of work done, but they reduce the effort needed to perform a task.

HOOK: During sports activities, a child gets the bike stuck in a muddy slope near the playground. Discuss types of the simple machines that can be used to remove the bike from the mud.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Wooden stick, wooden board, fiber rope, stone...
- ❖ **Artificial Materials:** Scissors, knife, door key, bottle opener, ramp, fiber rope, spanner, wheelbarrow, bicycle, car steering ...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's Roles:

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups.
- ❖ Provide the learners with various types of simple machines.
- ❖ Ask learners to identify types of simple machines

- ❖ Allow learners to demonstrate the application of the various types of simple machines.
- ❖ Guide learners to ensure safety
- ❖ Ask learners probing questions on simple machines used in homes, farms, industries, and transport.
- ❖ Provide constructive feedback and clear misconceptions



Learner's Tasks:

- ❖ Identifying types of simple machines
- ❖ Demonstrating the application of the various types of simple machines.
- ❖ Following safety protocols
- ❖ Responding to probing questions on simple machines used in homes, farms, industries, and transport.
- ❖ Receiving constructive feedback and clarity misconceptions



Assessment Guide

- ❖ Check if the learners are correctly demonstrating the application of the various types of simple machines.
- ❖ Check if learners are classifying types of simple machines.



Skills Developed

- ❖ Collaborating: Working in pairs or small groups
- ❖ Critical thinking: Demonstrating how simple machines works
- ❖ Analytical thinking: Identifying various types of simple machines
- ❖ Communicating: Explaining and presenting their ideas and gives reasons as to why simple machines makes work easier.



Key Learning Points

- ❖ Lever: Used in scissors, plows, crowbars (makes work easier by changing force/distance)
- ❖ Pulley: Used in cranes, hoists (lifts heavy loads with less effort)
- ❖ Wheel and Axle: Used in cars, bicycles (reduces friction for movement)
- ❖ Inclined Plane: Used in ramps, slides (reduces effort to lift objects).
- ❖ Wedge: Used in knives, axes (applies concentrated force)
- ❖ Screw: Used in jacks, bolts (converts rotation to linear force).



Learning Activity 2.1.3.3: Determining Mechanical Advantage (MA), and Velocity Ratio (VR) of a simple machine; $MA = \frac{\text{Load}}{\text{Effort}}$ and $VR = \frac{\text{distance moved by effort}}{\text{distance moved by load}}$.

In this activity, learners will determine the mechanical advantage (MA) and Velocity Ratio (VR) of a simple machine.

HOOK: The village water well has a heavy bucket. Pulling it up by hand is very tiring. A fellow learner suggested to use a wheel and an axle to lift a bucket from the bottom of the well. How would the wheel reduce the pulling effort and how much rope would be pulled compared to the vertical height the bucket rises?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** A fiber rope, water,
- ❖ **Artificial Materials:** A synthetic rope, tape measure, bucket...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's Roles

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups
- ❖ Provide the learners with the worksheets having variety of scenario questions on MA and VR of a simple machine
- ❖ Allow the learners to discuss and determine the MA and VR mathematically from the scenarios provided
- ❖ Facilitate the discussion on how the load and effort affect MA and VR of simple machines as depicted from scenarios provided in the worksheet
- ❖ Ask learners probing questions.
- ❖ Provide constructive feedback and emphasise on correct use of formulae
- ❖ Ask learners to present their solutions to the scenarios.



Learner's Tasks

- ❖ Discussing and determining MA and VR mathematically from the scenarios provided
- ❖ Discussing on how the load and effort affect MA and VR of simple machines as depicted from scenarios provided in the worksheet
- ❖ Responding to probing questions
- ❖ Receiving constructive feedback and emphasis on correct use of formulae
- ❖ Presenting their solutions to the scenarios



Assessment Guide

- ❖ Check the correct using of the formulas of MA and VR of a simple machine.
- ❖ Check for correct interpretation of the scenario questions and collection of data



Skills Developed

- ❖ Collaborating: Working in pairs or small groups as they discuss and do calculations on the questions provided.
- ❖ Critical thinking: Analysing the geometry of the simple machine as provided in the worksheet.
- ❖ Communicating: Explaining and presenting their ideas on calculations



Key Learning Points

- ❖ Mechanical Advantage shows how much a machine multiplies the effort applied.
- ❖ If MA is greater than 1, the machine makes work easier by increasing force.
- ❖ If MA is less than 1, the machine does not multiply force.
- ❖ Velocity Ratio (VR) is the ratio of the distance moved by the effort to the distance moved by the load.
- ❖ Velocity ratio depends on the design of the machine and does not change due to friction.



Learning Activity 4: Deriving the formula for efficiency of a simple machine as

$$\eta = \frac{MA}{VR} \times 100\%$$

In this activity, learners will derive the formula for efficiency of a simple machine as $\eta = \frac{MA}{VR} \times 100\%$ using an experiment.

HOOK: You are tasked with evaluating a pulley system used to lift heavy logs in your school yard. The pulley lifts a 200kg log with an effort of 500N, and the effort moves 10m while the load moves 2m. How would you derive the efficiency of a pulley system?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** O_x – chart
- ❖ **Artificial Materials:** A real simple machine that is available e.g. wheelbarrow, pulley,

Learning Environment set up

- ❖ Natural Environment such as school surroundings.
- ❖ Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's Roles

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups
- ❖ Provide the materials (various machines) to the learners in groups
- ❖ Allow learners to discuss definition for efficiency, $Efficiency = \frac{work\ output}{work\ input} \times 100\%$ in terms of load, effort and distance moved by load, and distance moved by effort respectively leading to the identification of MA and VR in the formula above.
- ❖ Guide the learners in identifying which part of the machine is inputting work and which one is outputting work.
- ❖ Allow the learners to present the procedure leading to the derivation of the formula.

$$\eta = \frac{MA}{VR} \times 100\%.$$
- ❖ Provide feedback



Learner's Task

- ❖ Discussing definition for efficiency, $Efficiency = \frac{work\ output}{work\ input} \times 100\%$ in terms of load, effort and distance moved by load, and distance moved by effort respectively leading to the identification of MA and VR in the formula above.
- ❖ Identifying part of the machine that is inputting work and that is outputting work.
- ❖ Presenting the procedure leading to the derivation of the formula. $\eta = \frac{MA}{VR} \times 100\%.$
- ❖ Receiving feedback



Assessment Guide

- ❖ Check if learners are able to relate work done by effort to input work
- ❖ Check if learners are able to relate work done by load to output work
- ❖ Check if learners are able to relate imaging ratios of load, effort, distance moved by load, and distance moved by effort to MA and VR respectively.



Skills Developed

- ❖ Problem solving: Following step-by-step logical process of deriving the formula of efficiency and apply the formula in new situations.
- ❖ Collaborating: Working in pairs or small groups as they discuss and do calculations on the questions provided.
- ❖ Communicating: Presenting their procedure leading to the derivation of the formula. Explaining and presenting their ideas on calculations
- ❖ Critical thinking: Identifying various work done (by effort, by load) by machines



Key Learning Points

- ❖ Efficiency is an important concept in our daily lives, helps us understand how well machines converts input (energy or effort) into useful output.
- ❖ Replace work with force x distance. Efficiency = $\frac{\text{load} \times \text{distance moved by load}}{\text{effort} \times \text{distance moved by effort}} \times 100\%$.
- ❖ Separate the fractions $(\text{load/effort}) \times (\text{distance moved by load/distance moved by effort}) \times 100\%$.
- ❖ The overall formula will be Efficiency = $\text{MA/VR} \times 100\%$.

EXPECTED STANDARD

SAMPLE QUESTIONS

1. A farmer in Kitwe wants to lift a 400 kg sack of fertilizer onto a truck. He uses a first-class lever made from a long pole with the fulcrum placed 1 m from the sack (load arm = 1 m) and the effort arm = 4 m. The weight of the load is 4000 N ($g = 10 \text{ m/s}^2$).
 - (a) Calculate the Mechanical Advantage (MA).
 - (b) If the farmer moves the effort end 4 m down, how far does the sack rise? Calculate the Velocity Ratio (VR).
 - (c) Assuming the machine is 80 % efficient, calculate the actual effort required.

2. Workers at a construction site in Copperbelt Province use a block-and-tackle pulley system with 4 ropes supporting the load to lift a 1200 N steel beam to the second floor. The effort applied is 400 N.
 - (a) Calculate the Mechanical Advantage.
 - (b) The effort moves 8 m while the beam rises 2 m. Calculate the Velocity Ratio.
 - (c) Calculate the efficiency of the system and explain why it is less than 100 %.
3. A mechanic uses an inclined plane (ramp) to push a 2000 N car up a 5 m long ramp to a height of 1 m instead of lifting it vertically.
 - (a) Calculate the Velocity Ratio of the ramp.
 - (b) If the actual effort needed is 600 N, calculate the Mechanical Advantage.
 - (c) Calculate the efficiency and state one real-life advantage of using the ramp.
4. screw jack is used to lift the front of a 10 000 N truck for tyre repair. The screw has a pitch of 5 mm and the effort arm (handle) is 0.4 m long. One complete turn of the handle raises the truck 5 mm.
 - (a) Calculate the Velocity Ratio.
 - (b) If the actual effort applied is 200 N, calculate the Mechanical Advantage.
 - (c) Calculate the efficiency of the screw jack.
5. A bicycle uses a gear system (wheel and axle combined with gears). The rear wheel radius is 0.35 m and the pedal sprocket radius is 0.08 m. When the cyclist applies force to the pedals, the rear wheel turns once for every 4.375 turns of the pedals.
 - (a) Calculate the Velocity Ratio of the gear system.
 - (b) If the load (resistance) at the wheel is 300 N and the effort at the pedal is 120 N, calculate MA.
 - (c) Calculate efficiency and explain how gears help cyclists on hills.

SAMPLE QUESTIONS ANSWERS

Answer 1

- (a) $MA = \text{Load} / \text{Effort}$ (ideal case)

Since effort arm / load arm = 4 m / 1 m, ideal $MA = 4$.

- (b) $VR = \text{distance moved by effort} / \text{distance moved by load}$

$VR = 4 \text{ m} / 1 \text{ m} = 4$.

(c) Actual effort = Load / (MA × efficiency)

Efficiency η = MA / VR × 100 % (but here we use given η = 80 %)

Actual effort = 4000 N / (4 × 0.8) = 4000 / 3.2 = 1250 N.

This lever solves the real-life problem of lifting heavy loads with less force – exactly as required by the syllabus learning activity.

Answer 2

(a) MA = Load / Effort = 1200 N / 400 N = 3.

(b) VR = distance moved by effort / distance moved by load = 8 m / 2 m = 4

(c) η = (MA / VR) × 100 % = (3 / 4) × 100 % = 75 %

The efficiency is less than 100 % because of friction in the pulleys and weight of the movable pulleys themselves – common in real-life pulley systems built to solve lifting problems.

Answer 3

(a) VR = length of ramp / height = 5 m / 1 m = 5

(b) MA = Load / Effort = 2000 N / 600 N $\approx$ 3.33.

(c) η = (MA / VR) × 100 % = (3.33 / 5) × 100 % $\approx$ 66.6 %.

Advantage: The ramp reduces the force required (from 2000 N to 600 N), making it easier and safer for one person to move the car – a classic application of the inclined plane.

Answer 4

(a) Distance moved by effort in one turn = circumference of handle = $2\pi \times 0.4$ m $\approx$ 2.513 m.

VR = distance moved by effort / distance moved by load = 2.513 m / 0.005 m = 502.6

(b) MA = Load / Effort = 10 000 N / 200 N = 50

(c) η = (MA / VR) × 100 % = (50 / 502.6) × 100 % $\approx$ 9.95 %

(The low efficiency is due to high friction in the screw thread – typical for screw jacks used in garages.)

Answer 5

(a) $VR = \text{radius of effort wheel} / \text{radius of load wheel (or turns ratio)} = 0.35 / 0.08 = 4.375$
(matches the given ratio).

(b) $MA = \text{Load} / \text{Effort} = 300 \text{ N} / 120 \text{ N} = 2.5$

(c) $\eta = (2.5 / 4.375) \times 100 \% \approx 57.1 \%$

Gears increase VR on hills so the cyclist can apply smaller force over a longer distance – exactly how gears solve real-life transportation problems.

SUB-TOPIC 2.1.4: PRESSURE



INTRODUCTION

Pressure is an important concept in Physics. It explains how forces act over surfaces and within fluids (liquids and gases). Acquiring knowledge on pressure helps learners to explain everyday phenomena such as why sharp knives easily cut through materials as opposed to blunt one; why dams are made thicker at the bottom, how juice or water move up through drinking straws into our mouths, and how hydraulic press machines operate. However, this module introduces pressure in solids and fluids and its application in our everyday-life.

Specific competence 2.1.4.1

- Apply the principles of pressure to solve real-life problems

Key terms

- ❖ **Pressure:** A force acting normally (at 90° angle) per unit area.
- ❖ **Force:** A push on an object.
- ❖ **Area:** Two-dimensional space.
- ❖ **Pascal (Pa):** A force of 1N acting normally per unit area (1m²). In short, it is $\frac{\text{N}}{\text{m}^2}$.
- ❖ **Density (ρ):** Mass (m) per unit volume (V), $\rho = \frac{m}{V}$
- ❖ **Upthrust:** Upward force due to fluid
- ❖ **Flotation:** Is when a body is suspended in a fluid (neither sinking down the fluid nor leaving the fluid body)
- ❖ **Law of Flotation:** A floating body displaces a fluid of weight equal to its own weight
- ❖ **Pascal's principle:** Pressure applied at one point in an enclosed fluid at rest is transmitted equally to all parts of the fluid.
- ❖ **Archimedes' principle:** When a body is wholly or partially immersed in a fluid, it experiences an upthrust which is equal to the weight of the displaced fluid.
- ❖

Learning activities

This sub-topic includes ten learning activities from the syllabus that emphasise hands-on learning and problem-solving to enhance learners' understanding."



Activity 2.1.4.1: Describing Pressure and assigning its associated Units/Establishing the relationship between force and area.

These activities are designed to help learners have a picture of what pressure is through observing the effects of a constant force acting perpendicularly on different measures of surface areas of a body.

In these activities, learners will use a concrete building block of known weight and dimensions to conduct an experiment.

HOOK: Imagine two learners are carrying the same heavy concrete block, one places it flat on its large face in soft wet clay near a construction site, while the other stands it upright on its small end. Which way do you think the block will sink deeper into the clay, and why?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Clay, sand or soil, ...
- ❖ **Artificial Materials:** Concrete block, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the concept of pressure using a hook
- ❖ Divide learners into small groups.
- ❖ Ask learners to demonstrate how units of pressure are derived and expressed as a Pascal (N/m^2) using an experiment.
- ❖ Ask open-ended questions to the learners
- ❖ Provide feedback on learner's methods and conclusions
- ❖ Supervise learners on safety
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Demonstrating how units of pressure are derived and expressed as a Pascal (N/m^2) using an experiment.
- ❖ Responding to open-ended questions
- ❖ Receiving feedback and conclusions
- ❖ Observing safety protocols
- ❖ Presenting their findings



Assessment Guide

- ❖ Assess ability to conduct an experiment, to collect and analyse data, and to draw conclusions.
- ❖ Assess understanding of concepts of pressure

- ❖ Assess learners' understanding of application of pressure to real life



Skills Developed

- ❖ **Scientific Inquiry:** Conducting experiments, collecting and analysing data, and drawing conclusions.
- ❖ **Critical Thinking:** Evaluating evidence, identifying patterns, and making connections between concepts.
- ❖ **Communication:** Presenting their findings
- ❖ **Collaboration:** Working in groups, sharing ideas, and building on each other's strengths.



Key Learning Points

- **Pressure** is defined as a force acting perpendicularly on a unit area ($P = \frac{F}{A}$).
- The SI unit of pressure is Pascal. 1 Pascal is equal to 1 N/m².



Activity 2.1.4.2: Measuring pressure using pressure sensors (manometer, barometer and other appropriate instruments) This activity is designed to help learners familiarise themselves with different sensors used to measure pressure in fluids

HOOK: A bicycle repair shop owner checks the tire pressure of a minibus using a pressure gauge and finds it's only 172 kPa much lower than normal. The driver says the tires felt 'soft' on the bumpy road. Why do under-inflated tires wear out faster and feel less safe?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Surrounding Air, ...
- ❖ **Artificial Materials:** Air compressor, Pressure gauge, U-tube, barometer, inflatable tube or tire, water pipe ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or car fitment centre, water and sewerage pumping sites, ...

Teacher's Role(s)

- ❖ Introduce the activity using the hook
- ❖ Put learners in small groups
- ❖ Provide pressure sensors such as pressure gauge, manometer, and barometer, to the learners.
- ❖ Guide learners on how to use pressure sensors.
- ❖ Ask probing questions
- ❖ Provide feedback to learners
- ❖ Ask learners present their findings



Learners' Tasks

- ❖ Receiving guidance on how to use pressure sensors.
- ❖ Responding to probing questions
- ❖ Receiving feedback
- ❖ Present their findings



Assessment Guide

- ❖ Assess learners' ability to set and measure pressure using sensors.
- ❖ Assess learners' ability to communicate their measurements of pressure correctly.
- ❖ Assess learner's ability to correctly identify different scales or units of measure of pressure



Skills Developed

- ❖ **Problem-Solving:** Setting sensors and measuring pressure
- ❖ **Critical Thinking:** Identifying appropriate sensors for measuring different types of pressure.
- ❖ **Communication:** Presenting their findings
- ❖ **Collaboration:** Working in pairs or small groups to set pressure sensors



Key Learning Points

- ❖ A sensor is quantity that detects and responds to a physical quantity (like pressure, temperature) by producing a measurable output
- ❖ Manometer: Measures pressure difference (U-tube liquid displacement)
- ❖ Barometer: Measures atmospheric pressure (mercury column height)
- ❖ Other sensors: Bourdon tube, digital pressure sensors (electronic output)



Activity 2.1.4.3: Determining pressure using the relationship between force and area, $P = \frac{F}{A}$

This activity is designed to help learners determine pressure using the relationship between force and

area $P = \frac{F}{A}$

HOOK: You are designing a presentation on sandals for a local fashion show. How can you make high-heeled sandals comfortable on Zambian gravel roads? Discuss how force and area affect pressure.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Sand, soil, grass (for testing pressure effects), ...
- ❖ **Artificial Materials:** Metal blocks, wooden planks, foam (for experiments), ...

Learning environment set up

- ❖ Natural Environment such as school surroundings.
- ❖ Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory)

Teacher's Role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners in small groups
- ❖ Guide learners on an experiment to vary force or area
- ❖ Ask learners to observe and record the effects of varying force or area.
- ❖ Ask learners to determine the pressure when force or area is varied using the formula, $P = \frac{F}{A}$
- ❖ Ask learners to compare the results determined at each variation
- ❖ Ask learners present their findings



Learners' Tasks

- ❖ Receiving guidance on an experiment to vary force or area
- ❖ Observing and recording the effects of varying force or area.
- ❖ Determining pressure when force or area is varied using the formula, $P = \frac{F}{A}$
- ❖ Comparing the results determined at each variation

- ❖ Presenting their findings



Assessment Guide

- ❖ Observe learners conducting an experiment correctly
- ❖ Check learners' accuracy on calculations of pressure



Skills Developed

- ❖ Problem-solving: Applying pressure concepts to real life.
- ❖ Measurement: Taking accurate force/area readings.
- ❖ Critical thinking: Relating area to comfort, pressure effects.
- ❖ Collaboration: Working together in groups doing an experiment
- ❖ Communication: Presenting their findings



Key Learning Points

- ❖ Pressure = Force / Area.
- ❖ Smaller area has a higher pressure (e.g., heels on gravel).
- ❖ Applications in design, engineering,



Activity 2.1.4.4: Deriving the equation for pressure in fluids (Pascal's Law); $P = \rho hg$ / Exploring factors affecting pressure in liquids (density, height and gravity).

Note that two activities in the syllabus i.e activity 5 and 7 have been merged.

In these activities, learners will design and conduct a practical task to derive an equation for pressure and discuss factors affecting pressure in liquids

HOOK: During the rainy season, learners notice that the base of a tall water storage tank made of concrete always develops cracks first, while the top remains fine. If the tank is filled to different heights with water, why does the pressure at the bottom increase so much more?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water,
- ❖ **Artificial Materials:** Glass Jar, metre rule, vernier calipers, measuring cylinder, electronic balance and an electronic calculator.

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Divide learners in small groups
- ❖ Provide guidance and support to learners on how to find the mass (m), height (h) of the water (liquid) in a jar and cross-section area (A) of the Jar.
- ❖ Ask learners to calculate volume of the column of water in the jar as, ($V = Ah$)
- ❖ Ask learners to calculate weight of water as $W = mg$
- ❖ Remind the learners that density (ρ) of a material is equal to mass per unit volume ($\rho = \frac{m}{V}$) and therefore, $F = \rho Vg$ or $F = \rho Ahg$
- ❖ Facilitate learners' derivation of an equation for pressure in fluids from the definition, $P = \frac{F}{A}$ to lead to $P = \rho hg$
- ❖ Provide feedback to learners
- ❖ Ask learners from each group to present how they derived the equation $P = \rho hg$ and factors affecting pressure in fluids



Learners' Tasks

- ❖ Receiving guidance and support on how to find the mass (m), height (h) of the water (liquid) in a jar and cross-section area (A) of the Jar.
- ❖ Calculating volume of the column of water in the jar as, ($V = Ah$)
- ❖ Calculating weight of water as $W = mg$
- ❖ Receiving a reminder that density (ρ) of a material is equal to mass per unit volume ($\rho = \frac{m}{V}$) and therefore, $F = \rho Vg$ or $F = \rho Ahg$
- ❖ Deriving an equation for pressure in fluids from the definition, $P = \frac{F}{A}$ to lead to $P = \rho hg$
- ❖ Receiving feedback

- ❖ Presenting how they derived the equation $P = \rho hg$ and factors affecting pressure in fluids.

Assessment Guide

- ❖ Assess learners' ability to relate cross-section area of a vessel and height of fill of a fluid to volume, mass and density.
- ❖ Assess learners' ability to analyse and identify variables and the relationships among them that lead to the definition of pressure in fluids.
- ❖ Assess learners' ability to communicate their findings from the practical task to the class.
- ❖ Ask open-ended question, did the learner demonstrate an understanding of the definition of pressure in fluids?



Skills Developed

- ❖ **Problem-Solving:** Deriving the solution to knowing the pressure of fluid at a given depth, height and how much pressure a liquid of particular density exerts at a given position in a fluid body.
- ❖ **Analytical Thinking:** Analysing effect of density and height on the pressure of a fluid.
- ❖ **Communication:** Presenting how they derived the equation $P = \rho hg$
- ❖ **Collaboration:** Working in pairs or small groups to set and conduct an experiment.



Key Learning Points

- ❖ Pressure in liquids depend on:
 - (a) the density of liquid (ρ)
 - (b) the height of column of liquid, h

The formula for pressure in a liquid is given by, $P = \rho hg$.



Activity 2.1.4.5: Calculating pressure in liquids and gas using $P = \rho hg$.

This activity is designed to help learners solve numerical problems on pressure in fluids.

HOOK: You are a diver working at a flooded copper mine and must descend to 10 meters below the water surface to inspect equipment. Why would you feel much stronger pressure on your ears at that depth compared to swimming in a shallow river, and roughly how many times greater is the pressure from the water column alone?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ...
- ❖ **Artificial Materials:** Rulers, whiteboard, markers, calculator, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using the scenario
- ❖ Divide class into small groups.
- ❖ Assign numerical scenario questions to each group using the worksheet
- ❖ Ask learners to solve numerical scenario questions on pressure in fluids using the formula, $P = \rho hg$.
- ❖ Lead a guided discussion to help learners connect the solutions to the questions on some real-life situation
- ❖ Ask learners to present their solutions
- ❖ Provide feedback



Learners' Tasks

- ❖ Solving numerical scenario questions on pressure in fluids using the formula, $P = \rho hg$.
- ❖ Connecting the solutions to the questions on some real-life situation with guidance
- ❖ Presenting their solutions
- ❖ Asking for clarity
- ❖ Receiving feedback



Assessment Guide

- ❖ Observe learners when calculating pressure in liquid and gas
- ❖ Administer a short- written test to assess learners' understanding of the pressure in fluids.



Skills Developed

- ❖ **Scientific Inquiry:** Relating questions on pressure in fluids to their everyday experiences.
- ❖ **Analytical Thinking:** Analysing data given in a worksheet and drawing conclusions on which variable(s) ought to be calculated in the question.
- ❖ **Communication:** Sharing their findings and explaining the concept of pressure in fluids.

- ❖ **Problem-Solving:** Applying the equation on pressure in fluids to real-world scenarios.



Key Learning Points

- ❖ Calculating pressure in liquids and gases, the formula is $P = \rho hg$ where P = pressure (pascal, Pa), ρ = fluid density (kg/m^3), g = gravitational acceleration (9.81m/s^2 on earth) and h = height or depth of the fluid (m)



Activity 2.1.4.5: Conducting an experiment to demonstrate principles of up thrust and floatation (Archimedes principle)

This learning activity will comprise two sub-activities



Activity 2.1.4.5.1: Demonstrating the principles of Upthrust

This activity is designed to help learners acquire skills to demonstrate principles of up-thrust.

HOOK: You lower a heavy stone tied to a string into the water. Even though the stone is still heavy, why does it suddenly feel much lighter once it's fully underwater, and what invisible force is helping to support it?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Stone and water, ...
- ❖ **Artificial Materials:** Graduated beaker or measuring cylinder and spring balance, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Put learners into small groups
- ❖ Provide materials to the learners in groups
- ❖ Instruct learners to weigh a stone in air and record the reading.

- ❖ Guide learners to lower the stone, while still attached to the spring balance, slowly in a beaker partially filled with water, and record the new reading.
- ❖ Ask learners to compare the two readings and then explain the reason for the difference
- ❖ Provide feedback to learners on their demonstration of upthrust.
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Weighing and recording the weight of the stone in air.
- ❖ Weighing and recording the weight of the stone immersed in water using a spring balance.
- ❖ Comparing the weight of the stone in air and in water and giving a reason for the difference
- ❖ Receiving feedback on the demonstration of upthrust.
- ❖ Presenting their findings



Assessment Guide

- ❖ Assess learners' ability to use spring balance.
- ❖ Assess learners' ability to interpret the difference in the readings.
- ❖ Assess learners' ability to draw conclusions on their observations



Skills Developed

- ❖ **Critical Thinking:** Interpretation of difference in the readings.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to weigh and record the readings



Activity 2.1.4.5.2: Demonstrating the principles of Floatation

This activity is designed to help learners acquire skills to demonstrate Archimedes principles.

HOOK: A large steel canoe floats easily on water. Yet a solid steel of the same weight sinks immediately. Why do some objects float while others sink, even if made of the same material?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Wooden Block and water, ...

- ❖ **Artificial Materials:** Electronic Balance, Eureka Can, Graduated beaker or measuring cylinder and spring balance, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's Role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups.
- ❖ Guide learners to weigh a wooden block in air and record the reading.
- ❖ Ask learners to fill the Eureka Can up to the spout and set it in such a way that any overflow collects into a graduated beaker or measuring cylinder.
- ❖ Instruct learners to lower the wooden block gently into the Eureka can until it is seen to float and some water collects into the beaker or measuring cylinder.
- ❖ Allow learners to find the weight of the collected water in the beaker or cylinder.
- ❖ Ask learners to compare the weight of the wooden block and the weight of the displaced water and make an inference.
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Weighing and recording weight of a wooden block in air.
- ❖ Filling the Eureka Can up to the spout and setting it in such a way that any overflow collects into a graduated beaker or measuring cylinder.
- ❖ Lowering the wooden block gently into the Eureka Can until it is seen to float and some water collects into the beaker or measuring cylinder.
- ❖ Finding the weight of the collected water in the beaker or cylinder.
- ❖ Comparing the weight of the wooden block and the weight of the displaced water and making an inference.
- ❖ Receiving feedback



Assessment Guide

- ❖ Assess learners' ability to set and use Eureka Can, spring balance, electronic balance and measuring cylinder or beaker.
- ❖ Assess learners' ability to observe the difference in the weight of the wooden block and the weight of the displaced water.
- ❖ Assess learners' ability to draw conclusions on their observations



Skills Developed

- ❖ **Critical Thinking:** Analysing data and drawing conclusions about how objects float on different media.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to weigh and record the readings.



Key Learning Points

- ❖ **Pascal's principle** states that the pressure applied at one point in an enclosed fluid at rest is transmitted equally to all parts of the fluid.
- ❖ The upward force due to a fluid is called **Upthrust**.
Upthrust, $U_{\text{upthrust}} = W_{\text{air}} - W_{\text{fluid}}$, where W_{air} is weight of object in air and W_{fluid} is weight of the same object in a fluid denser than air e.g water.
- ❖ The magnitude of upthrust depends on: The density of the fluid and the volume of body immersed.
- ❖ **Archimedes' principle** states that when a body is wholly or partially immersed in a fluid, it experiences an upthrust which is equal to the weight of the fluid displaced.
- ❖ **Law of flotation** states that a floating body displaces a fluid of weight equal to its own weight.



Activity 2.1.4.6: Creating a model that uses the principle of pressure (hydraulic press/brake and car jack, simple manometer, a mercury barometer)



Activity 2.1.4.6.1: Creating a Hydraulic Press System Model

This activity is designed to help learners acquire skills in Creating a hydraulic Press system.

HOOK: In a small garage in school, a mechanic uses a simple car jack to lift a heavy school bus with just a few pumps of a handle. How would such a small force from one person lift several tons of metal?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Coloured water,
- ❖ **Artificial Materials:** Two plastic syringes (one large, one small-without needles), transparent plastic tube, rubber bands/tape, and small weights or books

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos,

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners in small groups.
- ❖ Distribute materials to the learners.
- ❖ Instruct learners to fill both syringes and connecting tube completely with coloured water (ensure no air bubbles in the system)
- ❖ Instruct learners to connect the syringes firmly using the plastic tube.
- ❖ Instruct learners to place a small load on the plunger of the larger syringe.
- ❖ Ask learners to press on the plunger of a small syringe gently
- ❖ Ask learners to observe what happens to the larger plunger and the load.
- ❖ Provide feedback to learners on their creation.



Learners' Tasks

- ❖ Filling both syringes and connecting tube completely with coloured water (ensure no air bubbles in the system)
- ❖ Connecting the syringes firmly using the plastic tube.
- ❖ Placing a small load on the plunger of the larger syringe.
- ❖ Pressing on the plunger of a small syringe gently
- ❖ Observing what happens to the larger plunger and the load.
- ❖ Receiving feedback on their creation.



Assessment Guide

- ❖ Assess learners' ability to create a hydraulic press model
- ❖ Assess learners' ability to observe and analyse data obtained using a created hydraulic press model.
- ❖ Assess learners' ability to link hydraulic press to real life.
- ❖ Assess learners' ability to observe the magnitude of input force and output force in the hydraulic press system.

- ❖ Assess learners' ability to communicate their observation to the class
- ❖ Assess learners' ability to draw conclusions on their observations.



Skills Developed

- ❖ **Real world application:** Application of hydraulic press system in lifting of heavy loads, drilling, bending, crushing, ...
- ❖ **Creativity and innovation:** Connecting the two syringes by a plastic tube.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to weigh and record the readings.



Activity 2.1.4.6.2: Creating a Simple Manometer Model

This activity is designed to help learners acquire skills in creating a manometer.

HOOK: A welder in a school workshop notices that when he/she blows air into one end of a U-shaped tube filled with colored water, the water level on the other side rises dramatically. How would this simple setup help mechanics check tire pressure?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Coloured water, ...
- ❖ **Artificial Materials:** Transparent plastic tubing (U-shaped), Ruler, Tape and Cardboard or wooden board, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Put learners into small groups.
- ❖ Distribute materials to the learners.
- ❖ Instruct learners to bend the transparent tube into a U-shape and tape it to a board.

- ❖ Instruct learners to pour coloured water into the tube until it is about halfway full but leave one end open to the atmosphere.
- ❖ Instruct learners to gently blow air into the open end or connect it to an inflated balloon.
- ❖ Ask learners to observe what happens to the level of water on both sides of the manometer.
- ❖ Provide feedback to learners on their creation of the manometer.
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Bending the transparent tube into a U-shape and tape it to a board
- ❖ Pouring coloured water into the tube until it is about halfway full but leave one end open to the atmosphere.
- ❖ Blowing air gently into the open end or connecting it to a balloon.
- ❖ Observing what happens to the level of water on both sides of the manometer upon blowing or connecting it to an inflated balloon.
- ❖ Receiving feedback on their creation of the manometer.
- ❖ Presenting their findings



Assessment Guide

- ❖ Assess learners' ability to create a manometer model
- ❖ Assess learners' ability to observe and analyse data obtained using a created manometer model.
- ❖ Assess learners' ability to link manometer press to real life
- ❖ Assess learners' ability to observe the change in level of the water on both side of the arms of the manometer.
- ❖ Assess learners' ability to draw conclusions on their observations.



Skills Developed

- ❖ **Real world application:** Application of manometers to real life problem
- ❖ **Creativity and innovation:** Creating a model of a manometer using plastic tube
- ❖ **Critical Thinking:** Analysing the observations.
- ❖ **Communication:** Presenting their findings or observations to the class.
- ❖ **Collaboration:** Working in small groups.



Activity 2.1.4.6.3: Creating a Mercury Barometer Model.

This activity is designed to help learners acquire skills in creating a Barometer.

HOOK: Before a big storm hits, a weather station reports that the mercury level in their barometer tube has dropped lower than usual. Why would a mercury level in a sealed tube drop?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Coloured water, and a finger,
- ❖ **Artificial Materials:** Transparent plastic tube (about 80cm to 100cm) closed at one end, Shallow bowl, and a Ruler,

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Put learners into small groups.
- ❖ Distribute materials to the learners.
- ❖ Instruct learners to fill the tube completely with coloured water.
- ❖ Instruct learners to cover the open end with a finger and invert the tube and place the covered end into a bowl of water.
- ❖ Instruct learners to gently release the finger while keeping the tube vertical.
- ❖ Ask learners to observe the water level in the tube.
- ❖ Provide feedback to learners on their creation of the barometer.
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Filling the tube completely with coloured water.
- ❖ Covering the open end with a finger and thereafter inverting the tube and placing the covered end into a bowl of water.
- ❖ Releasing the finger gently while keeping the tube vertical.
- ❖ Observing the water level in the tube.
- ❖ Receiving feedback on their creation of the barometer.
- ❖ Present their findings



Assessment Guide

- ❖ Assess learners' ability to create a barometer model
- ❖ Assess learners' ability to observe and analyse data obtained using a created barometer model.
- ❖ Assess learners' ability to link created barometer to the mercury barometer
- ❖ Assess learners' ability to communicate their observation to the class
- ❖ Assess learners' ability to draw conclusions on their observations.



Skills Developed

- ❖ **Real world application:** Application of barometers to real life situation
- ❖ **Creativity and innovation:** Making a barometer using a plastic tube and water
- ❖ **Critical Thinking:** Analysing the observations and inferencing.
- ❖ **Communication:** Presenting their findings/ observations to the class.
- ❖ **Collaboration:** Working in small groups.



Key Learning Points

Air pressure has many applications in everyday life situations. e.g. in a bicycle pump, siphon, drinking straw, syringe, force and lift pumps, etc.

EXPECTED STANDARD

SAMPLE QUESTIONS

1. A miner carries a 120 kg load on his head using a flat board (area 0.15 m²). Calculate the pressure on his head. ($g = 9.8 \text{ m/s}^2$)
2. During rainy season, a farmer notices that a tractor with narrow tires sinks deeper into muddy soil than one with wide tires, even though both carry the same weight. Explain this using the concept of pressure.
3. A woman wearing high-heeled shoes (heel area 1 cm² = 0.0001 m²) and weighing 600 N stands on soft ground. Calculate the pressure under each heel and explain why she might sink more than if wearing flat shoes (heel area 100 cm²).
4. In a Zambian market, a heavy sack of mealie meal (weight 500 N) is placed on a wooden pallet with base area 0.8 m². Calculate the pressure on the floor. If the sack is tipped onto a smaller area of 0.2 m², what happens to the pressure and why might the floor crack?
5. A sharp knife cuts through meat easily, but a blunt one does not, even with the same force. Explain this difference in terms of pressure, and suggest how a chef in Lusaka could improve cutting efficiency without increasing force

SAMPLE QUESTIONS ANSWERS

Answer 1:

$$\text{Force} = \text{weight} = \text{mass} \times g = 120 \text{ kg} \times 9.8 \text{ m/s}^2 = 1176 \text{ N}$$

$$\text{Pressure} = F / A = 1176 \text{ N} / 0.15 \text{ m}^2 = 7840 \text{ Pa (or 7.84 kPa)}$$

(Working must be shown for full marks.)

Answer 2:

Pressure = Force / Area. Both tractors have the same weight (force), but narrow tires have smaller contact area → higher pressure → greater force per unit area on the soft mud → tractor sinks deeper. Wide tires spread the weight over larger area → lower pressure → less sinking.

Answer 3:

Assuming weight on two heels: Force per heel = $600 \text{ N} / 2 = 300 \text{ N}$

Pressure (high heel) = $300 \text{ N} / 0.0001 \text{ m}^2 = 3,000,000 \text{ Pa}$ (3 MPa)

Pressure (flat shoe) = $300 \text{ N} / 0.1 \text{ m}^2 = 3,000 \text{ Pa}$

The high heel produces ~1000 times higher pressure, much greater tendency to sink into soft ground.

Answer 4:

Original pressure = $500 \text{ N} / 0.8 \text{ m}^2 = 625 \text{ Pa}$

New pressure = $500 \text{ N} / 0.2 \text{ m}^2 = 2500 \text{ Pa}$ (4 times higher)

Pressure increases because the same force is concentrated over a smaller area, higher risk of cracking weak wooden floors.

Answer 5

Pressure = Force / Area. Sharp knife has very small contact area at the edge, very high pressure → cuts easily. Blunt knife has larger contact area → lower pressure → does not cut well.

Suggestion: Sharpen the knife (reduce area) or use a serrated knife to concentrate force.

PHYSICS TEACHING MODULE

TERM 2

TOPIC 2.2: THERMAL PHYSICS



INTRODUCTION

Thermal Physics is the branch of Physics that studies the thermal properties of matter, including temperature, heat, and the behaviour of systems in relation to thermal energy. This topic introduces learners to important concepts that explain how heat affects matter and how thermal energy is transferred and utilised in everyday life. Knowledge of Thermal Physics is important in many practical applications such as cooking, food preservation, engine operation, energy production, and industrial processes. Thermal Physics covers six sub-topics: Simple Kinetic Theory of Matter, Measurement of Temperature, Expansion of Solids, Liquids and Gases, the Internal Combustion Engine, Heat Transfer, and Measurement of Heat. These sub-topics help learners develop an understanding of how thermal energy influences the behaviour of matter and how this knowledge can be applied in science, technology, and everyday life.



Key Competence(s):

- Analytical Thinking
- Collaborative
- Communication
- Critical Thinking
- Problem Solving
- Real world application
- Environmental sustainability
- Creativity and innovation

SUB-TOPIC 2.2.1 SIMPLE KINETIC THEORY OF MATTER



INTRODUCTION

Matter is made up of tiny particles (atoms or molecules) that are in constant random motion, with spaces between them. The particle's motion and interactions explain the properties of solids, liquids and gases.

Specific competence

- Analyse the impact of simple kinetic theory of matter on technological innovations



Key terms

- **Kinetic theory of matter:** Is a scientific model that explains the behavior of matter by stating that particles (atoms or molecules) are in constant motion, and their energy relates to temperature and phase changes.

- **Gas leak detector:** Is a device that senses the presence of gas (e.g., methane, propane) in the air to alert users of a potential leak, preventing explosions or poisoning.
- **Thermal insulators:** These are materials that reduce the transfer of heat between objects or environments, keeping heat in or out to improve energy efficiency (examples include foam, fiberglass, or aerogel).
- **Refrigeration system:** Is a mechanical setup that removes heat from a space or substance to lower its temperature, typically using a refrigerant that cycles through compression, condensation, evaporation, and expansion.
- **Molecular model of matter:** Is a representation that describes substances as composed of molecules with specific arrangements and interactions, used to predict physical and chemical properties.
- **Brownian motion:** Is the random, erratic movement of microscopic particles suspended in a fluid (liquid or gas), caused by collisions with fast-moving molecules in the surrounding medium.
- **Diffusion:** Is the process by which particles spread from an area of high concentration to an area of low concentration, driven by random molecular motion until equilibrium is reached.
- **Evaporation:** Is the phase transition where liquid molecules gain enough energy to escape the surface and become vapor, typically cooling the remaining liquid.
- **Cooling effect of evaporation:** Is the temperature drop that occurs when a liquid evaporates, because the highest-energy molecules leave the surface, reducing the average energy (temperature) of the remaining liquid.

Learning activities

This module has four learning activities from the syllabus which have been expanded into nine activities, focusing on competence-based learning through hands-on activities and problem-solving tasks.



Activity 2.2.1.1: Describing the kinetic theory of matter

This activity is designed to help learners describe the kinetic theory.

HOOK: You notice that when boiling water for nshima, steam rises from the pot and spreads around the kitchen and you wonder what is happening to the particles in the water as it heats up.

a) Using the kinetic theory of matter, explain what happens to the particles of water when it is heated.

b) Why do the particles move faster as the temperature increases?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Stones, water, leaves, ...
- ❖ **Artificial Materials:** Balloon, plastic bottle, metal spoon, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce kinetic theory of matter using a hook
- ❖ Put learners in small groups
- ❖ Provide the materials for them to describe kinetic theory of matter
- ❖ Facilitate group work on kinetic theory of matter
- ❖ Ask probing questions such as:
 1. How do particles move when heated?
 2. How would you explain the kinetic theory of matter to predict the behavior of gas particles in a sealed container when temperature rises?
 3. In what ways does the kinetic theory of matter help you analyse real-world phenomena like pressure changes in tires?
- ❖ Encourage explanation using kinetic theory of matter
- ❖ Ask learners to present their findings.



Learners' Tasks

- ❖ Participating actively during the discussion from the hook
- ❖ Responding to probing questions such as:
 1. How do particles move when heated?
 2. How would you explain the kinetic theory of matter to predict the behavior of gas particles in a sealed container when temperature rises?
 3. In what ways does the kinetic theory of matter help you analyse real-world phenomena like pressure changes in tires?
- ❖ Explaining the concept of kinetic theory of matter
- ❖ Presenting their findings.



Assessment Guide

- ❖ Observe learner's participation during the discussion
- ❖ Review written explanations
- ❖ Evaluate understanding through feedback
- ❖ Check application of kinetic theory



Skills Developed

- ❖ Critical Thinking: Explaining particle's behavior using the kinetic theory of matter
- ❖ Communication: Presenting their findings
- ❖ Collaboration: Working in small groups
- ❖ Scientific Inquiry: Observing and interpreting data



Key Learning Points

- ❖ Particles in Gases: Widely spaced, moving rapidly
- ❖ Temperature Effects: Heat increases particle movement
- ❖ Kinetic Theory of Matter: Explains particle behavior changes



Activity 2.2.1.2: Exploring the molecular model of matter.

This activity is designed to help learners explore molecular model of matter through experiments.

HOOK: In a science lesson, learners were provided with three substances: a stone, water in a cup, and air in a balloon. Discuss how would you show the arrangement of particles in these three states of matter?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Sugar, water, clay, ...
- ❖ **Artificial Materials:** Food coloring, plastic cups, stirring rod, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce an activity of molecular model using a scenario.
- ❖ Put learners in small groups
- ❖ Provide them with the materials for them to explore on molecular model of matter
- ❖ Facilitate group work
- ❖ Ask guiding questions:
 1. How do molecules move when hot and when cold?
 2. How can you use the molecular model to compare the arrangement of particles in solids, liquids and gases?

- ❖ Encourage explanation using molecular model
- ❖ Allow them to present their findings



Learners' Tasks

- ❖ Exploring molecular model of matter
- ❖ Discussing in small group
- ❖ Responding to guiding questions:
 1. How do molecules move when hot and when cold?
 2. How can you use the molecular model to compare the arrangement of particles in solids, liquids and gases?
- ❖ Explaining using the molecular model
- ❖ Present their findings



Assessment Guide

- ❖ Observe learners explore molecular model of matter
- ❖ Review written explanations
- ❖ Evaluate learners understanding through feedback
- ❖ Check learners understanding of application of molecular model



Skills Developed

- ❖ Critical Thinking: Explaining molecular behavior using a model
- ❖ Communication: Presenting their findings
- ❖ Collaboration: Working in small groups
- ❖ Analytical thinking: Breaking down molecular structures under different states of matter to understand their properties.
- ❖ Scientific reasoning: Using evidence to infer relationships between molecular arrangement and material behavior.



Key Learning Points

- ❖ Molecular Motion: Molecules are in constant random motion
- ❖ Temperature Effects: Heat increases molecular speed of particles
- ❖ Molecular Arrangement: Affects substance properties (e.g., dissolving, melting, ...)
- ❖ Model Application: Explains behavior of matter in terms particle arrangement.



Activity 2.2.1.3: Demonstrating kinetic theory of matter such as Brownian motion, diffusion, evaporation, cooling effect of evaporation.

This activity is designed to help learners demonstrate kinetic theory of matter such as Brownian motion, diffusion, evaporation, cooling effect of evaporation. These demonstrations must be done in a safe place.



Activity 2.2.1.3.1: Demonstrating kinetic theory of matter using Brownian motion.

This activity is designed to help learners demonstrate kinetic theory of matter using Brownian motion by observing smoke or pollen particles.

HOOK: While studying in the school laboratory, you observe tiny pollen grains moving randomly on the surface of water using a microscope.

a) Name the phenomenon observed in this experiment.

b) Explain why the pollen grains move randomly.

Suggested Teaching and Learning Materials

Natural Materials: Smoke particles in air, pollen grains in water,

Artificial Materials: Microscope, magnifying glass, phones, Brownian motion kit,

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity Brownian Motion using a hook
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Ask learners to set up Brownian Motion experiment using smoke or pollen grains and observe using a microscope, phone, magnifying glass, ...
- ❖ Facilitate discussions on the findings
- ❖ Ask learners to relate Brownian motion to kinetic theory of matter principles
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Setting up Brownian Motion experiment using smoke or pollen grains and observe microscope, phone, magnifying glass, ...
- ❖ Discussing the findings
- ❖ Relating Brownian motion to kinetic theory of matter principles
- ❖ Presenting their findings
- ❖ Receiving feedback



Assessment Guide

- ❖ Evaluate observation accuracy, explanations of motion causes, and linking to kinetic theory



Skills Developed

- ❖ Observation: Handling of equipment to observe random motion of particles
- ❖ Inference: Connecting random motion to molecular collisions
- ❖ Collaboration: Presenting their findings
- ❖ Communication: Working in small groups
- ❖ Scientific reasoning: Using theory to explain phenomena



Key Learning Points

- ❖ Random motion of particles in fluids due to molecular bombardment.
- ❖ Evidence supporting kinetic theory of matter.



Activity 2.2.1.3.2: Demonstrating kinetic theory of matter using diffusion.

This activity is designed to help learners demonstrate kinetic theory of matter using diffusion.

HOOK: During break time, a learner sprays perfume in one corner of the classroom. After a few minutes, learners sitting far away from that corner can smell it.

a) How would the smell of perfume reach other learners?

b) State one factor that can increase the rate of this process.

Suggested Teaching and Learning Materials

Natural Materials: Ink from plants, water, scent from flowers, ...

Artificial Materials: Ink, perfume, droppers, jar, beaker, diffusion apparatus, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners in small groups
- ❖ Provide the learners with materials
- ❖ Ask learners to set up diffusion experiment using the materials available
- ❖ Ask learners to discuss diffusion.
- ❖ Guides learners to compare diffusion rates in gases and in liquids
- ❖ Ask learners probing questions
- ❖ Ask learners to relate diffusion to kinetic theory of matter principles
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Setting up diffusion experiment using the materials available
- ❖ Discussing diffusion in small groups.
- ❖ Comparing diffusion rates in gases and in liquids
- ❖ Responding to probing questions
- ❖ Relating diffusion to kinetic theory of matter principles
- ❖ Presenting their findings
- ❖ Receiving feedback on the findings



Assessment Guide

- ❖ Check explanations of diffusion process
- ❖ Check learner's ability to compare the rate of diffusion in liquids and gases
- ❖ Check learner's ability to apply diffusion to kinetic theory of matter.



Skills Developed

- ❖ Creativity and Innovation: Setting up an experiment on diffusion

- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings
- ❖ Analysis: Linking diffusion to kinetic theory of matter
- ❖ Psychomotor: Handling apparatus to observe particle movement
- ❖ Application: Relating diffusion to real-life like observing spreading of scent, contamination of water, ...



Key Learning Points

- ❖ Diffusion is random particle movement driven by concentration gradients.
- ❖ Faster in gases than liquids; affected by temperature and particle size.
- ❖ Important in biological processes (gas exchange, nutrient transport).



Activity 2.2.1.3.3: Demonstrating kinetic theory of matter using evaporation.

This activity is designed to help learners demonstrate kinetic theory of matter using evaporation.

HOOK: You notice that on a hot day, wet shirt hung outside dries faster than on a cool day.

- How you explain this cause using the kinetic theory of matter?
- Give one other factor that can increase the rate of drying.

Suggested Teaching and Learning Materials

Natural Materials: Water, cloth, sunlight, ...

Artificial Materials: Thermometer, evaporation dish, stopwatch, crucible, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technology Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners in small groups
- ❖ Provide the materials to the learners in groups
- ❖ Ask learners to set up evaporation experiment using the materials available

- ❖ Ask learners to conduct experiment.
- ❖ Ask learners to measure evaporation rates and record data.
- ❖ Facilitate discussions
- ❖ Ask learners probing questions
- ❖ Guides investigation of factors affecting evaporation rate.
- ❖ Ask learners to relate evaporation to kinetic theory of matter principles
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Setting up evaporation experiment using the materials available
- ❖ Conducting experiment on evaporation.
- ❖ Measuring evaporation rates and recording data.
- ❖ Discussing on evaporation
- ❖ Responding to probing questions
- ❖ Investigating the factors affecting rate of evaporation.
- ❖ Relating evaporation to kinetic theory of matter principles
- ❖ Presenting their findings



Assessment Guide

- ❖ Check explanations of evaporation process
- ❖ Check learner's ability to compare the rate of evaporation
- ❖ Check learner's ability to apply evaporation to kinetic theory of matter.



Skills Developed

- ❖ Creativity and Innovation: Setting up an experiment on evaporation
- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings
- ❖ Analysis: Linking evaporation to kinetic theory of matter
- ❖ Application: Relating evaporation to real-life, like drying of clothes, cooling effect of evaporation, ...



Key Learning Points

- ❖ Evaporation involves high-energy molecules escaping through the liquid surface.
- ❖ Rate increases with temperature, surface area, and airflow.
- ❖ Used in processes like drying and cooling systems.



Activity 2.2.1.3.4: Demonstrating kinetic theory of matter using cooling effect of evaporation.

This activity is designed to help learners demonstrate kinetic theory of matter using cooling effect of evaporation by evaporating liquids.

HOOK: After playing football, you pour some methylated spirit on your hand and notice that the hand feels cold. How would explain the cause of the cooling on your hand?

Suggested Teaching and Learning Materials

Natural Materials: Water, wet cotton wool, ...

Artificial Materials: Wet-bulb hygrometer, thermometer, electric fan, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide all the necessary materials
- ❖ Ask learners to conduct experiment on cooling effect of evaporation using the materials provided
- ❖ Ask learners to observe temperature change during evaporation
- ❖ Ask learners probing questions
- ❖ Facilitate discussions on cooling effects of evaporation
- ❖ Ask learners to relate cooling effects of evaporation to kinetic theory of matter principles
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Conducting experiment on cooling effect of evaporation
- ❖ Observing temperature change during evaporation experiment
- ❖ Responding to probing questions
- ❖ Discussing cooling effects of evaporation
- ❖ Relating cooling effects of evaporation to kinetic theory of matter principles
- ❖ Presenting their findings



Assessment Guide

- ❖
- ❖ Check explanations of cooling effect of evaporation
- ❖ Check learner's ability to observe temperature change during evaporation
- ❖ Check learner's ability to apply cooling effect of evaporation



Skills Developed

- ❖ Creativity and Innovation: Setting up an experiment on cooling effect of evaporation
- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings
- ❖ Analysis: Linking cooling effect of evaporation to kinetic theory of matter



Key Learning Points

- ❖ Evaporation removes energy, causing temperature drop.
- ❖ Biological use in sweating, ... and industrial use in cooling systems,
- ❖ Effectiveness of cooling, depends on evaporation rate and environmental conditions e.g ambient temperature, humidity, drought, ...



Activity 2.2.1.1.4: Developing an innovation on simple kinetic theory of matter to solve a real-world problem (*gas leak detector, thermal insulators, refrigeration system, ...*)

This activity is designed to help learners develop an innovation on simple kinetic theory of matter concept to solve a real-world problem (*gas leak detector, thermal insulators, refrigeration system...*).



Activity 2.2.1.1.4.1: Developing an innovation on simple kinetic theory of matter to solve a real-world problem using gas leak detector.

This activity is designed to help learners develop an innovation based on the simple kinetic theory of matter to solve a real-world problem using gas leak detector.

HOOK: A learner notices that sometimes gas leaks from a bunsen burner but it is difficult to detect. Discuss how diffusion of gas particles can help detect a gas leak. Suggest a simple device that could help detect leaking gas.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Wood for casing, plant-based fibers, ...

- ❖ **Artificial Materials:** Semiconductor sensors (MQ-type), micro-controller board, batteries, wiring, plastic casing, LED/alarm buzzer, video, computer simulation, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the learners with a video or a simulation on gas leak.
- ❖ Ask learners to identify real-world problems from a video or a simulation
- ❖ Ask learners to research on gas leak detection
- ❖ Ask learners to construct a model of a gas leak detector
- ❖ Allow the learners to discuss how they would install the gas leak detector and ensure safety protocols.
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Watching a video or a simulation on gas leak.
- ❖ Identifying real-world problems from a video or a simulation
- ❖ Researching on gas leak detector
- ❖ Constructing a model of a gas leak detector
- ❖ Discussing how they would install the gas leak detector and ensuring safety protocols are followed.
- ❖ Receiving feedback on a gas leak detector installation



Assessment Guide

- ❖ Evaluate the constructed a model of a gas leak detector
- ❖ Assess learner's ability to choose the right position, and ability to install a gas leak detector.



Skills Developed

- ❖ Technical design: Ability to construct a model of a gas leak detector.
- ❖ Problem-solving: Choice of the right position, and ability to install a gas leak detector.
- ❖ Collaboration: Working in groups
- ❖ Scientific inquiry: Conducting research on gas leak detector.
- ❖ Application: Applying kinetic theory of matter knowledge to design a gas leak detector



Key Learning Points

- ❖ Kinetic theory explains gas particle motion and diffusion, which underlies leak detection.
- ❖ Sensors exploit molecular collisions to indicate gas presence.
- ❖ Safety protocols are essential when handling potential gas hazards.
- ❖ Innovation can solve real-world problems by applying basic scientific principles.



Activity 2.2.1.1.4.2: Developing an innovation on simple kinetic theory of matter to solve a real-world problem using thermal insulators.

This activity is designed to help learners develop an innovation based on the simple kinetic theory of matter to solve a real-world problem using thermal insulators.

HOOK: During a science project, learners want to design a simple cooler that keeps drinks cold without electricity. How would you design a simple cooler using locally available materials?

Suggested Teaching and Learning Materials

Natural Materials: Cotton wool, cork from wood, bamboo, ...

Artificial Materials: Foam sheets, fiberglass, aluminum foil, plastic wraps, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide learners with the materials
- ❖ Ask learners to research on thermal insulator e.g a cooler box, food warmer, thermal flask,
- ❖ Ask learners to construct a model of thermal insulator using materials provided
- ❖ Ask learners probing questions
- ❖ Provide feedback to the learners
- ❖ Ask learner to present their findings on the model



Learners' Tasks

- ❖ Researching on thermal insulator e.g a cooler box, food warmer, thermal flask,
- ❖ Constructing a model of thermal insulator using materials provided
- ❖ Responding to probing questions
- ❖ Receiving feedback from the teacher
- ❖ Presenting their findings on the model



Assessment Guide

- ❖ Evaluate experimental setup
- ❖ Assess learner's ability to select appropriate materials for the model
- ❖ Assess the ability to construct a thermal insulator model
- ❖ Check teamwork and analytical skills



Skills Developed

- ❖ Creativity: Constructing a model of thermal insulator
- ❖ Innovation: Linking kinetic theory to the created model
- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings on the model constructed.



Key Learning Points

- ❖ Kinetic theory links particle motion to heat transfer in materials.
- ❖ Insulation reduces conduction & convection by limiting molecular energy exchange.
- ❖ Natural materials often have porous structures that trap air, lowering thermal conductivity.
- ❖ Optimising thickness and layering improve insulation performance.
- ❖ Real-world applications include building thermal efficiency and energy saving.



Activity 2.2.1.4.3: Developing an innovation on simple kinetic theory of matter to solve a real-world problem using refrigeration System

This activity is designed to help learners develop an innovation based on the simple kinetic theory of matter to solve a real-world problem using refrigeration System

HOOK: A learner notices that fish and milk spoil quickly during very hot days because some traders do not have proper cooling facilities. A learner decided to develop a simple innovation based on the kinetic theory of matter and the idea used in a refrigeration system to help keep the fish and milk fresh for a longer time. How would you develop an innovation of a refrigeration system?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water,
- ❖ **Artificial Materials:** Compressor or simple evaporator coil, refrigerant (safe non-hazardous type), tubing, foams, clay (for construction of a pot), cotton clothes (chitenge material), sisal sack, thermometers, plastic container, insulated box, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulations, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide learners with the materials
- ❖ Ask learners to research on application of kinetic theory of matter regarding evaporation and cooling effects.
- ❖ Ask learners to construct a model of refrigeration system (Joe sack) using materials provided
- ❖ Assist learners in analysing efficiency and optimisation
- ❖ Ask learners probing questions
- ❖ Provide feedback to the learners
- ❖ Ask learner to present their findings



Learners' Tasks

- ❖ Researching on application of kinetic theory of matter regarding evaporation and cooling effects.
- ❖ Constructing a model of refrigeration system (Joe sack) using materials provided
- ❖ Analysing efficiency and optimisation of the model constructed
- ❖ Responding to probing questions
- ❖ Receiving feedback from the teacher
- ❖ Presenting their findings



Assessment Guide

- ❖ Check learners understanding on functionality of the cooling effect and temperature measurements.
- ❖ Assess learner's ability to choose appropriate materials to construct a refrigeration system model.
- ❖ Assess understanding of evaporation and kinetic theory of matter.
- ❖ Evaluate problem-solving in optimising refrigerant flow or insulation.



Skills Developed

- ❖ Creativity: Constructing a model of refrigeration system (Joe sack)
- ❖ Critical thinking: Constructing a refrigeration system (Joe sack) using local materials
- ❖ Collaboration: Working with others to construct a refrigeration system model
- ❖ Communication: Presenting their findings
- ❖ Analytical thinking: Interpreting temperature variation.
- ❖ Innovation: Using local materials in constructing a model based on simple kinetic theory of matter (cooling effects of evaporation).



Key Learning Points

- ❖ Kinetic theory of matter: Cooling effects of evaporation
- ❖ Refrigeration cycles use evaporation and condensation to transfer heat.
- ❖ Efficient systems require good insulation and optimal refrigerant flow.
- ❖ Measuring temperature variation cold chamber and evaporator (where the cold chamber is the inside part of the container and the evaporator is the sack wrapped outside the container).
- ❖ Simple refrigeration models (Joe sack, clay pot, ...) can be used in real-world applications like food preservation.
- ❖ Innovation in design can improve energy efficiency and cooling performance.

EXPECTED STANDARD

SAMPLE QUESTIONS

Question 1

A Zambian company is manufacturing affordable double-walled stainless steel vacuum flasks (thermos) for households and street vendors to keep tea or nshima warm for many hours without reheating.

- a) Using the simple kinetic theory of matter, explain why heat loss is greatly reduced in a vacuum flask compared to an ordinary single-walled container.
- b) Analyse how this technological innovation improves daily life and the economy in areas with limited electricity or firewood.
- c) Suggest one limitation of this technology.

Question 2

Engineers in Lusaka are designing a more efficient compressed natural gas (CNG) system for public minibuses to replace diesel use.

- a) Explain, using the kinetic theory of matter, why compressing natural gas into high-pressure cylinders allows a large volume of fuel to be stored in a small space.
- b) Analyse the impact of this technology on Zambia's transport sector and environment.
- c) Describe what happens to the temperature of the gas during rapid filling according to kinetic theory, and why cooling is sometimes needed.

Question 3

A food processing plant in Zambia is installing industrial spray dryers to turn fruit pulp (mango, banana) into powder for longer shelf life and export.

- a) Using kinetic theory, explain how the rapid evaporation in the spray dryer lowers the temperature of the tiny liquid droplets.
- b) Analyse how this innovation helps Zambian farmers and the export economy.
- c) Explain why the drying air must be hot according to the kinetic theory.

Clue: Need to take an educational tour to a processing plant to have experiential link to the application of kinetic theory of matter.

Question 4

A factory is producing aerosol mosquito repellents using compressed gas propellants for household and agricultural use.

- a) Using the simple kinetic theory of matter, explain how pressing the nozzle releases the liquid as a fine mist.
- b) Analyse the positive and negative impacts of this technology on public health in Zambia.
- c) Suggest how kinetic theory helps manufacturers make the product safer.

Question 5

Solar-powered water pasteurisation units are being distributed in rural Zambian communities to kill bacteria in drinking water without electricity or fuel.

- a) Explain, using kinetic theory, why heating water to approximately 65 to 70 °C for a few minutes is sufficient to make it safe, even though it is not boiling.
- b) Analyse the broader impact of this technology on community health and education.
- c) Describe what happens to particle motion when the water cools after pasteurisation.

ANSWERS TO SAMPLE QUESTIONS

Question 1

- a) According to the simple kinetic theory, heat transfer by conduction occurs when fast-moving particles in a hot substance collide with slower particles and transfer kinetic energy. Convection occurs when heated gas/liquid particles rise, carrying kinetic energy. In the vacuum flask, the space between the double walls has no (or very few) gas particles, almost no particles to collide and transfer energy, conduction and convection are virtually eliminated. Only slow radiation remains (often reduced by shiny surfaces).
- b) The innovation allows people to carry hot food/drinks safely for long periods, reduces need to re-boil water or re-heat food, saves charcoal, electricity, or paraffin. It lowers household costs and deforestation pressure. Street vendors keep food hot longer, sell more, increase income. In clinics, it helps keep water hot for hygiene.
- c) Limitation: Radiation still occurs slowly (cannot be completely stopped), so contents eventually cool. Very high initial cost for quality flasks may limit access in poorest households.

Question 2

- a) Kinetic theory states gas pressure is caused by particles colliding frequently and forcefully with container walls. Compressing the gas forces particles closer together, more frequent and harder collisions per unit area, much higher pressure. At high pressure, the same number of particles occupies far less volume, large amount of gas stored in small cylinders.
- b) Impact: Lower fuel cost per kilometer, cheaper public transport fares, benefits low-income passengers. Cleaner burning than diesel, reduces particulate matter and CO₂ in urban areas like Lusaka, improves air quality and public health. Supports local gas industry growth.
- c) Rapid compression does work on the gas, increases average kinetic energy of particles, temperature rises significantly (can become dangerously hot). Cooling systems (fans or heat exchangers) are used to remove excess heat and prevent damage to cylinders or valves.

Question 3

- a) In liquids, particles have a range of kinetic energies. The fastest-moving particles at the surface escape (evaporate), average kinetic energy of remaining particles decreases, temperature of droplets drops (evaporative cooling).
- b) Powder has very long shelf life (no water for microbes), farmers sell surplus fruit instead of losing it to spoilage, higher income. Export of dried fruit powder brings foreign currency, supports national economy and job creation in processing plants.
- c) Hot air provides high kinetic energy to gas particles, increases collision rate and energy transfer to liquid droplets, speeds up evaporation rate so drying happens quickly before droplets spoil or stick.

Question 4

- a) Inside the sealed can, gas particles are under high pressure (frequent, forceful wall collisions). When the nozzle opens, gas particles rush out rapidly, dragging liquid particles with them → liquid breaks into tiny droplets (aerosol mist).
- b) Positive: Widespread use reduces malaria transmission, fewer hospital cases, healthier workforce and school attendance. Negative: Inhaling propellants or overuse can cause respiratory irritation; improper disposal of cans risks explosion or environmental harm.
- c) Manufacturers control pressure (by adjusting gas/liquid ratio) so it is high enough for good spray but not dangerously high, reduces explosion risk. Use safer propellants with lower flammability.

Question 5

- a) Bacteria and pathogens are proteins/enzymes. Heating increases kinetic energy of water particles, more frequent and energetic collisions with pathogen molecules, denatures (unfolds) proteins, kills/inactivates them. 65 to 70 °C for minutes achieves this without needing 100 °C boiling (saves energy/time).
- b) Safer drinking water, sharp drop in diarrhea, cholera, typhoid, fewer child deaths and school absences, better education and productivity. Reduces firewood/charcoal use, less deforestation and smoke-related lung disease.
- c) As water cools, average kinetic energy of particles decreases, particles move slower, fewer collisions, lower pressure and slower diffusion. Dissolved gases may come out of solution if cooled rapidly.

SUB-TOPIC 2.2.2: MEASUREMENT OF TEMPERATURE



INTRODUCTION

Temperature is a measure of the average kinetic energy of the particles of a substance (the degree of hotness or coldness of a body). The S.I unit of temperature is a kelvin (K). Temperature is measured using a **thermometer**.

Specific competence

- Measure temperature using appropriate instruments



Key terms

Lower fixed point: This is the temperature of pure melting ice at standard atmospheric pressure.

Upper fixed point: This is the temperature at which pure water boils under normal atmospheric pressure

Ice point: Is the temperature of pure melting ice at one atmosphere.

Steam point: Is the temperature of pure boiling water at one atmosphere.

Thermometer: Is an instrument used to measure temperature.

Boiling point: Is the temperature at which saturated vapour pressure of a liquid is equal to the external pressure.

Melting point: Is the temperature at which a substance changes from solid to liquid.

Calibration: Is the process of standardising unmarked thermometer to give accurate temperature readings.

Temperature scale: Is a system for measuring temperature in Celsius, Kelvin scales,

Physical properties: These are characteristics such as volume, density, electrical resistance, and gas pressure that change with temperature.

Impurities effect: Is the influence of impurities on the boiling and melting points of substances, typically raising or lowering them.

Pressure effect: Is the influence of external pressure on the boiling and melting points of substances.

Thermometric Liquids: These are liquids used in Liquid-in-Glass Thermometers.

Learning activities

This module has nine learning activities from the syllabus which are focusing on competence-based learning through hands-on activities and problem-solving tasks.



Activity 2.2.2.1.1: Measuring temperature using appropriate instruments.

This activity is designed to help learners measure temperature using appropriate instruments.

HOOK: During a science lesson, a learner is asked to measure the temperature of hot water, room air, and a metal object heated over a burner in the laboratory. Identify the most appropriate instrument you would use to measure the temperature of each item.

Suggested Teaching and Learning Materials

Natural Materials: Ice, water, ...

Artificial Materials: Liquid-in-glass thermometer, metal, thermocouple, infrared scanner, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners in small groups
- ❖ Provide the learners with the materials
- ❖ Ask learners to select the appropriate thermometer type to use.
- ❖ Ask learners to measure and record temperature of substances provided using appropriate thermometers
- ❖ Ask probing questions such as:
 1. What factors would you consider in selecting the most suitable thermometer for measuring the temperature of some substances?
 2. What steps would you take to ensure accurate temperature readings when using a thermocouple?
- ❖ Ensure safety when handling thermometers and measuring temperatures (e.g. avoiding breakage and coming into contact with mercury and excessive temperatures).
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Selecting the appropriate thermometer type to use.

- ❖ Measuring and recording temperature of substances provided using appropriate thermometers
- ❖ Responding to probing questions such as:
 3. What factors would you consider in selecting the most suitable thermometer for measuring the temperature of some substances?
 4. What steps would you take to ensure accurate temperature readings when using a thermocouple?
- ❖ Following safety protocols
- ❖ Presenting their findings



Assessment Guide

- ❖ Evaluate learner's ability to select appropriate thermometer
- ❖ Assess learner's ability to read accurately
- ❖ Check learner's ability in data presentation
- ❖ Assess learner's adherence to safety protocols



Skills Developed

- ❖ Psychomotor: Ability to handle thermometers correctly in different situations
- ❖ Analytical thinking: Interpreting instrument readings and assessing measurement validity.
- ❖ Observation: Correct reading of temperature
- ❖ Communication: Presenting their findings
- ❖ Collaboration: Working in small groups



Key Learning Points

- ❖ Types of thermometers and suitable applications.
- ❖ Proper immersion and reading techniques.
- ❖ Accurate data recording and interpretation.
- ❖ Temperature is a physical quantity
- ❖ Different thermometers (instruments) suit different contexts



Activity 2.2.2.1.2: Interpreting temperature data from various sources (*experiments...*)

This activity is designed to help learners interpret temperature data collected from various experiments.

HOOK: In a school experiment, a learner records temperature of water from different sources such as a river, hot spring, cold spring, boiling pot, ... The recorded temperatures are 25°C, 70°C, 10°C, 100°C, ... respectively. How would you interpret the above temperature data from various sources?

Suggested Teaching and Learning Materials

Natural Materials: Water, ...

Artificial Materials: Thermometers, boiling pot, ...

Learning environment set up

- Natural Environment such as school surroundings, hot spring, cold spring, river, dam, streams, ...
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce an activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide learners with materials.
- ❖ Take learners to selected water sources atleast a minimum of four.
- ❖ Ask learners to measure temperature of water at various sources
- ❖ Facilitate group discussion on data interpretation from the experiment.
- ❖ Provide learners with similar data from sources not known to them and ask them to interpret.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners.



Learners' Tasks

- ❖ Selecting water sources
- ❖ Measuring temperature of water at various sources
- ❖ Interpreting data from the experiment through discussion.
- ❖ Receiving data from sources not known to them and interpreting data using inference.
- ❖ Presenting data findings
- ❖ Receiving feedback



Assessment Guide

- ❖ Evaluate learner's ability to interpret data accurately
- ❖ Check logical measurement of temperature



Skills Developed

- ❖ Analytical thinking: Extracting meaning from data
- ❖ Communication: Presenting data findings

- ❖ Collaboration: Working in small groups
- ❖ Critical thinking: interpreting data collected through inferences.



Key Learning Points

- ❖ Sources of temperature data
- ❖ Validation methods for experimental readings
- ❖ Effects of environmental conditions on measurements
- ❖ Reporting and interpreting data accurately
- ❖ Data collected from various sources represents physical phenomena like hot or cold
- ❖ Trends show thermal changes
- ❖ Interpretation informs scientific decisions



Activity 2.2.2.3: Recognising various types of thermometers: (*liquid in glass, thermocouple, thermo scanners (infrared scanners)*)

This activity is designed to help learners recognise various types of thermometers such as liquid in glass, thermocouple, thermo scanners (infrared scanners), ... by checking their construction and operation.

HOOK: While visiting a clinic, a learner notices that the nurse uses a digital thermometer to check a patient's temperature, while in the laboratory her teacher uses a liquid-in-glass thermometer, and at a copper mine smelter the technician uses a thermocouple or infrared scanner. Explain how a learner would recognise each type of the thermometer.

Suggested Teaching and Learning Materials

Natural Materials: None

Artificial Materials: Liquid-in-glass, thermocouple, infrared scanner, demonstration samples, ...

Learning environment set up

- Natural Environment such as school surroundings, healthy post, ...
- Artificial Environment such as the school laboratory, mines smelter, clinics, hospital, ...

Teacher's Role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the materials to the learners
- ❖ Ask the learners to discuss the construction and characteristics of each thermometer

- ❖ Ask learners to present different characteristics of the thermometers discussed.
- ❖ Ask learners to demonstrate safe operation and handling for every type
- ❖ As open-ended questions such as:
 1. What are the operational limits of each and every thermometer?
 2. How would you decide which thermometer to use for a precise measurement?
- ❖ Provide feedback to the learners
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Discussing the construction and characteristics of each thermometer
- ❖ Presenting different characteristics of the different thermometers through a discussion
- ❖ Demonstrating safe operation and handling of each and every thermometer
- ❖ Responding to pen-ended questions such as:
 1. What are the operational limits of each and every thermometer?
 2. How would you decide which thermometer to use for a precise measurement?
- ❖ Receiving feedback from the teacher
- ❖ Presenting their findings



Assessment Guide

- ❖ Check learner's ability to recognise each type of thermometer
- ❖ Assess learner's ability to understand construction and applications of each type of the thermometer



Skills Developed

- ❖ Classification: Grouping or sorting thermometer in terms of construction and application.
- ❖ Observation: Identifying various types of thermometers.
- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings.



Key Learning Points

- ❖ Liquid-in-glass uses expansion
- ❖ Thermocouples use electrical signals
- ❖ Infrared scanners measure surface radiation.
- ❖ Appropriate industrial or lab applications
- ❖ Specific safety precautions for each thermometer



Activity 2.2.2.1.4: Determining the boiling and melting points of different substances.

This activity is designed to help learners determine the boiling and melting points of different substances by conducting a practical experiment.

HOOK: In a laboratory, a learner heats a beaker containing ice until temperature remains constant. He continues heating until the next constant temperature is reached. What would you observe during the constant temperatures?

Suggested Teaching and Learning Materials

Natural Materials: Water, wax, metal (lead, aluminium, ...), ice, ...

Artificial Materials: Heating apparatus (bunsen burner, candle, water bath, spirit lamps, ...), thermometer, safety gear, stopwatch, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can use mobile laboratory.
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide learners with materials for the experiment.
- ❖ Ask learners to determine the melting and boiling points of substances using an experiment (record the temperature and time during the experiment).
- ❖ Ask learners to determine the melting and boiling points of substances graphically using the data collected from experiment
- ❖ Ask learners to report their findings
- ❖ Provide feedback to the learners.



Learners' Tasks

- ❖ Determining melting and boiling points of substances using an experiment
- ❖ Determining melting and boiling points of substances graphically using the data collected from experiment
- ❖ Reporting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's ability to conduct an experiment in a safe environment
- ❖ Assess learner's acquisition of experimental data accurately

- ❖ Check learners understanding of constant temperature at melting and boiling points
- ❖ Assess learner's ability on graphical analysis
- ❖ Assess learner's ability to draw conclusion from the analysis



Skills Developed

- ❖ Experimental design: Setting up an experiment correctly
- ❖ Collaboration: Working together in small groups
- ❖ Communication: Presenting their findings
- ❖ Precision: Achieving reproducible measurements with minimal error
- ❖ Observation: Ability to observe the constant temperature at melting and boiling.
- ❖ Critical thinking: Ability to interpret the experimental data collected



Key Learning Points

- ❖ Definitions of melting and boiling points
- ❖ Applications in material science and cooking



Activity 2.2.2.5: Experimenting on the effects of pressure and impurities on the melting and boiling points of substances.

This activity is designed to help learners experiment on the effects of pressure and impurities on the melting and boiling points of substances.

HOOK: Musa heats a pot of pure water and it boils within a short time. He then adds salt to another pot of water of the same amount and heats it. The salt water takes longer to boil.

a) What would be reason for this observation?

b) Predict what would happen to the boiling point of water at the top of a mountain.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, salt, ice, ...
- ❖ **Artificial Materials:** Pressure vessel, pure & impure substances, thermometer, pressure gauge, ...

Learning environment set up

- ❖ Natural Environment such as school surroundings.

- ❖ Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory
- ❖ Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide all the materials to the learners in groups
- ❖ Ask learners to conduct experiments by varying pressure and adding impurities (record the temperature and time during the experiments).
- ❖ Ask learners to compare results on melting and boiling temperatures.
- ❖ Ask learners to draw graphs for effects of impurities and pressure on the melting and boiling points of a substances from the data collected from experiment
- ❖ Ask learners to discuss the effects of impurities and pressure on melting and boiling points of substances.
- ❖ Supervises safe operation of pressure equipment.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Conducting experiments by varying pressure and adding impurities (record the temperature and time during the experiments).
- ❖ Comparing results on the collected data on melting and boiling temperatures.
- ❖ Drawing graphs for effects of impurities and pressure on the melting and boiling points of a substances from the data collected from experiment
- ❖ Discussing the effects of impurities and pressure on melting and boiling points of substances.
- ❖ Following safe operation of pressure equipment.
- ❖ Present their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Evaluate learner's experimental safety protocols
- ❖ Assess learners' data acquisition accuracy during the experiment
- ❖ Assess learner's quality of explanations on the effects of pressure and impurity on the melting and boiling points



Skills Developed

- ❖ Scientific inquiry: Investigating how pressure and impurities affects boiling and melting point of substances.
- ❖ Critical thinking: Linking pressure and impurities to phase change

- ❖ Collaboration: Working in small groups
- ❖ Communication: Presenting their findings
- ❖ Safety management: Handling pressurised systems safely



Key Learning Point

- ❖ Pressure and impurities effects on phase changes of substances



Activity 2.2.2.6: Calibrating unmarked thermometer.

This activity is designed to help learners calibrate unmarked thermometer using fixed points.

HOOK: In the laboratory, a learner is given a thermometer without markings. The teacher asks a learner to calibrate it using ice and boiling water. State how you would calibrate the unmarked thermometer using the ice and boiling water?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ...
- ❖ **Artificial Materials:** Unmarked liquid-in-glass thermometer, reference calibrated thermometer, ice-water bath, boiling-water bath, beaker, stirring rod,

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g mobile laboratory.
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the materials to the learners
- ❖ Ask the learners to discuss the purpose of calibration
- ❖ Ask learners to place the bulb of unmarked thermometer in a water bath containing pure melting ice and observe the mercury thread dropping to a constant point and mark this point as 0°C.
- ❖ Ask learners to place to expose the bulb of unmarked thermometer to steam in the hypsometer or hot water bath and observe the rising of the mercury thread to a constant point and mark this point as 100 °C
- ❖ Ask learners determine the range between 0°C and 100°C using a ruler

- ❖ Ask the learners to divide the range into 100 equal parts and spreads the parts throughout the range.
- ❖ Ask the learners to compare the readings of the calibrated to the referenced calibrated thermometer tested at lower fixed point and upper fixed points respectively.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners.



Learners' Tasks

- ❖ Discussing the purpose of calibration
- ❖ Placing the bulb of unmarked thermometer in a water bath containing pure melting ice and observing the mercury thread dropping to a constant point and mark this point as 0°C.
- ❖ Placing to expose the bulb of unmarked thermometer to steam in the hypsometer or hot water bath and observing the rising of the mercury thread to a constant point and mark this point as 100 °C
- ❖ Determining the range between 0°C and 100°C using a ruler
- ❖ Dividing the range into 100 equal parts and spreading the parts throughout the range.
- ❖ Comparing the readings of the calibrated to the referenced calibrated thermometer tested at lower fixed point and upper fixed points respectively.
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Evaluate correct safety setup
- ❖ Assess learner's ability to determine the fixed points
- ❖ Assess learner's ability to determine the spread over the range



Skills Developed

- ❖ Technical skill: Executing specific procedures to determine the fixed points as well the range and spread.
- ❖ Quality control: Testing the calibrated against the referenced thermometer.
- ❖ Instrumentation: Understanding calibration processes
- ❖ Collaboration: Working in groups to determine the fixed points
- ❖ Communication: Presenting the findings



Key Learning Points

- ❖ Principles of thermometer calibration and its necessity in science.
- ❖ Use of fixed points (0°C and 100°C) and their physical basis.
- ❖ Importance of calibration for trustworthy experimental data



Activity 2.2.2.1.7: Experimenting on the physical properties that change with temperature (*volume, density, electrical resistance, gas pressure, ...*)

This activity is designed to help learners experiment on the physical properties that change with temperature (*volume, density, electrical resistance, gas pressure...*)

HOOK: A learner heats a fixed mass of air trapped in a sealed, flexible cylinder fitted with a pressure gauge and a resistive wire embedded in the wall. As the temperature rises, how would the volume, density, gas pressure (assuming volume was held constant), and electrical resistance change?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Air, ...
- ❖ **Artificial Materials:** Flexible gas cylinder fitted with pressure gauge, resistance wire with ohmmeter, thermometer, heating apparatus (Bunsen burner or hot plate), safety goggles, gloves, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the materials to the learners
- ❖ Ask learners to research on the physical laws governing temperature effects on volume, density, electrical resistance, and gas pressure.
- ❖ Guide learners to set up the apparatus (as in the scenario above) and do the experiment on physical properties that change with temperature (volume, density, electrical resistance, gas pressure, ...)
- ❖ Instructs precise temperature recording and corresponding property measurement in the experiment
- ❖ Ask learners present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Researching on the physical laws governing temperature effects on volume, density, electrical resistance, and gas pressure.
- ❖ Setting up the apparatus (as in the scenario above) and doing the experiment on physical properties that change with temperature (volume, density, electrical resistance, gas pressure, ...)
- ❖ Receiving Instruction on precise temperature recording and corresponding property measurement in the experiment
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess safety of setup,
- ❖ Assess learner's understanding of effects of temperature on density, volume, electrical resistance and gas pressure.



Skills Developed

- ❖ Experimental technique: Controlling variables (temperature) and safe handling
- ❖ Collaboration: Working together in small groups
- ❖ Communication: Presenting their findings
- ❖ Analytical thinking: Relating observed changes to theoretical principles.
- ❖ Observation: Observing change in volume, density, electrical resistance and gas pressure when heat was supplied.



Key Learning Points

- ❖ Temperature effect on density and how it relates to volume change
- ❖ Relationship between electrical resistance and temperature in conductors
- ❖ Increase of gas pressure as the temperature increases.
- ❖ Practical application in engineering and everyday phenomena



Activity 2.2.2.8: Experimenting on suitability of alcohol and mercury for use in liquid-in-glass thermometers.

This activity is designed to help learners experiment on suitability of alcohol and mercury for use in liquid-in-glass thermometers.

HOOK: During a science lesson, a learner compares two thermometers: one containing mercury and the other containing alcohol. How would you compare the suitability of the two thermometric liquids for use in thermometers?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Test liquids (water, oil), ...
- ❖ **Artificial Materials:** Alcohol (ethanol) liquid-in-glass thermometer, mercury liquid-in-glass thermometer, temperature source (hot water bath), safety gear (gloves, fume hood for mercury), comparison chart, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e, g. mobile laboratory
- Technological Environment such as virtual lab, simulation, video, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the materials to the learners
- ❖ Ask learners to immerse each thermometer in test liquids and read temperatures accurately
- ❖ Instructs learners to note response time, measurement range and readability for each thermometer
- ❖ Ask learners to discuss their observations linking to physical properties of alcohol & mercury (expansion characteristics, temperature range, visibility)
- ❖ Provide the learners with feedback
- ❖ Ask learners to present their findings



Learners' Tasks

- ❖ Immersing each thermometer in test liquids and reading temperatures accurately
- ❖ Instructing learners to note response time, measurement range and readability for each thermometer
- ❖ Discussing observations linking to physical properties of alcohol & mercury (expansion characteristics, temperature range, visibility)
- ❖ Receiving feedback from the teacher
- ❖ Presenting their findings



Assessment Guide

- ❖ Evaluate learners correct identification of thermometric properties
- ❖ Assess learners safe handling of thermometers
- ❖ Assess learners understanding of accuracy and stability of each thermometric liquids for its suitability for use in thermometers



Skills Developed

- ❖ Comparative analysis: Evaluating two thermometric liquids for use in thermometers.
- ❖ Safety awareness: Recognising which thermometric liquids is toxic to humans and environment.
- ❖ Collaboration: Working together in small groups
- ❖ Communication: Presenting their findings
- ❖ Observation: Noting physical responses of each thermometric liquids.



Key Learning Points

- ❖ Physical characteristics of alcohol and mercury (expansion rate, toxicity, visibility)
- ❖ Advantages and limitations of alcohol versus mercury thermometers (temperature range, response)
- ❖ Safety protocols for handling mercury and alcohol in laboratory
- ❖ Application criteria for choosing thermometer type in experiments or industry. Note the Choice depends on temperature range and safety



Activity 2.2.2.9: Exploring the relationship between Celsius and Kelvin scales.

This activity is designed to help learners explore the relationship between Celsius and Kelvin scales through conversions.

HOOK: During a Physics lesson on temperature measurement, a learner uses a thermometer to measure the temperature of a substance and records it as 27°C. The teacher then asks the learner to express the same temperature using the Kelvin scale, which is commonly used in scientific measurements. State the relationship between the Celsius and Kelvin scales

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** None
- ❖ **Artificial Materials:** Thermometer with Celsius scale, reference tables, calculator, conversion worksheet, Kelvin-Celsius chart, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide them with the all the necessary materials
- ❖ Ask learners to discuss Celsius and Kelvin scales, emphasising on absolute zero and thermodynamic temperature.
- ❖ Ask learners to use the worksheet with scenario questions to show conversion formula: $K = ^\circ C + 273.15$
- ❖ Supervise learners solving conversion problems and checking calculations
- ❖ Facilitates understanding of why Kelvin is used in Physics
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Discussing Celsius and Kelvin scales, emphasising on absolute zero and thermodynamic temperature.
- ❖ Showing conversion formula: $K = ^\circ C + 273.15$ using worksheet with scenarios questions
- ❖ Solving conversion problems and checking calculations
- ❖ Discussing why Kelvin is used in Physics
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Check learner's accuracy of conversion of units
- ❖ Check learner's understanding of absolute zero temperature.



Skills Developed

- ❖ Unit conversion: Applying mathematical formula to explore the relationship between Celsius and Kelvin scales
- ❖ Collaboration: Working together in small groups
- ❖ Communication: Presenting their findings
- ❖ Scientific reasoning: Understanding thermodynamic scales
- ❖ Data application: Using Kelvin in calculations
- ❖ Conceptual linking: Relating scales to physical phenomena



Key Learning Points

- ❖ Definitions of Celsius and Kelvin temperature scales

- ❖ Conversion formulas and mathematical basis
- ❖ Significance of absolute zero (0 K) and its physical meaning
- ❖ Applications of Kelvin in thermodynamics, gas laws and scientific research

EXPECTED STANDARD

SAMPLE QUESTIONS

Question 1

A rural health clinic in Mongu District receives a shipment of mercury clinical thermometers and alcohol-in-glass laboratory thermometers. During a malaria outbreak, nurses need to measure body temperature of patients quickly and safely.

- State the most appropriate thermometer to use for measuring human body temperature in this clinic and explain why it is suitable according to its design features.
- Describe the correct procedure for measuring body temperature using the chosen thermometer.
- Analyse one advantage and one limitation of using this thermometer compared to an infrared (thermo scanner) thermometer in a busy clinic.

Question 2

A farmer in Chipata is monitoring soil and air temperature to decide the best time for planting maize. He has a Six's maximum and minimum thermometer installed in a Stevenson screen on his farm.

- Explain how the Six's maximum and minimum thermometer measures and records both the highest and lowest temperatures over 24 hours.
- Describe how the farmer should reset the thermometer after reading it.
- Analyse why this instrument is more appropriate than a simple liquid-in-glass thermometer for this agricultural purpose.

Question 3

In a school laboratory in Mufulira, learners are determining the boiling point of water at local atmospheric pressure and comparing it to 100 °C (standard at sea level).

- Name the most appropriate thermometer for accurately measuring the temperature of boiling water up to about 110 °C.
- Outline the steps to use the thermometer safely and accurately in this experiment.
- Explain why an alcohol-in-glass thermometer might be preferred over a mercury-in-glass one for this experiment in a school setting.

Question 4

During the COVID-19 period, a supermarket in Chitambo installed non-contact infrared thermometers (thermo scanners) at the entrance to screen customers' forehead temperatures quickly.

- Explain the principle by which an infrared thermometer measures temperature without contact.
- Describe how to use it correctly to get an accurate reading.
- Analyse two advantages and one limitation of using infrared thermometers in a public place like a supermarket compared to clinical thermometers.

Question 5

A small food processing factory in Chipangali uses a thermocouple thermometer connected to a digital display to monitor the temperature inside ovens during biscuit baking (around 180 to 220 °C).

- State why a thermocouple thermometer is the most appropriate instrument for this high-temperature industrial application.
- Briefly describe how a thermocouple works to measure temperature.
- Analyse the impact of using this instrument on product quality and worker safety compared to using a liquid-in-glass thermometer.

SAMPLE QUESTIONS ANSWERS

Question 1

- The most appropriate is the clinical thermometer (mercury-in-glass type with a constriction/kink). It is designed for the narrow range 35 °C to 42 °C, has a constriction to prevent mercury from falling back after removal, and gives accurate readings to 0.1 °C.
- Procedure: Shake the thermometer to bring mercury below 35 °C; place under the tongue (or armpit); leave for 3 to 5 minutes; remove and read the scale at eye level; disinfect after use.
- Advantage: Cheap, no batteries needed, reliable in power-outage areas. Limitation: Contains toxic mercury (risk if broken), slow reading (3 to 5 min), requires close contact (infection risk in outbreaks), infrared is faster and non-contact.

Question 2

- It has two bulbs: alcohol (expands/contracts) connected to mercury thread. As temperature rises, alcohol pushes mercury up (records max on one scale); as it falls, alcohol contracts and mercury is

pulled back (records min on the other scale). Metal indices are pushed to extremes and stay at maximum and minimum positions.

b) Reset by using a magnet or tilting to move indices back to current mercury levels.

c) It records extremes automatically over time (useful for daily/weekly monitoring without constant checking); simple thermometer only gives instant reading. Helps farmer avoid frost damage or heat stress to crops.

Question 3

a) Laboratory thermometer (alcohol-in-glass or mercury-in-glass) with range -10°C to 110°C (or higher).

b) Steps: Secure thermometer in cork with bulb above water but below steams level; heat water steadily; stir; read temperature when steady (boiling); read meniscus at eye level; avoid touching hot bulb.

c) Alcohol preferred: lower freezing point (measures below 0°C if needed), non-toxic (safer if broken), cheaper; mercury is toxic and has higher freezing point but better for high precision in some ranges.

Question 4

a) It detects infrared radiation emitted by the skin (all objects emit IR proportional to temperature); converts radiation intensity to temperature reading using calibration.

b) Hold 3 to 5 cm from forehead (or point to temporal artery); ensure no hair or sweat obstruction; press trigger; read display instantly.

c) Advantages: Fast (seconds), non-contact (hygiene, reduces infection risk), no consumables. Limitation: Affected by external factors (sweat, air conditioning, direct sun) and less accurate than clinical; needs batteries or charging.

Question 5

a) Thermocouples measure very high temperatures (up to 1000°C), respond quickly, are rugged, and can be used in harsh environments (steam, vibration).

b) Two dissimilar metals joined at a junction, temperature difference between measuring junction (in oven) and reference junction produces small voltage; voltage is measured and converted to temperature.

c) Impact: Precise continuous monitoring, consistent baking, better quality biscuits, less waste. Workers avoid opening hot ovens repeatedly, safer; liquid-in-glass cannot handle high temperatures, would break, and is not remote or readable.

SUB-TOPIC 2.2.3: EXPANSION OF SOLIDS, LIQUIDS AND GASES



INTRODUCTION

Thermal expansion is the increase in size (length, area, or volume) of a substance when its temperature rises. When matter is heated, its particles gain kinetic energy, move faster, and spread farther apart. This causes the substance to expand. When cooled, the particles lose kinetic energy, move closer together, and the substance contracts. However, when water is heated at 0° C it contracts until the temperature reaches 4°C, beyond that point water expands like any other substances.

Specific Competence: Demonstrate expansion of solids, liquids and gases.



Key terms

- ❖ Anomalous Expansion: This is the abnormal expansion experienced by water only.
- ❖ Kinetic energy: Is the energy that an object or particles have because of motion.
- ❖ Boyle's law states that for a fixed mass of gas at constant temperature, the pressure (P) of the gas is inversely proportional to its volume (V), $PV = \text{Constant}$
- ❖ Charles's law states that for a fixed mass of gas at constant pressure, the volume (V) of the gas is directly proportional to its absolute temperature (T), $\frac{V}{T} = \text{Constant}$
- ❖ Gay-Lussac's law states that for a fixed mass of gas at constant volume (V), the pressure of the gas is directly proportional to its absolute temperature (T), $\frac{P}{T} = \text{Constant}$
- ❖ The Ideal Gas Law states that for a fixed mass of gas the product of pressure (P) and volume (V) is direct proportional to the absolute temperature (T), expressed as $\frac{PV}{T} = \text{Constant}$

Learning Activities

This sub-topic, has 5 learning activities but two have already been covered in sub-topic 2.2.2.



Learning Activity 2.2.3.1.1: Demonstrating thermal expansion of solids, liquids (including anomalous expansion of water) and gases.

In this learning activity, learners will carry activities to observe how different states of matter expand when heated. These practical demonstrations will help learners understand that thermal expansion is a physical change caused by temperature variation.

HOOK: One hot afternoon, the school caretaker tried to close the metal gate at the entrance of the school. In the morning the gate closed easily, but in the afternoon, it failed to fit into the latch. Discuss why the metal gate become difficult to close in the afternoon?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ice cubes...
- ❖ **Artificial Materials:** Conical flask, rubber stopper with capillary tube, coloured water, beaker ice cubes, thermometer, Retort stand, transparent container, balloons, stop watch, refrigerator...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can use Mobile laboratory.



Learning Activity 2.2.3.1.1: Demonstrating thermal expansion of solids, liquids (including anomalous expansion of water) and gases.

In this learning activity, learners will carry activities to observe how different states of matter expand when heated. These practical demonstrations will help learners understand that thermal expansion is a physical change caused by temperature variation.

HOOK: One hot afternoon, the school caretaker tried to close the hinged metal gate at the entrance of the school. In the morning the gate closed easily, but in the afternoon, it failed to fit. Discuss why the metal gate failed to close in the afternoon?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ice, ...
- ❖ **Artificial Materials:** Conical flask, rubber stopper with capillary tube, coloured water, beaker, thermometer, Retort stand, transparent container, ballons, refrigerator, ...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can use mobile laboratory

Teacher's Roles

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups.
- ❖ Provide the learners with the materials
- ❖ Ask learners demonstrate thermal expansion of solids, liquids (anomalous expansion of water), and gases using an experiment.
- ❖ Ask learners to discuss and write down their observation from the experiment
- ❖ Ask learners to discuss the application of expansion in real life.
- ❖ Ask learners to present their findings.

- ❖ Provide feedback to the learners



Learner's Tasks

- ❖ Demonstrating thermal expansion of solids, liquids (anomalous expansion of water), and gases using an experiment.
- ❖ Discussing and writing down their observation from the experiment
- ❖ Discussing the application of expansion in real life.
- ❖ Presenting their findings.
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Check if the learners are correctly conducting the practical activities safely.
- ❖ Check learner's understanding of expansion of solids, liquids (anomalous expansion of water), and gases
- ❖ Check learner's participation in making conclusions on expansions and anomalous expansion of water.



Skills Developed

- ❖ Collaborating: Working in small groups as they demonstrate thermal expansion.
- ❖ Critical thinking: Observing expansion and making conclusion
- ❖ Communicating: Presenting their findings on expansion.



Key Learning Points

- ❖ When matter is heated, its particles gain kinetic energy, move faster, and spread farther apart. This causes solids, liquids, and gases to expand.
- ❖ When cooled, the particles lose kinetic energy, move closer together, and the substance contracts
- ❖ Understanding expansion is important in science and engineering. This explains why bridges have expansion gaps, why thermometers work, and why railway lines are laid with small spaces between rails.



Learning Activity 2.2.3.1.2: Demonstrating that various solids, liquids, and gases expands at different rates.

In this activity, learners will do experiments to observe how different states of matter expand at different rates when heated.

HOOK: During a science lesson at a secondary school, learners in groups were asked to investigate how different substances expand when heated. Each group had three materials: an iron rod (solid), cooking oil (liquid) in a bottle with a narrow tube, and air (gas) trapped inside a balloon. They placed iron rod near a flame, the bottle of oil in hot water, and the balloon in warm water. Discuss what you observed to each substance after some time.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ice, ...
- ❖ **Artificial Materials:** Conical flask, rubber stopper with capillary tube, coloured water, beaker, ice cubes, ball and ring, bimetallic strip, thermometer, source of heat, transparent container, balloons, stop watch, refrigerator...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.

Teacher's Roles:

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups.
- ❖ Provide the learners with the materials
- ❖ Ask learners to demonstrate how various solids, liquids, and gases expands at different rates using an experiment.
- ❖ Ask learners to discuss and write down their observation
- ❖ Ask learners to discuss the application of expansion in various solids, liquids and gases.
- ❖ Ask probing questions.
- ❖ Allow learners to present their findings.
- ❖ Provide constructive feedback and clear misconceptions



Learner's Tasks

- ❖ Demonstrating how various solids, liquids, and gases expands at different rates using an experiment.
- ❖ Discussing and writing down their observation
- ❖ Discussing the application of expansion in various solids, liquids and gases.
- ❖ Responding to probing questions from the teacher
- ❖ Presenting their findings.

- ❖ Receiving constructive feedback and clearing misconceptions



Assessment Guide

- ❖ Check correct setting up of the experiment.
- ❖ Check if the learners are correctly conducting the practical activities safely.
- ❖ Check learner's ability to demonstrate the expansion of solids, liquids, and gases.
- ❖ Assess the learner's ability to explain which state of matter expands the most when heated.



Skills Developed

- ❖ Collaborating: Working in small groups
- ❖ Analytical thinking: Observing solids, liquids, and gases expanding at different rates
- ❖ Communicating: Presenting their findings.



Key Learning Points

- ❖ Heating causes a solids liquid and gas to expand
- ❖ Solids expand the least because their particles are tightly packed.
- ❖ Liquids expand more than solids because their particles are less tightly packed.
- ❖ Gases expand the most because their particles are far apart and move freely.



Learning Activity 2.2.3.1.3: Determining the boiling and melting points of different substances.

Note that this learning activity has been covered under sub-topic 2.2.2, learning activity 2.2.2.1.4



Learning Activity 2.2.3.1.4: Experimenting on the effects of pressure and impurities on the boiling and melting points of substances.

Note that this learning activity has been covered under sub-topic 2.2.2, learning activity 2.2.2.1.5



Learning Activity 2.2.3.1.5: Demonstrating the use of equations on gas laws to solve numerical problems (Boyle's Law, Charles' Law and Gay-Lussac's and Ideal gas equation)

In this activity we will use mathematical equations from gas laws to solve numerical problems Understanding these equations will help the learners predict how gases behave when conditions change.

HOOK: Imagine you are helping to organise a school science fair. A large balloon filled with air is kept in the classroom at temperature of 20°C . Later it is moved outside where the temperature is 35°C . Discuss what would happen to the volume of the balloon and give the reasons as to why?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water...
- ❖ **Artificial Materials:** Balloon, source of heat, bottle, worksheets, hot water bath, thermometer, pressure sensor, ...

Learning Environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.

Teacher's Roles

- ❖ Introduce the activity to the learners using the hook.
- ❖ Put learners into small groups
- ❖ Provide learners with the materials
- ❖ Ask learners to demonstrate the use of equations on gas laws to solve numerical problems using scenarios on the worksheets.
- ❖ Ask learners to identify the correct gas law equation to use
- ❖ Ask learners to present their findings.
- ❖ Provide constructive feedback.
- ❖ Give assessment test based on scenario questions



Learner's Tasks

- ❖ Demonstrating the use of equations on gas laws to solve numerical problems using scenarios on the worksheets.
- ❖ Identifying the correct gas law equation to use
- ❖ Presenting their findings.
- ❖ Receiving constructive feedback.
- ❖ Writing assessment test based on scenario questions



Assessment Guide

- ❖ Check learners' ability to identify an appropriate equation for a scenario given.
- ❖ Assess learner's ability to apply gas laws to real life situations.



Skills Developed

- ❖ Problem solving: Solving numerical problems analytically with correct units.
- ❖ Collaboration: Working in pairs or small groups as they discuss and do calculations on the scenario questions provided.
- ❖ Communication: Presenting their solutions on gas laws
- ❖ Critical thinking: Linking the gas laws to real life.



Key Learning Points

- ❖ Boyle's Law: At constant temperature the pressure of a gas is inversely proportional to its volume.

$$P_1V_1 = P_2V_2$$

- ❖ Charle's Law: At constant pressure, volume is directly proportional to temperature in kelvin.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- ❖ Gay Lussac's Law - At constant volume, pressure is directly proportional to temperature in kelvin

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- ❖ The Ideal Gas Law states that for a fixed mass of gas the product of pressure (P) and volume (V) is direct proportional to the absolute temperature (T), expressed as;

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

- ❖ Application of gas laws in everyday life.

1. Boyle's Law used in human respiration and pumping the bicycle. Pressure, and volume changes.
2. Charles' Law used in hot air balloons rise into air because it becomes less-denser.
3. Gay-Lussac's Law used in tyres. and aerosol Spray Cans. Pressure changes with temperature.
4. Gas Law used in Cooling systems and pressure cookers depend on gas compression and expansion.

EXPECTED STANDARD

SAMPLE QUESTIONS

1. In a school laboratory, a teacher uses a ball-and-ring apparatus made of brass. When both are at room temperature, the ball passes through the ring. The ball is then heated with a Bunsen burner for 2 minutes.
 - a) Describe step-by-step how this apparatus demonstrates thermal expansion of solids.
 - b) If the same ball is made of steel (linear expansion coefficient $\alpha = 1.2 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$) and the ring is made of aluminium ($\alpha = 2.3 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$), predict what happens when both are heated equally and explain why.
2. A flask is completely filled with water at $0 \text{ }^{\circ}\text{C}$, fitted with a narrow glass tube, and heated slowly. The water level in the tube first falls slightly then rises. A separate Hope's apparatus experiment shows water freezing only at the surface of a lake.
 - a) Describe how the flask experiment demonstrates expansion of liquids.
 - b) Using the anomalous expansion of water, explain with a sketch description why fish survive in a frozen lake in winter. At what temperature is water densest?
3. A bicycle tyre contains 600 cm^3 of air at a pressure of 1.0 atm (atmospheric pressure). The rider pumps more air, increasing the pressure to 2.5 atm while the temperature remains constant.

Calculate the new volume of air inside the tyre using Boyle's Law. Explain why the tyre feels harder.
4. A weather balloon is filled with 4.0 L of helium at $27 \text{ }^{\circ}\text{C}$ on the ground. It rises to a height where the temperature drops to $-73 \text{ }^{\circ}\text{C}$. Pressure is assumed constant (balloon expands freely).

Calculate the new volume of the balloon using Charles's Law. Convert temperatures to Kelvin first.
5. A sealed rigid metal can contains gas at a pressure of 1.5 atm and temperature of $27 \text{ }^{\circ}\text{C}$. The can is placed in boiling water at $100 \text{ }^{\circ}\text{C}$.

Calculate the new pressure inside the can using Gay-Lussac's Law.

SAMPLE QUESTIONS ANSWERS

Answers 1

a) The metal ball expands in all directions when heated (linear expansion). Its diameter becomes larger than the inner diameter of the ring, so it no longer passes through. On cooling, it contracts and passes again. This shows solids expand on heating and contract on cooling.

Answers2

a) When heated, water expands, so the level in the narrow tube rises. This shows liquids expand on heating (real expansion greater than apparent because glass also expands slightly).

b) Water contracts when cooled from 4 °C to 0 °C (anomalous behaviour). At 4 °C water is densest and sinks; colder water (0–4 °C) stays at the top and freezes. Ice (less dense) floats, insulating the water below at ~4 °C where fish can survive. Water is densest at 4 °C.

Answers 3

Boyle's Law: $P_1V_1 = P_2V_2$ (constant temperature)

$$1.0 \text{ atm} \times 600 \text{ cm}^3 = 2.5 \text{ atm} \times V_2$$

$$V_2 = (1.0 \times 600) / 2.5 = 240 \text{ cm}^3$$

The tyre feels harder because the same amount of air is now squeezed into a smaller volume, increasing pressure on the walls.

Answers 4

Convert: $T_1 = 27 + 273 = 300 \text{ K}$; $T_2 = -73 + 273 = 200 \text{ K}$

Charles's Law: $V_1 / T_1 = V_2 / T_2$

$$V_2 = V_1 \times (T_2 / T_1) = 4.0 \times (200 / 300) = 2.67 \text{ L (approx.)}$$

The balloon shrinks because temperature decreased.

Answers 5

Convert: $T_1 = 300 \text{ K}$; $T_2 = 373 \text{ K}$

Gay-Lussac's Law: $P_1 / T_1 = P_2 / T_2$

$$P_2 = P_1 \times (T_2 / T_1) = 1.5 \times (373 / 300) \approx 1.865 \text{ atm}$$

The pressure increases, which is why aerosol cans warn "do not heat".

SUB-TOPIC 2.2.4 INTERNAL COMBUSTION ENGINE (I.C.E)



INTRODUCTION

An **internal combustion engine (I.C.E.)** is a machine that converts chemical energy in fuel into mechanical energy. This happens when fuel burns inside the engine, producing heat and expanding gases that move parts of the engine to produce motion.

- ❖ **Specific Competence:** Demonstrate how various internal combustion engine types operate (four stroke engine).



Key terms

- ❖ **Spark Ignition (SI) Engine:** An internal combustion engine where the air–fuel mixture is ignited by a spark plug to produce power.
- ❖ **Compression Ignition (CI) Engine:** Air is compressed to a very high pressure and temperature, causing it to ignite automatically without a spark plug.
- ❖ **A rotary engine:** Is an internal combustion engine that uses a rotating triangular motor instead of pistons to convert fuel energy into mechanical motion.
- ❖ **Hybrid Engine:** A hybrid engine is a system that combines a conventional internal combustion engine (ICE) with an electric motor and battery.
- ❖ **Homogeneous Charge Compression Ignition (HCCI):** A premixed air–fuel mixture (like in petrol engines) is compressed until it auto-ignites (like in diesel engines).
- ❖ **Fuel Cell Engine:** A fuel cell engine generates electricity through a chemical reaction between hydrogen and oxygen instead of burning fuel.

Learning Activities

These learning activities will help learners to demonstrate how various internal combustion engines operate (four stroke engine).

This sub-topic, has 5 learning activities as outlined in the syllabus.



Learning Activity 2.2.4.1.1 Recognising different types of internal combustion engines in terms of spark ignition, compression ignition, rotary, ...

In this lesson, learners will carry out activities to recognise different types of internal combustion engines in terms of spark ignition, compression ignition, rotary, ...

HOOK: You are a mechanics student and removes a spark plug from a small petrol motorcycle engine. When you try to start it, the engine fails to start. Why did the engine fail to start? Discuss with your friends the engine that might still work without a spark plug.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** None
- ❖ **Artificial Materials:** Engine models, charts showing the four-stroke engine cycle, Worksheets.

Learning Environment set up

- ❖ Natural Environment: Such as school surroundings, ...
- ❖ Artificial Environment: Classroom, Laboratory with engine models, outdoor space where learners can observe vehicles engines, motorcycles or petrol or diesel run generators, ...
- ❖ Technological Environment: Virtual lab, video clips/animations, laptop/projector or smartphone,

Teacher's Roles

- ❖ Introduce the concept of internal combustion engines using the hook.
- ❖ Put learners in small groups
- ❖ Ask learners to research on different types of internal combustion engines in terms of spark ignition, compression ignition, rotary,
- ❖ Provides learners charts with model diagrams, a real model, simulation videos of types of internal combustion engines.
- ❖ Ask learners to discuss the differences between the engines
- ❖ Ask learners to demonstrate ignition methods e.g. a generator, ...
- ❖ Ask guiding questions to check learners understanding of different types of engines in terms of ignition.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learner's Tasks

- ❖ Researching on different types of internal combustion engines in terms of spark ignition, compression ignition, rotary,
- ❖ Viewing charts with model diagrams and real model, and watching simulation videos of types of internal combustion engines.
- ❖ Discussing the differences between the internal combustion engines in terms of ignition
- ❖ Demonstrating ignition methods e.g a generator, ...
- ❖ Responding to guiding questions on different types of engines in terms of ignition.
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's participation in groups

- ❖ Assess learner's ability to differentiate between the types of engines in terms of ignition
- ❖ Assess learner's ability to demonstrate ignition.



Skills Developed

- ❖ Collaboration: Working in small groups to recognise types of internal combustion engine in terms of ignition.
- ❖ Critical thinking: Comparing different engines by ignition type
- ❖ Communication: Presenting their findings on internal combustion engine.
- ❖ Observation: Identifying engine types by some parts



Key Learning Points

- ❖ Internal combustion engines produce power by burning fuel inside the engine cylinder.
- ❖ The three main types of internal combustion engines are:
- ❖ Spark Ignition Engine (SI) – Uses a spark plug to ignite fuel (e.g., petrol engines)
- ❖ Compression Ignition Engine (CI) – Uses high compression to ignite fuel (e.g., diesel engines).
- ❖ Rotary Engine – Uses a rotating rotor instead of pistons.
- ❖ Each engine type has different ignition methods and mechanical design.



Learning Activity 2.2.4.1.2 Exploring the operation of the internal combustion engine (ICE)

This activity is designed to help learners explore the operation of the internal combustion engine (ICE)

HOOK: During the school's Technology and Innovation Exhibition you notice that many machines used such as motorcycles, cars, generators and water pumps produce power from engines. Explain how these machines produce power that makes them move or operate.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:**
- ❖ **Artificial Materials:** Engine diagrams or charts, Model engines or toy, printed worksheets, Pictures of different engine, Multimedia videos or animations, ...

Learning Environment Set Up

- ❖ **Natural Environment:** Such as school surroundings, ...
- ❖ **Artificial Environment:** Laboratory with engine models of engines, outdoor space where learners can observe vehicle engines, motor vehicles garage,

- ❖ **Technological Environment:** Computer laboratory, virtual laboratory,

Teacher's Roles

- ❖ Introduce the concept of internal combustion engines using the hook.
- ❖ Put learners in small groups
- ❖ Provide diagrams or models or video of the internal combustion engine.
- ❖ Guide learners to identify and describe major engine components such as piston, cylinder, crankshaft, spark plug, and valves.
- ❖ Ask learners to watch the virtual or manipulate the models of the internal combustion engine to observe its operation.
- ❖ Ask learners to discuss working principle of the four-stroke cycle engine: intake, compression, power, and exhaust.
- ❖ Ask learners probing questions
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learner's Task

- ❖ Identifying and describing major engine components such as piston, cylinder, crankshaft, spark plug, and valves.
- ❖ Watching the virtual or manipulating the models of the internal combustion engine to observe its operation.
- ❖ Discussing working principle of the four-stroke cycle engine: intake, compression, power, and exhaust.
- ❖ Responding to probing questions
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's participation in group discussions
- ❖ Assess learner's ability to identify the main parts of an internal combustion engine
- ❖ Assess learner's ability to explore the operation of the internal combustion engine.



Skills Developed

- ❖ Problem-solving: Understanding how energy is produced.
- ❖ Critical Thinking: The ability to analyse how engine parts work together
- ❖ Collaboration: The ability to work effectively with others in a group to solve problems or complete tasks.
- ❖ Communication: The ability to express ideas clearly through speaking or writing.

- ❖ **Observation:** The ability to carefully examine diagrams, models, and demonstrations.



Key Learning Point

- ❖ Main parts of an internal combustion engine (cylinder, piston, spark plug, valves, and crankshaft).
- ❖ The burning fuel produces high-pressure gases that push the piston.
- ❖ Internal combustion engines power many machines such as cars and generators.
- ❖ The four-stroke cycle: Intake stroke, Compression stroke, Power stroke, and Exhaust stroke
- ❖ How fuel and air mixture produces energy. e.g.
 1. Intake: Fuel and air enter the cylinder
 2. Compression: Piston compresses the mixture.
 3. Power: Spark plug ignites fuel pushing piston down
 4. Exhaust: Burnt gases leave the pipe.



Learning Activity 2.2.4.1.3 Creating a model of an internal combustion engine

In this learning activity, learners will be able to create a model of an internal combustion engine.

HOOK: A local mechanic in the community wants to teach learners how a car engine works, but real engines are expensive and complicated to bring into a classroom. He asks the learners to design a simple model of an internal combustion engine using materials available in the environment so that learners can easily understand how it works. Discuss how would create a model of an internal combustion engine

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Clay soil for modelling engine shapes,
- ❖ **Artificial Materials:** Engine model or diagram, Charts showing the four-stroke engine cycle, Worksheets, Cardboard or plastic bottles, Syringes to represent pistons, Straws to represent pipes, Glue, scissors, markers, rulers, planks, ...

Learning Environment set up

- ❖ **Natural Environment:** Such as school surroundings,
- ❖ **Artificial Environment:** Classroom, Laboratory with or without engine models, outdoor space where learners can observe vehicles, motorcycles or generators.
- ❖ **Technological Environment:** Video clips/animations, Laptop/projector or smartphone,

Teacher's Roles

- ❖ Introduce the activity using a hook.
- ❖ Put learners in small groups
- ❖ Ask learners to design and construct a simple engine model.
- ❖ Ask learners to use materials such as cardboard, syringes, balloons, or plastic bottles, ... to create a simple engine model
- ❖ Ask learners to label the different parts of the model.
- ❖ Ask learners to demonstrate how the model represents the engine cycle.
- ❖ Ask learners to present the completed model to the class.
- ❖ Provide feedback to the learners



Learner's Task

- ❖ Designing and constructing a simple engine model.
- ❖ Creating a simple engine model using materials such as cardboard, syringes, balloons, or plastic bottles, ...
- ❖ Labelling different parts of the simple engine model.
- ❖ Demonstrating how the model represents the engine cycle.
- ❖ Presenting the completed model to the class.
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's ability to create and design a model that clearly represent the engine structure.
- ❖ Assess learner's accuracy in labelling the parts of the model correctly
- ❖ Assess learner's ability to clearly explain how their model demonstrates engine operation.
- ❖ Assess learner's cooperation during group work.



Skills Developed

- ❖ Creativity: The ability to generate new ideas or designs. Example: designing a simple working model.
- ❖ Collaboration: Working together to construct and present a model.
- ❖ Problem Solving: Identifying ways to represent engine components using available materials.
- ❖ Communication: Explaining how the model works.
- ❖ Critical thinking: Explaining the concept correctly
- ❖ Communicating: Presenting their findings



Key Learning Points

- ❖ Models simplify complex mechanical systems.
- ❖ Engine parts work together to convert fuel energy into motion.
- ❖ Hands-on activities improve understanding of engineering systems.
- ❖ Using locally available materials encourages innovation and resourcefulness.



Learning Activity 2.2.4.1.4 Comparing efficiency of diesel and petrol engine

In this learning activity, learners will be able to compare efficiency of diesel and petrol engine.

HOOK: A farmer is choosing between a diesel tractor and a petrol tractor.

- a) How would you identify the engine type that is generally more fuel efficient?**
- b) Explain one reason for the higher efficiency.**
- c) Suggest why diesel engines are commonly used in heavy machines**

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Air for demonstrating combustion concepts, Water for demonstrating cooling systems, ...
- ❖ **Artificial Materials:** Charts comparing diesel and petrol engines, Videos showing engine operation, Worksheets, calculators for efficiency calculations,

Learning Environment set up

- ❖ **Natural Environment:** Such as school surroundings, ...
- ❖ **Artificial Environment:** Classroom, Laboratory, shelter, car repair garage,
- ❖ **Technological Environment:** Virtual laboratory, simulations, videos, ...

Teacher's Roles

- ❖ Introduce the activity using a hook.
- ❖ Put learners in small groups
- ❖ Provide the learners with a video showing engine operation for both diesel and petrol engines
- ❖ Ask learners to compare the working principles of diesel and petrol engines.
- ❖ Ask learners to discuss which engine is more efficient and why.
- ❖ Ask learners to analyse examples of machines that use each engine type.
- ❖ Present their findings in groups.



Learner's Task

- ❖ Watching video showing engine operation for both diesel and petrol engines
- ❖ Comparing the working principles of diesel and petrol engines.
- ❖ Discussing which engine is more efficient and why.
- ❖ Analysing examples of machines that use each engine type.
- ❖ Presenting their findings in groups.



Assessment Guide

- ❖ Assess learner's understanding engine efficiency
- ❖ Assess learner's ability to compare the differences between diesel and petrol engines.
- ❖ Assess learner's ability apply each type of engine suitably



Skills Developed

- ❖ Critical Thinking: Analysing which engine performs better in different situations.
- ❖ Analytical Skills: Interpreting fuel consumption data.
- ❖ Communication: Presenting comparisons clearly.
- ❖ Collaboration: Working in groups



Key Learning Points

- ❖ Diesel engines generally have higher fuel efficiency than petrol engines.
- ❖ Petrol engines use spark ignition.
- ❖ Diesel engines use compression ignition.
- ❖ Different engines are suitable for different applications.



Learning Activity 2.2.4.1.5 Exploring emerging engine technologies such as hybrid, homogeneous charge compression ignition, ...

In this learning activity, learners will be able to explore emerging engine technologies such as hybrid, homogeneous charge compression ignition,

HOOK: Your team is tasked with developing a Hybrid Engine for a new ecologically friendly car. The goal is to improve fuel efficiency by 30%.

- a) What emerging engine technologies (like HCCL, Hybridisation, ...) would you explore?
- b) How would you balance efficiency and performance?
- c) What challenges might arise with hybrid tech?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** None
- ❖ **Artificial Materials.** Charts of hybrid engines, Videos explaining modern engine technologies, Internet research materials, ...

Learning Environment set up

- ❖ **Natural Environment:** Such as school surroundings, ...
- ❖ **Artificial Environment:** Classroom, Laboratory, shelter, car repair garage,
- ❖ **Technological Environment:** Virtual laboratory, simulations, videos, ...

Teacher's Roles

- ❖ Introduce the activity using a hook.
- ❖ Put learners in small groups
- ❖ Ask learners to research on engine technologies such as hybrid engines and homogeneous charge compression ignition using videos explaining modern engine technologies, internet research materials, ...
- ❖ Ask learners to discuss advantages of modern engines compared to traditional engines that is how modern engines improve efficiency and reduce pollution.
- ❖ Ask learners to present research findings to the class.
- ❖ Provide feedback to the class



Learner's Task

- ❖ Researching on engine technologies such as hybrid engines and homogeneous charge compression ignition using videos explaining modern engine technologies, internet research materials, ...
- ❖ Discussing advantages of modern engines compared to traditional engines that is how modern engines improve efficiency and reduce pollution.
- ❖ Presenting their research findings to the class.
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's ability to research i.e gather relevant and accurate information.
- ❖ Assess learner's understanding how modern engines improve efficiency.
- ❖ Evaluate learner's ability to clearly communicate research findings



Skills Developed

- ❖ Research Skills: The ability to gather and interpret information from different sources.
- ❖ Critical Thinking: Evaluating advantages and limitations of new technologies.
- ❖ Communication: Presenting their findings
- ❖ Collaboration: Working together during research and presentations.



Key Learning Points

- ❖ Hybrid engines combine electric motors with petrol or diesel engines.
- ❖ Modern technologies aim to reduce fuel consumption and emissions.
- ❖ Homogeneous charge compression ignition improves combustion efficiency.
- ❖ Innovation in engine technology supports environmental sustainability.

EXPECTED STANDARDS

SAMPLE QUESTIONS

1. A driver starts his minibus every morning. The engine makes a loud noise and the vehicle moves after he turns the key. The fuel tank contains petrol or diesel that is burned inside the engine.
 - (a) What type of engine is this?
 - (b) Why is the internal combustion engine important for transport in Zambia?
2. A farmer uses a small generator to pump water for irrigation. The generator has an engine that burns diesel.

Explain in one sentence why this engine is called an “internal” combustion engine.
3. Your uncle drives a Toyota Hilux pickup. He says the engine “eats” fuel to make the wheels turn.

Name two common fuels used in Zambian internal combustion engines.
4. A mechanic is teaching apprentices using a cut-away model of a car engine. He says the engine completes four strokes in one cycle.

Describe the four strokes of a four-stroke petrol engine in the correct order. Use the scenario of a piston moving up and down.

5. A diesel truck suddenly loses power. The driver knows it is a compression-ignition engine.

(a) How does ignition occur in a diesel engine (different from petrol)?

(b) Name one advantage of diesel engines in heavy mining vehicles.

6. A motorcycle mechanic shows a small two-stroke engine and a car four-stroke engine.

Although the syllabus focuses on four-stroke, explain briefly why most cars use four-stroke engines instead of two-stroke.

7. You are asked to create a simple working model of an internal combustion engine for a school science exhibition.

List the main parts you would include and label them.

SAMPLE QUESTIONS ANSWERS

Answer 1

(a) This is an Internal Combustion Engine (ICE) because the fuel burns (combusts) inside the cylinders to produce power.

(b) It converts chemical energy in fuel into mechanical energy to move vehicles. Most buses, trucks, cars, and mining vehicles in Zambia use ICEs. Without them, public transport, delivery of goods, and mining operations would stop.

Answer 2

It is called an internal combustion engine because the fuel burns (combusts) **inside** the engine cylinders, not outside like in a steam engine.

Answer 3

Petrol (for spark-ignition engines) and diesel (for compression-ignition engines).

Answer 4

1. Intake stroke: Piston moves down, intake valve opens → air-fuel mixture enters the cylinder.

2. Compression stroke: Piston moves up, both valves closed → mixture is compressed.

3. Power (ignition) stroke: Spark plug produces a spark → mixture explodes, pushing piston down (this turns the crankshaft).

4. Exhaust stroke: Piston moves up, exhaust valve opens → waste gases leave the cylinder.

The cycle then repeats. This is how the engine produces continuous power.

Answer 5

(a) Air is compressed so much that its temperature rises very high. Diesel fuel is injected at the top of the compression stroke and ignites automatically (no spark plug).

(b) Diesel engines produce more torque (pulling power), which is needed for heavy loads in mines.

Answer 6

Four-stroke engines are more efficient, produce less pollution, and last longer than two-stroke engines. They complete the cycle in four separate strokes, allowing better control of fuel and air.

Answer 7

- Cylinder and piston
- Crankshaft and connecting rod
- Intake and exhaust valves
- Spark plug (petrol) or fuel injector (diesel)
- Camshaft
- Flywheel

SUB-TOPIC 2.2.5: HEAT TRANSFER



INTRODUCTION

Heat Transfer covers how heat moves through conduction, convection, and radiation, with applications in daily life and technology.

Specific competence

- Apply the concepts of heat transfer.



Key terms

- ❖ **Conduction:** Process of heat transfer through a medium (material e.g solids, liquids, and gases).
- ❖ **Radiation:** Process of heat transfer as electromagnetic waves through space (e.g heat from the stove, sun, ...).
- ❖ **Convection:** Process of heat transfer due to the movement of fluid (e.g water, cooking oil, air,.)



Learning activities

This subtopic has seven (7) learning activities from the syllabus focusing on competence-based learning through hands-on activities and problem-solving tasks.

Activity 2.2.5.1.1: Describing Heat Transfer

In this activity learners will describe heat transfer through conduction, convection, and radiation by research and discussion.

HOOK: A learner pours hot tea into a metal cup. After a few minutes, the cup becomes very hot to touch, and steam rises from the tea. The learner wonders how the heat moved from the tea to the cup and then into the surrounding air. Discuss how heat would move from tea to the cup and to the surrounding.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, air, ...
- ❖ **Artificial Materials:** Whiteboard/markers for brainstorming, projector or printed diagrams/videos (optional), learner journal/textbooks, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g mobile laboratory

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Divide learners into small groups.
- ❖ Ask learners research on the concept of heat transfer; conduction, radiation, and convection.
- ❖ Ask learners to discuss the concepts of heat transfer
- ❖ Ask learners to present their findings
- ❖ Provide feedback and clearing misconceptions



Learners' Tasks

- ❖ Researching on the concept of heat transfer; conduction, radiation, and convection.
- ❖ Discussing the concepts of heat transfer
- ❖ Presenting their findings
- ❖ Receiving feedback and clearing misconceptions



Assessment Guide

- ❖ Assess learner's ability in group discussion
- ❖ Assess learners understanding of concepts
- ❖ Assess learners critical thinking through differentiating methods of heat transfer
- ❖ Assess learner's ability to communicate during group discussion.



Skills Developed

- ❖ **Critical Thinking:** Evaluating evidence, identifying patterns, and making connections between concepts of heat transfer.
- ❖ **Problem-Solving:** Discussing real life applications.
- ❖ **Communication:** Presenting their findings.
- ❖ **Collaboration:** Working in groups, sharing ideas, and discussing the concepts.



Key Learning Points

Heat is transferred from hotter to colder regions in three main ways:

- ❖ **Conduction:** Heat travels through solids as vibrating particles pass energy to neighbouring particles (no overall movement of matter). Metals (e.g. copper) are good conductors; wood, plastic and air are poor conductors (insulators).
- ❖ **Convection:** Heat travels in fluids (liquids and gases) by the movement of the fluid itself. Hotter, less dense fluid rises and cooler fluid sinks, forming convection currents (e.g. sea/land breezes, refrigerators).
- ❖ **Radiation:** Heat travels as electromagnetic waves (infrared) through empty space. No medium is needed.



Activity 2.2.5.1.2: Demonstrating heat transfer by conduction, convection, and radiation

In this activity, learners will demonstrate heat transfer by conduction, convection, and radiation through practical tasks.

HOOK: Learners are cooking food on a charcoal stove in a Home Economics Laboratory. A metal spoon in the pot becomes hot, the water in the pot begins circulating and boiling, and one learner standing nearby feels heat without touching the fire. Discuss whether these methods of heat transfers are the same or different

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Wood, water, ...
- ❖ **Artificial Materials:** Metal rods, candles/heat lamps, bunsen burners, hot plates, heat lamps, stopwatches, thermometers, safety gloves/goggles, worksheet, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the classroom, laboratory, ...
- Technological Environment such as virtual laboratory, simulation, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Divide learners into small groups.
- ❖ Provide learners with materials

- ❖ Guide learners on how to setup various apparatus to demonstrate heat transfers.
- ❖ Ask learners to conduct experiments on heat transfers.
- ❖ Ask open-ended questions to encourage learners to think critically about the observations they are making.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to learners on their activities, experiment results and analysis.



Learners' Tasks

- ❖ Setting up various apparatus to demonstrate heat transfers under teacher's guidance
- ❖ Conducting experiments on heat transfers.
- ❖ Responding to open-ended questions
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learners' ability to set apparatus to demonstrate different forms of heat transfer.
- ❖ Assess learners' ability to use apparatus to demonstrate different forms of heat transfer.
- ❖ Assess learners' ability to communicate their observations of the experiments on heat transfer.



Skills Developed

- ❖ **Problem-Solving:** Setting and conducting experiments.
- ❖ **Critical Thinking:** Analysing the observations of different forms of heat transfers.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in groups to set apparatus to demonstrate heat transfer.



Key Learning Points

Heat is transferred from hotter to colder regions in three main ways:

- ❖ **Conduction:** Heat travels through solids as vibrating particles pass energy to neighbouring particles (no overall movement of matter). Metals (e.g. copper) are good conductors; wood, plastic and air are poor conductors (insulators).

- ❖ **Convection:** Heat travels in fluids (liquids and gases) by the movement of the fluid itself. Hotter, less dense fluid rises and cooler fluid sinks, forming convection currents (e.g. sea/land breezes, refrigerators).
- ❖ **Radiation:** Heat travels as electromagnetic waves (infrared) through empty space. No medium is needed.



Activity 2.2.5.1.3: Applying heat transfer in everyday life such as food warmers, flasks, textile industry, refrigerators, sea and land breeze, heating elements, car radiator, ...

In this activity, learners will apply heat transfer in everyday life such as food warmers, flasks, textile industry, refrigerators, sea and land breeze, heating elements, car radiator, ... through research.

HOOK: Two learners sit outside on a hot afternoon. One sits under a tree and feels cool while the other sits on a metal chair in the sun and feels very hot. Discuss why some materials and places feel hotter than others.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Hot water, air, trees, ...
- ❖ **Artificial Materials:** Recycled materials (e.g., cardboard, foil, cotton wool), thermometers, heat sources for testing, presentation aids (e.g., chart paper, markers), books, metal, flask, food warmer, ...

Learning environment set up

- Natural Environment such as school surroundings, sea and land, ...
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory
- Technological environment such as virtual laboratory, simulations, videos, ...

Teacher's Role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide learners in small groups

- ❖ Ask learners to research on the application of conduction, convection, and radiation in everyday life.
- ❖ Ask learners to discuss the application of conduction, convection, and radiation in everyday life.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners
- ❖ Ask learners to design an engineering project (e.g., a model insulator for keeping food hot) using recycled materials.



Learners' Tasks

- ❖ Researching on the application of conduction, convection, and radiation in everyday life.
- ❖ Discussing the application of conduction, convection, and radiation in everyday life.
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher
- ❖ Designing an engineering project (e.g., a model insulator for keeping food hot) using recycled materials.



Assessment Guide

- ❖ Assess learners' ability to apply heat transfer in everyday life.
- ❖ Assess learners' ability to analyse and identify the relationships between different materials and heat transfer.
- ❖ Assess learners' ability to communicate their findings from the research task to the class.
- ❖ Assess learners' ability to design a model of heat transfer application in everyday life using recycled materials.



Skills Developed

- ❖ **Problem-Solving:** Designing a model that applies heat transfer concepts
- ❖ **Critical Thinking:** Analysing effect of heat transfer in everyday life
- ❖ **Communication:** Presenting the research findings to the class.
- ❖ **Collaboration:** Working in pairs or small groups to design a thermal model



Key Learning Points

Application of heat transfer in our everyday life such as food warmers, flasks, textile industry, refrigerators, sea and land breeze, heating elements, car radiators, ...



Activity 2.2.5.1.4: Exploring the relationship between kinetic theory and heat transfer.

in this activity, learners will be able to explore the relationship between kinetic theory and heat transfer through discussions.

HOOK: A teacher sprays perfume in one corner of the classroom. Within a short time, learners sitting far away can smell it. Discuss how the smell would spread through the room and what this would tell us about particle motion and heat.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Scent from a plant, air, water, ...
- ❖ **Artificial Materials:** Whiteboard/markers, projector for animations (optional), textbooks, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.
- Technological Environment such as virtual laboratory, simulation, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide class into small groups.
- ❖ Provide learners with the materials
- ❖ Assign each group questions for discussions on relationship between kinetic theory of matter and heat transfer
- ❖ Ask learners to present their findings from the discussion.
- ❖ Provide feedback to the learners.



Learners' Tasks

- ❖ Discussing on the relationship between kinetic theory of matter and heat transfer
- ❖ Presenting their findings from the discussion.
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's level of participation during the discussion.
- ❖ Assess learner/s ability to understand the relationship between kinetic theory of matter and heat transfer.



Skills Developed

- ❖ **Scientific Inquiry:** Relating questions on heat transfer to kinetic theory of matter
- ❖ **Critical Thinking:** Relating kinetic theory of matter to heat transfer.
- ❖ **Communication:** Sharing their findings in the discussions



Key Learning Points

- ❖ Particles in hotter regions move faster and transfer energy.
- ❖ Technological applications include thermos flasks (reduce all three transfers), car radiators (convection), solar panels (good absorbers) and insulation in buildings (reduces energy waste and helps environmental sustainability).



Learning activities

Activity 2.2.5.1.5: Demonstrating the use of bad and good conductors of heat.

This activity is designed to help learners demonstrate the use of bad and good conductors of heat. In this activity, learners will use hands- on to test different materials ability to conduct heat.

HOOK: A learner puts a metal pot with a wooden handle on a hot metal plate of a stove. After sometime, the pot feels hotter than the handle. Discuss why this was the observation.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Wood, ...

- ❖ **Artificial Materials:** Copper wire, glass rods, plastic, rubber, copper wire, bunsen burner, thermometers, stopwatches, safety equipment, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.
- Technological Environment such as virtual laboratory, simulation, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide learners into small groups
- ❖ Provide learners with the materials
- ❖ Ask learners to setup apparatus to demonstrate good and bad conductors of heat
- ❖ Ask learners to demonstrate how different materials conduct heat at different rates.
- ❖ Ensure learners adhere to safety during the demonstration
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Setting up apparatus demonstrating good and bad conductors of heat
- ❖ Demonstrating how different materials conduct heat at different rates.
- ❖ Adhering to safety protocols during the demonstration
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's ability to conduct a demonstration on bad and good conductors of heat.
- ❖ Assess learner's ability to classify good and bad conductors of heat
- ❖ Assess learners on safety protocols
- ❖ Assess learner's ability to apply the concepts of good and bad conductors of heat



Skills Developed

- ❖ **Scientific Inquiry:** Conducting an experiment to demonstrate the use of good and bad conductors of heat.
- ❖ **Critical Thinking:** Classifying the bad and good conductors of heat.
- ❖ **Problem-Solving:** Choosing the right material for the application, for example a cooking pot must be made up of good conductor while the handle of bad conductor (insulator).
- ❖ **Communication:** Presenting their findings to a class.
- ❖ **Collaboration:** Working in groups, sharing ideas, and building on each other's strengths.



Key Learning Points

- ❖ Good conductors of heat are materials that easily allow the passage of heat. For examples; metals (copper, silver, aluminium, ...), graphite, ...
- ❖ Bad conductors (insulators) of heat are materials that do not easily allow the passage of heat. For examples; plastics, rubber, glass, air, ...



Activity 2.2.5.1.6: Distinguishing good from bad absorbers/emitters of radiant energy.

This activity is designed to help learners distinguish good from bad absorbers/emitters of radiant energy. In this activity, learners will investigate how fast temperature rises on dull and shine cans comparatively when exposed to the same radiant energy (e.g heat from the Sun).

HOOK: Two identical containers were filled with water and placed in the sun. One container was painted black while the other was shiny silver. After sometime, the water in the black container became hotter than in the shiny silver. Discuss what caused a difference in temperature?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, sun, wax, ...
- ❖ **Artificial Materials:** Two identical soda cans, black and silver paint, stopwatch, thermometer, and bunsen burner, charcoal brazier, candle, candle wax, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory.

Teacher's role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide learners into small groups
- ❖ Distribute materials for the experiment to the learners.
- ❖ Ask learners to setup apparatus for the experiment.
- ❖ Ask the learners to conduct an experiment
- ❖ Ensure safety precautions are followed by learners
- ❖ Ask open-ended questions to encourage learners to think critically about the data they are collecting
- ❖ Provide feedback to learners on their activities, observation, data collection, and analysis.



Learners' Tasks

- ❖ Setting up apparatus for the experiment.
- ❖ Conducting an experiment
- ❖ Adhering to safety protocols
- ❖ Responding to open-ended questions
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learners' ability to setup apparatus an experiment.
- ❖ Assess learners' ability to distinguish between bad absorbers and good absorbers of radiant heat.
- ❖ Assess learners' ability to communicate their measurements of temperature rise against time.



Skills Developed

- ❖ **Problem-Solving:** Setting and conducting measurements to compare radiant absorption rate of the two differently painted cans.
- ❖ **Critical Thinking:** Analysing the rates of radiant absorption of the two cans.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in groups to set the apparatus.



Key Learning Points

- ❖ **Radiation:** Heat travels as electromagnetic waves (infrared) through empty space.
- ❖ No medium is needed for transmission of radiant energy
- ❖ Black, dull surfaces are good absorbers and emitters of heat; shiny, white surfaces are poor absorbers and emitters of heat.



Activity 2.2.5.1.7: Demonstrating greenhouse effects.

This activity is designed to help learners demonstrate greenhouse effects. In this activity, learners will demonstrate greenhouse effects using an experiment.

HOOK: A learner has noticed that temperatures in the school greenhouses are much higher than outside, and scientists warn that rising carbon dioxide levels are making the whole country hotter, affecting rainfall and crops. Design and carry out a simple experiment to demonstrate that certain gases trap heat like a greenhouse.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Air, soil, water, sun, ...
- ❖ **Artificial Materials:** Clear plastic bottles, thermometers, baking soda, balloons, corks with small holes (to seal bottles), lamp / desk light (if no sunlight), black construction paper or dark soil (to line bottle bottoms), stopwatch, ruler, notebook, safety gloves, goggles, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory, classroom, ...
- Technological Environment such as virtual laboratory, simulation, videos, ...

Teacher's Role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide learners into small groups
- ❖ Distribute materials for the experiment to the learners.
- ❖ Ask learners to follow safety rules
- ❖ Guide learners to setup and conduct an experiment to demonstrate the greenhouse effects.
- ❖ Facilitate whole-class discussion after data collection.
- ❖ Guide learners to link findings to real-life climate action
- ❖ Ask higher-order questions:
 - a) Why is the temperature rising faster in one bottle?
 - b) How does this relate to climate change in Zambia?
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Following safety protocols
- ❖ Setting up and conducting an experiment to demonstrate the greenhouse effects.
- ❖ Discussing data collected
- ❖ Linking findings to real-life climate action
- ❖ Responding to higher-order questions:
 - a) Why is the temperature rising faster in one bottle?
 - b) How does this relate to climate change in Zambia?
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's ability to follow safety protocols
- ❖ Assess learner's participation during discussion
- ❖ Assess learner's ability to set and handle apparatus
- ❖ Assess learner's ability to observe and collect accurate data



Skills Developed

- ❖ Critical Thinking: Analysing temperature data, compare bottles, and evaluate real-world implications
- ❖ Collaboration: Working effectively with others
- ❖ Scientific Inquiry: Investigating and drawing evidence-based conclusions.
- ❖ Communication: Clearly explaining scientific ideas orally and in writing.
- ❖ Problem-Solving: Linking laboratory results to local climate challenges and suggesting solutions.



Key Learning Points

- ❖ Greenhouse gases (especially CO₂) absorb and trap infrared heat and raising temperature.
- ❖ Enhanced greenhouse effect is caused by human activities (burning fossil fuels, deforestation); it leads to global warming, changed rainfall patterns and threats to Zambian agriculture.

EXPECTED STANDARD

SAMPLE QUESTIONS

1. A learner is preparing a cup of hot tea using a metal spoon and a plastic spoon. After stirring for a few seconds, the metal spoon feels very hot to touch, but the plastic spoon remains cool.
 - (a) Explain why this happens using the concept of conduction.
 - (b) Suggest one reason why cooking pots have wooden or plastic handles.
2. During a sunny day at the beach, a learner notices that dark-coloured beach towels get much hotter than light-coloured ones, while a shiny metal water bottle stays cooler.
 - (a) Identify the mode of heat transfer involved in heating the towels and bottle.
 - (b) Explain why dark surfaces become hotter and how this relates to absorbers and emitters of radiant energy.
3. In a village near Kitwe, people often cook nshima in large pots over open fires. Hot air rises from the fire, while cooler air sinks, helping to circulate heat around the cooking area.
 - (a) Name the process responsible for this heat circulation.
 - (b) Draw a simple labelled diagram to show the convection currents in the air above the fire.
4. A family wants to keep food hot during a long trip in a car. They wrap the food in aluminium foil

and place it in a cardboard box lined with old newspapers and cotton cloth.

(a) Identify which heat transfer process(es) each material helps to reduce and explain how.

(b) Suggest one improvement they could make using recycled materials to further reduce heat loss.

5. In a refrigerator, cold air near the cooling coils sinks, while warmer air rises, creating a continuous flow that keeps the entire fridge cool.

Explain how convection helps maintain a uniform low temperature inside the refrigerator.

SAMPLE QUESTION ANSWERS

1. (a) The metal spoon is a good conductor of heat. Its particles vibrate vigorously and quickly pass kinetic energy to the hand. The plastic spoon is a poor conductor (insulator), so energy transfer is very slow.

(b) Wooden or plastic handles are poor conductors; they prevent heat from reaching the hand and avoid burns.

2. (a) **Radiation (infrared rays from the sun).**

(b) Dark (dull/black) surfaces are good absorbers of radiant heat and therefore become hotter.

Shiny surfaces are poor absorbers (good reflectors) of radiation, so they stay cooler. Good absorbers are also good emitters.

3. (a) **Convection.**

(b) Diagram: arrows showing hot air rising from the fire, spreading out, cooling and sinking back down to the fire in a continuous loop (convection current).

4. (a) Aluminium foil (shiny) – reduces radiation (reflects heat).

Newspapers and cotton cloth (poor conductors) – reduce conduction.

Cardboard box – reduces convection by limiting air movement.

(b) Add a tight-fitting lid (stops convection at the top) or extra layers of polystyrene foam/old cloth for better insulation.

5. Convection currents circulate air inside the refrigerator: cold air near the cooling coils sinks (being denser), warmer air rises, and the continuous flow keeps the temperature even throughout the fridge.

SUB-TOPIC 2.2.6: MEASUREMENT OF HEAT



INTRODUCTION

Measurement of heat focuses on quantifying heat using concepts like heat capacity and latent heat, emphasizing practical experiments and numerical problem-solving. This subtopic builds on kinetic theory from earlier sections and prepare learners for advanced topics like renewable energy in Form 4.

Specific competence

- Solve practical and numerical problems involving measurement of heat.



Key terms

- ❖ **Heat Capacity:** Is the amount of heat energy required to change the temperature of a substance by one degree Celsius (1°C) or one Kelvin (1K)
- ❖ **Specific Heat Capacity:** Is the amount of heat energy required to change the temperature of 1 kg of a substance by 1°C or 1K.
- ❖ **Latent Heat:** Is the amount of heat energy required to change the state (solid, liquid, and gas) of a substance without change in temperature.
- ❖ **Specific Latent Heat of Fusion:** Is the amount of energy required to change 1 kg of a substance from solid to liquid without changing its temperature.
- ❖ **Specific Latent heat of vaporization:** Is the amount of energy required to change 1 kg of a substance from liquid to gas at its boiling point without changing its temperature.

Learning Activities

This sub-topic, has 5 learning activities as outlined in the syllabus focusing on hands-on-activities and problem-solving tasks.



Activity 2.2.6.1.1: Exploring terms used in the measurement of heat (heat capacity, specific heat capacity, latent heat of fusion/vapourisation ...)

In this activity, a learner will explore terms used in the measurement of heat (heat capacity, specific heat capacity, latent heat of fusion/vapourisation ...) by conducting a practical task.

HOOK: On a hot afternoon in Lusaka, a learner decides to make a cold drink to cool down. The learner places 200 g of ice cubes at 0°C into a clean metal cup that has a heat capacity of $120 \text{ J}/^{\circ}\text{C}$. The learner then slowly pours 300 g of hot water at 90°C from a kettle into the cup. Assume that no heat is lost to the surroundings and that evaporation can be ignored. After some time, the ice melts completely and the mixture reaches a steady final temperature of 30°C , making the drink comfortably warm. Using this situation, answer the following questions:

- When the ice at 0°C changes into liquid water without a change in temperature, it absorbs a large amount of heat. Identify the heat concept involved and explain why the temperature remains constant during this change of state.
- After the ice has melted, the temperature of the water begins to rise. What concept would you use to determine the amount of heat required to raise the temperature of the water? Explain how this concept is used in calculations.
- The metal cup also absorbs heat as its temperature increases. Which concept is used to describe the amount of heat absorbed by the cup per degree rise in temperature (measured in $\text{J}/^{\circ}\text{C}$)? Explain how this concept differs from the one used to calculate the heat required to raise the temperature of water.
- Suppose the learner continues heating the mixture until the water reaches 100°C and begins to change into steam (vapour). Identify the heat concept that explains the large amount of heat required for this change of state without any further increase in temperature. Explain why the temperature remains constant during this process.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ice, ...
- ❖ **Artificial Materials:** Metal cup, bunsen burner, thermometer, electronic balance, beaker, tripod stand, kettle, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory.

Teacher's Role(s)

- ❖ Introduce the activity using a hook
- ❖ Divide learners in small groups.

- ❖ Provide learners with materials for the experiment
- ❖ Ask learners to research on the following terms; heat capacity, specific heat capacity, latent heat of fusion/vapourisation
- ❖ Ask learners explore the terms used in the measurement of heat by conducting an experiment (Heat capacity, specific heat capacity, latent heat of fusion/vapourisation)
- ❖ Ask learners to report their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Researching on the following terms; heat capacity, specific heat capacity, latent heat of fusion/vapourisation
- ❖ Exploring the terms used in the measurement of heat by conducting an experiment (Heat capacity, specific heat capacity, latent heat of fusion/vapourisation)
- ❖ Reporting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learners' ability to distinguish between Heat capacity and specific Heat capacity
- ❖ Assess learners' ability to distinguish between latent heat of fusion and latent heat of vapourisation.
- ❖ Assess learners' ability to communicate their findings from the practical task to the class.
- ❖ Assess learners' ability to relate terms on heat quantities to everyday life.



Skills Developed

- ❖ **Problem-Solving:** Applying knowledge of heat capacity to everyday life
- ❖ **Analytical Thinking:** Comparing heat capacity and specific heat capacity, and latent heat of fusion and latent heat of vapourisation.
- ❖ **Communication:** Presenting their discussions to the class.
- ❖ **Collaboration:** Working in pairs or small groups



Key Learning Points

- ❖ **Heat capacity (C):** Heat needed to raise the temperature of an object by 1 °C: ($C = \frac{Q}{\Delta T}$)
- ❖ **Specific heat capacity (c):** Heat needed to raise the temperature of 1 kg of a substance by 1 °C: ($c = \frac{Q}{m\Delta T}$) (water has a high value of 4180 J kg⁻¹ °C⁻¹).

- ❖ **Latent heat:** Heat absorbed or released during a change of state without temperature change. Latent heat of fusion ($Q = mL_f$) for melting or freezing; latent heat of vaporization ($Q = mL_v$) for boiling/condensing.



Activity 2.2.6.1.2: Solving Practical and Numerical Problems Involving Measurement of Heat in Everyday Life.

In this activity, learners will be able to solve practical and numerical problems involving measurements of heat in everyday life.

HOOK: Imagine it is a cool morning and you are making morning tea for your family. You pour 1.5kg of cold tap water (20°C) into two identical kettles: In kettle 1, you heat for 3 minutes and temperature reaches 60 °C (perfect for mixing with milk and tea leaves). In kettle 2, you keep heating the same amount of water for 6 minutes until it reaches near boiling at 95°C. Both kettles use the same amount of electricity per minute, we ignore heat losses.

Why does the water in kettle 2 end up much hotter than kettle 1, even though both start with the same water?

Using the formula, $Q = mc\Delta T$ and the specific heat capacity of water, $C = 4200\text{J/kg}^\circ\text{C}$. Calculate roughly how much more heat energy the water in kettle 2 gains as compared to

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water, ...
- ❖ **Artificial Materials:** Electric kettle, thermometers, stopwatch, electronic balance, calculators, worksheet, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be e.g. mobile laboratory

Teacher's Role(s)

- ❖ Introduce the activity using a scenario
- ❖ Divide class into small groups.
- ❖ Provide learners with materials for the experiment.
- ❖ Ask learners to conduct experiments involving measurement of heat.
- ❖ Ask learners to collect the data during the experiment using the materials provided

- ❖ Assign a worksheet with numerical questions to each group.
- ❖ Ask learners to solve numerical questions on measurements of heat from the collected data and the worksheet.
- ❖ Lead a guided discussion to help learners connect the solutions to the questions on some real-life situation
- ❖ Provide feedback to the learners on the solutions



Learners' Tasks

- ❖ Conducting experiments involving measurement of heat.
- ❖ Collecting data during the experiment using the materials provided
- ❖ Solving numerical questions on measurements of heat from the experimental data and the worksheet.
- ❖ Connecting the solutions to the questions on real-life situation through a guided discussion
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learner's participation during the experimental and numerical problem solving.
- ❖ Observe learners during the discussion leading into solving numerical problems.
- ❖ Assess learner's ability to communicate their findings through discussion.
- ❖ Assess learner's understanding on the measurements of heat.



Skills Developed

- ❖ **Scientific Inquiry:** Relating questions on measurements of heat to their everyday experiences.
- ❖ **Critical Thinking:** Analysing data obtained from the experiment.
- ❖ **Communication:** Sharing their findings and explaining the concept measurements of heat to the class.
- ❖ **Problem-Solving:** Applying the equations on different forms of heat quantities.



Key Learning Points

- ❖ **Heat capacity (C):** Heat needed to raise the temperature of an object by 1 °C: ($C = \frac{Q}{\Delta T}$)
- ❖ **Specific heat capacity (c):** Heat needed to raise the temperature of 1 kg of a substance by 1 °C: ($c = \frac{Q}{m\Delta T}$) (water has a high value of 4180 J kg⁻¹ °C⁻¹).

- ❖ **Latent heat:** Heat absorbed or released during a change of state without temperature change. Latent heat of fusion ($Q = mL_f$) for melting or freezing; latent heat of vaporization ($Q = mL_v$) for boiling/condensing



Activity 2.2.6.1.3: Exploring the Differences Between Temperature and Heat Energy.

In this activity, learners will explore the differences between temperature and heat energy using an experiment.

HOOK: A cup of boiling water and a large bucket of warm water are compared. The cup has a higher temperature but the bucket contains more total heat energy. Why would water in a bucket contain more heat energy despite being at lower temperature than the boiling water in the cup?

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Cooking oil and water (Equal masses e.g., 100 g each).
- ❖ **Artificial Materials:** Beakers, buckets, heat source (hot plate), thermometers, safety equipment, ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups.
- ❖ Provide learners with the materials for the experiment.
- ❖ Ask learners to conduct an experiment using a beaker, bucket and hot plat (source of heat) to explore the difference between temperature and heat energy.
- ❖ Ensure learners safety during the experiment.
- ❖ Guide learners in observing temperature rise in water or oil in a beaker and in a bucket.
- ❖ Ask learners to present their findings on the difference temperature and heat energy
- ❖ Provide feedback to learners on their exploration of the difference between temperature and heat.



Learners' Tasks

- ❖ Conducting an experiment using a beaker, bucket and hot plat (source of heat) to explore the difference between temperature and heat energy.
- ❖ Following safety protocols during the experiment.

- ❖ Observing temperature rise in water or oil in a beaker and in a bucket.
- ❖ Presenting their findings on the difference temperature and heat energy
- ❖ Receiving feedback on their exploration of the difference between temperature and heat.



Assessment Guide

- ❖ Assess learners' ability to set and use apparatus to explore the difference between heat and temperature.
- ❖ Assess learners' ability to collect and analyse data obtained in the task.
- ❖ Assess learners' ability to communicate their observation to the class
- ❖ Assess learners' ability to draw conclusions on their observations



Skills Developed

- ❖ **Critical Thinking:** Analysing data and drawing conclusions on the different between temperature and heat energy of a substances.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to setup apparatus and record the readings.



Key Learning Points

- ❖ Temperature is a measure of how hot/cold something is ($^{\circ}\text{C}$, K, $^{\circ}\text{F}$)
- ❖ Heat energy is the energy transferred because of a temperature difference (joules, J)



Activity 2.2.6.1.4: Measuring Heat Capacity ($C = \frac{Q}{\Delta T}$) and Specific Heat Capacity

($c = \frac{Q}{m\Delta T}$) of solids and liquids.

In this activity, learners will be measuring heat capacity and specific heat capacity of solids and liquids using an experiment.

HOOK: Two identical heaters are used to heat water and sand. After the same time the sand becomes hotter than the water. Discuss what caused the temperature difference in the two substances.

Suggested Teaching and Learning Materials

- ❖ **Natural Materials:** Water (cold or hot), ...

- ❖ **Artificial Materials:** Calorimeters, metal samples (aluminum, copper), thermometers, electronic balances, stopwatches, hot water bath, bunsen burner, source of electricity, insulation materials (e.g., polystyrene cups), ...

Learning environment set up

- Natural Environment such as school surroundings.
- Artificial Environment such as the laboratory or any room or place where learners can be (mobile laboratory).

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide learners with the materials for the experiment.
- ❖ Ask learners to discuss on how to use a calorimeter to do heat measurements
- ❖ Ask learners to setup a calorimeter and conduct an experiment to measure heat capacity and specific heat capacity of solids (metal) and liquids (water)
- ❖ Ask learners to discuss how they would use experimental data to calculate heat capacity and specific heat capacity.
- ❖ Ask learners to present their findings
- ❖ Provide feedback to the learners



Learners' Tasks

- ❖ Discussing on how to use a calorimeter to do heat measurements
- ❖ Setting up a calorimeter and conducting an experiment to measure heat capacity and specific heat capacity of solids (metal) and liquids (water)
- ❖ Calculating heat capacity and specific heat capacity.
- ❖ Presenting their findings
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learners' ability to take basic measurements of temperature change and mass of a metal and water.
- ❖ Assess learners' ability to connect temperature change to property of a material called heat capacity.
- ❖ Assess learner's ability to determine heat capacity and specific heat capacity experimentally and mathematically.
- ❖ Assess learners' ability to collect and analyse data obtained in the task.
- ❖ Assess learners' ability to communicate their observations to the class
- ❖ Assess learners' ability to draw conclusions on their observations



Skills Developed

- ❖ **Real world application:** Comparing which material heats up faster than the other
- ❖ **Critical Thinking:** Analysing data and drawing conclusions about how objects heat up or lose heat easily.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to weigh and record the mass and temperature readings.
- ❖ **Analytical thinking:** Choosing an appropriate formula to calculate heat capacity and specific heat capacity according to experimental set up (heat loss = heat gained)



Key Learning Points

- ❖ **Heat capacity (C):** Heat needed to raise the temperature of an object by 1 °C: ($C = \frac{Q}{\Delta T}$)
- ❖ **Specific heat capacity (c):** Heat needed to raise the temperature of 1 kg of a substance by 1 °C: ($c = \frac{Q}{m\Delta T}$) (water has a high value of 4180 J kg⁻¹ °C⁻¹).



Activity 2.2.6.1.5: Determining the Latent Heat of Fusion ($Q = mL_f$) and Latent Heat of Vaporization ($Q = mL_v$) of Substances.

In this activity learners will determine the Latent Heat of Fusion and Latent Heat of Vaporization of Substances by melting ice in warm water to measure quantity of radiation for fusion or boiling water to measure heat quantity of radiation for vapourisation.

HOOK: Ice is heated in a pot and learners observe that while the ice melts, the temperature stays the same even though heat is still being supplied. Later the boiling water also remains at a constant temperature while turning into steam. Why would temperature remain constant during these changes?

Stage 2

- ❖ **Natural Materials:** Hot water, Ice, cold water, ...
- ❖ **Artificial Materials:** Beakers, thermometers, electronic balances, heat source (for vaporization), calorimeters, and timers, ...

Learning environment set up

- **Natural Environment** such as school surroundings.

- **Artificial Environment** such as the laboratory or any room or place where learners can be (mobile laboratory).
- **Technological Environment** such as virtual laboratory, simulation, videos, ...

Teacher's role(s)

- ❖ Introduce the activity using a scenario
- ❖ Put learners into small groups
- ❖ Provide the learners with materials for the experiment
- ❖ Ask learners to use a calorimeter to collect the data for determining latent heats.
- ❖ Encourage learners to think critically about their observations and make inferences.
- ❖ Ask learners to calculate the latent heat of fusion and latent heat of vapourisation using experimental data collected.
- ❖ Ask learners probing questions
- ❖ Ask learners to present their findings
- ❖ Ask learners to apply latent heats e.g in steam engines
- ❖ Provide feedback to learners



Learners' Tasks

- ❖ Collecting the data for determining latent heats.
- ❖ Observing and making inferences on the experiment
- ❖ Calculating latent heat of fusion and latent heat of vapourisation using experimental data collected.
- ❖ Responding to probing questions
- ❖ Presenting their findings
- ❖ Applying latent heats in real life situations e.g in steam engines
- ❖ Receiving feedback from the teacher



Assessment Guide

- ❖ Assess learners' ability to determine latent heats experimentary.
- ❖ Assess learners' ability to observe and analyse data obtained using an experiment on latent heats.
- ❖ Assess learners' ability to communicate their observation to the class about latent heat and its application e.g. steam engine.
- ❖ Assess learners' ability to draw conclusions on their observations.



Skills Developed

- ❖ **Problem solving:** Applying latent heats to real world problems e.g. transportations using steam
- ❖ **Critical Thinking:** Analysing data and drawing conclusions about how ice melts and water boils without change in temperature.
- ❖ **Communication:** Presenting their findings to the class.
- ❖ **Collaboration:** Working in small groups to set and conduct experiments



Key Learning Points

- ❖ **Latent heat:** Heat absorbed or released during a change of state without temperature change. Latent heat of fusion ($Q = mL_f$) for melting or freezing; latent heat of vaporization ($Q = mL_v$) for boiling/condensing.
- ❖ Real world application of latent heat e.g. weather patterns, steam engines, thermal energy storage, refrigeration,

EXPECTED STANDARD

SAMPLE QUESTIONS

1. A learner heats 0.5 kg of cooking oil (specific heat capacity $\approx 2000 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$) and 0.5 kg of water (specific heat capacity $= 4180 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$) using the same heat source for the same amount of time. Both start at $25 \text{ }^\circ\text{C}$.

(a) Which liquid will have a higher temperature after heating? Explain your answer using the concept of specific heat capacity.

(b) Relate this to why water is often used in cooling systems (e.g., car radiators) instead of oil.

2. During cooking, a pot of water is brought to the boil at $100 \text{ }^\circ\text{C}$. Even though the stove continues to supply heat, the temperature of the water stays at $100 \text{ }^\circ\text{C}$ while it boils away.

(a) Name the type of heat involved here and explain why the temperature does not rise.

(b) Calculate the heat energy required to completely vaporize 0.2 kg of boiling water at $100 \text{ }^\circ\text{C}$ (specific latent heat of vaporization of water $= 2\,260\,000 \text{ J kg}^{-1}$). Show your working.

3. A learner adds 0.1 kg of ice at $0 \text{ }^\circ\text{C}$ to 0.4 kg of warm water at $40 \text{ }^\circ\text{C}$ in an insulated cup. After the ice melts completely, the final temperature is $28 \text{ }^\circ\text{C}$.

(a) Explain why heat is absorbed even though the temperature of the ice stays at $0 \text{ }^\circ\text{C}$ during melting.

(b) Name the two processes involved in the heat transfer to the ice (one for melting and one for warming the resulting water).

4. In a Zambian household, people often place hot charcoal in a brazier to warm a room during cold evenings. The room heats up gradually, but the metal brazier becomes very hot quickly.

(a) Identify the main heat transfer process that heats the brazier itself and the process that warms the air in the room.

(b) If the brazier has a mass of 2 kg and is made of iron (specific heat capacity $\approx 450 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$), calculate the heat absorbed by the brazier when its temperature rises from $25 \text{ }^\circ\text{C}$ to $200 \text{ }^\circ\text{C}$.

5. A solar water heater panel painted black is placed in direct sunlight. It absorbs heat quickly and transfers it to water flowing through pipes.

(a) Explain why black paint is used instead of shiny silver paint.

(b) Discuss how understanding specific heat capacity and latent heat could help improve the design to store heat for use at night.

SAMPLE QUESTIONS ANSWERS

1. (a) Cooking oil will have the higher temperature. It has a lower specific heat capacity, so the same amount of heat produces a larger temperature rise ($\Delta T = Q / (mc)$).

(b) Water's high specific heat capacity means it can absorb a lot of heat with only a small temperature increase – ideal for carrying away heat in car radiators without boiling quickly.

2. (a) Latent heat of vaporization. The heat energy is used to overcome forces between particles and change liquid water into vapour (change of state), not to increase the kinetic energy of the particles, so temperature remains constant at 100 °C.

(b) $Q = m L_v$

$$Q = 0.2 \text{ kg} \times 2\,260\,000 \text{ J kg}^{-1} = 452\,000 \text{ J}$$

3. (a) The heat is used as latent heat of fusion to break the bonds between ice particles and change it into liquid water without raising the temperature.

(b) (i) Latent heat of fusion (melting the ice at 0 °C)

(ii) Specific heat capacity of water (warming the melted water from 0 °C to 28 °C).

4. (a) Conduction heats the metal brazier quickly. Convection then carries the heat from the hot brazier to warm the air in the room.

(b) $Q = m c \Delta T$

$$\Delta T = 200 \text{ °C} - 25 \text{ °C} = 175 \text{ °C}$$

$$Q = 2 \text{ kg} \times 450 \text{ J kg}^{-1} \text{ °C}^{-1} \times 175 \text{ °C} = 157\,500 \text{ J}$$

5. (a) Black paint is a good absorber of radiant heat (infrared) from the sun, so the panel heats up faster. Shiny silver paint reflects most radiation and absorbs very little.

(b) Use materials with high specific heat capacity (or phase-change materials with high latent heat) inside the panel or in a storage tank. These can absorb large amounts of heat during the day and release it slowly at night without large temperature drops.

BIBLIOGRAPHY

1. Abbot, A. F. (2003). Ordinary Level Physics. Longman.
2. Giancoli, D. C. (2014). Physics: Principles with Applications (7th ed.). Pearson Education.
3. Halliday, D., Resnick, R., & Walker, J. (2013). Fundamentals of Physics. John Wiley & Sons.
4. Jones, M., & Jones, G. (2010). Cambridge IGCSE Physics Coursebook. Cambridge University Press.
5. Khan Academy. (n.d.). Fluids, Pressure, and Buoyancy.
6. Mvula, A. K. (2015). Excel and Advance in Physics Grade 10. Longhorn.
7. Nelkon, M., & Parker, P. (2008). Advanced Level Physics (8th ed.). Heinemann Educational Publishers
8. Open University. (2018). Teaching Secondary Physics: Forces and Pressure. Open Educational Resources (OER).
9. Physics Classroom. (n.d.). Pressure in Solids and Fluids.
10. Sears, F. W., Zemansky, M. W., & Young, H. D. (2013). University Physics with Modern Physics (13th ed.). Pearson Education.
11. Sears, F. W., Zemansky, M. W., & Young, H. D. (1987). University Physics. Addison-Wesley.
12. Serway, R. A., & Jewett, J. W. (2018). Physics for Scientists and Engineers. Cengage Learning.
13. Tipler, P. A., & Mosca, G. (2008). Physics for Scientists and Engineers. W.H. Freeman and Company.
14. UNESCO. (2017). Pedagogy of Science Education for Secondary Schools. UNESCO Publishing.
15. Verma, H. C. (2011). Concepts of Physics, Volume 1. Bharati Bhawan.
16. Young, H. D., & Freedman, R. A. (2019). University Physics. Pearson Education.
17. Curriculum Development Centre (CDC), (2024). Physics Syllabus Secondary Education Ordinary Level Form 1-4. Lusaka, Zambia.

BIBLIOGRAPHY

1. Abbot, A. F. (2003). Ordinary Level Physics. Longman.
2. Curriculum Development Centre (CDC), (2024). Physics Syllabus Secondary Education Ordinary Level Form 1-4. Lusaka, Zambia.
3. Giancoli, D. C. (2014). Physics: Principles with Applications (7th ed.). Pearson Education.
4. Halliday, D., Resnick, R., & Walker, J. (2013). Fundamentals of Physics. John Wiley & Sons.
5. Jones, M., & Jones, G. (2010). Cambridge IGCSE Physics Coursebook. Cambridge University Press.
6. Khan Academy. (n.d.). Fluids, Pressure, and Buoyancy.
7. Ministry of Education, Curriculum Development Centre, Environmental Health and Pollution Management Education Framework (2023)., Lusaka: Zambia
8. Ministry of Education, Curriculum Development Centre, Physics Syllabus (2013), Lusaka, Zambia.
9. Ministry of Education, Zambia Education Curriculum Framework (2023), Lusaka, Zambia.
10. Ministry of Finance and National Development, (2022). Eighth National Development Plan, Republic of Zambia
11. Mvula, A. K. (2015). Excel and Advance in Physics Grade 10. Longhorn.
12. Nelkon, M., & Parker, P. (2008). Advanced Level Physics (8th ed.). Heinemann Educational Publishers
13. Open University. (2018). Teaching Secondary Physics: Forces and Pressure. Open Educational Resources (OER).
14. Physics Classroom. (n.d.). Pressure in Solids and Fluids.
15. Sears, F. W., Zemansky, M. W., & Young, H. D. (1987). University Physics. Addison-Wesley.
16. Sears, F. W., Zemansky, M. W., & Young, H. D. (2013). University Physics with Modern Physics (13th ed.). Pearson Education.
17. Serway, R. A., & Jewett, J. W. (2018). Physics for Scientists and Engineers. Cengage Learning.
18. Tipler, P. A., & Mosca, G. (2008). Physics for Scientists and Engineers. W.H. Freeman and Company.
19. UNESCO. (2017). Pedagogy of Science Education for Secondary Schools. UNESCO Publishing.
20. Verma, H. C. (2011). Concepts of Physics, Volume 1. Bharati Bhawan.
21. Young, H. D., & Freedman, R. A. (2019). University Physics. Pearson Education.